

Received 10 January 2025, accepted 25 January 2025, date of publication 29 January 2025, date of current version 5 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3535943

SURVEY

Comprehensive Analysis of DDoS Anomaly Detection in Software-Defined Networks

ABDINASIR HIRSI¹, (Graduate Student Member, IEEE),
MOHAMMED A. ALHARTOMI^{2,3}, (Member, IEEE), **LUKMAN AUDAH^{1,4}**, (Member, IEEE),
ADEB SALH⁵, (Member, IEEE), **NAN MAD SAHAR⁴**,
SALMAN AHMED⁶, (Graduate Student Member, IEEE),
GODWIN OKON ANSA⁷, (Member, IEEE), AND **ABDULLAHI FARAH⁸**

¹Advanced Telecommunication Research Center, Faculty of Electrical and Electronic Engineering, Universiti Tun Hussein Onn Malaysia, Parit Raja 86400, Malaysia

²Department of Electrical Engineering, University of Tabuk, Tabuk 71491, Saudi Arabia

³Innovation and Entrepreneurship Centre (IEC), University of Tabuk, Tabuk 71491, Saudi Arabia

⁴Faculty of Electrical and Electronic Engineering, Universiti Tun Hussein Onn Malaysia, Parit Raja 86400, Malaysia

⁵Faculty of Information and Communication Technology, University Tunku Abdul Rahman (UTAR), Kampar 31900, Malaysia

⁶VLSI and Embedded Technology (VEST) Focus Group, Faculty of Electrical and Electronic Engineering, Universiti Tun Hussein Onn Malaysia, Parit Raja 86400, Malaysia

⁷Department of Computer Science, Faculty of Physical Sciences, Akwa Ibom State University, Akwa Ibom 520101, Nigeria

⁸Engineering Department, Somtel Telecommunication Company, Bosaso, Bari 25290, Somalia

Corresponding authors: Lukman Audah (hanif@uthm.edu.my) and Adeb Salh (adebali@utar.edu.my)

This work was supported by the Ministry of Higher Education (MOHE) through Fundamental Research Grant Scheme under Grant FRGS/1/2022/TK07/UTHM/02/25.

ABSTRACT Software-Defined Networking (SDN) offers significant advantages for modern networks, including flexibility, centralized control, and reduced dependency on vendor-specific hardware. However, these benefits introduce security vulnerabilities, particularly from Distributed Denial-of-Service (DDoS) attacks, which represent some of the most disruptive threats to SDN environments. A review of the literature shows that while various techniques have been proposed to counteract DDoS threats, many studies have focused on single detection methods, with only a few utilizing multiple approaches. This fragmented focus limits a comprehensive approach to addressing DDoS threats across the SDN layers. To bridge this gap, this paper presents the first comprehensive review of DDoS anomaly detection in SDN, examining over 165 primary research articles published between 2020 and 2024. A novel taxonomy of DDoS attacks is introduced, categorizing them by distinct characteristics, and mapping each attack type to relevant detection methods within specific SDN layers. The survey provides a layer-by-layer analysis of DDoS detection techniques, covering the application, control, and infrastructure layers, and offers a structured overview that clarifies the applicability and effectiveness of each method. The paper concludes by synthesizing key findings, identifying unresolved challenges, and outlining future research directions to advance DDoS detection mechanisms in SDN. This roadmap is designed to guide researchers in addressing security vulnerabilities and enhancing SDN resilience against evolving DDoS threats.

INDEX TERMS Amplification attacks, anomaly detection, botnet attacks, DDoS detection, flooding attacks, protocol attacks, reflection attacks, SDN security, software-defined networking (SDN), taxonomy of DDoS attacks.

The associate editor coordinating the review of this manuscript and approving it for publication was Derek Abbott¹.

I. INTRODUCTION

The rapid advancement of next-generation technologies demands the frequent reconfiguration of network policies and regular device upgrades to accommodate increasingly complex requirements [1]. However, conventional networks

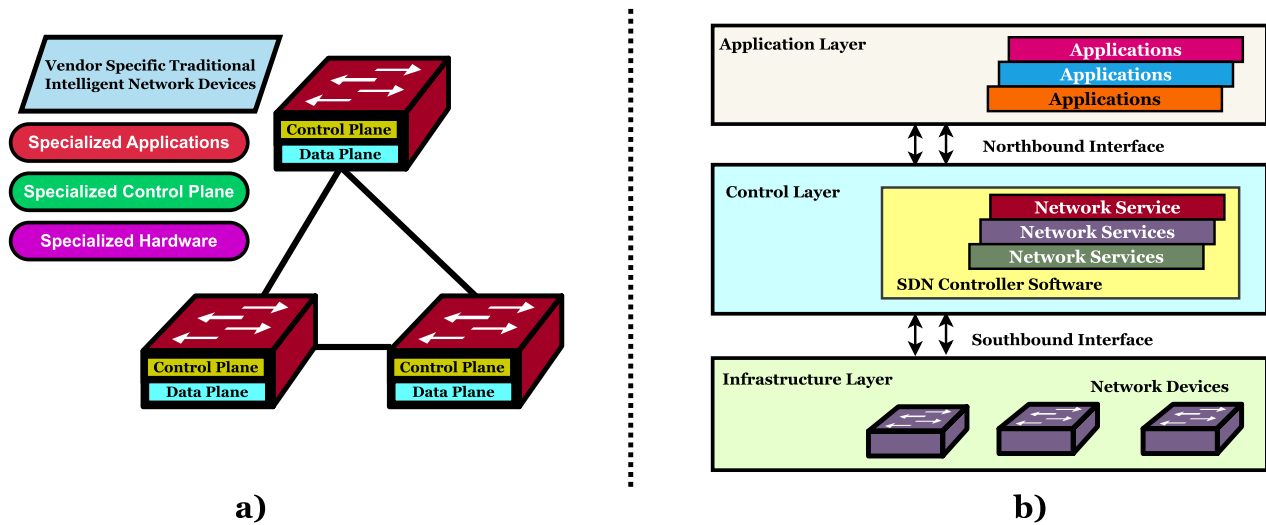


FIGURE 1. Comparison between traditional network architecture and SDN architecture. Traditional networks rely on vendor-specific, tightly coupled planes, with specialized applications, control, and data planes embedded within each device. In contrast, SDN decouples these layers, enabling centralized control, simplified management, and flexibility through separate application, control, and infrastructure layers.

struggle to meet these demands owing to their rigid architectures and limited flexibilities [2]. Traditional network setups are tightly coupled across all network planes, including the application plane, which hosts critical security components such as Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) [3]. This interdependence complicates the integration of new features or upgrades, as modifications in one layer often require extensive adjustments throughout the entire network (see Figure 1 for a comparison of traditional network architectures and SDN architecture) [4].

Moreover, updating or reconfiguring conventional networks is both complex and costly [5]. These networks rely heavily on hardware-specific configurations, requiring administrators to manage disparate components from multiple vendors individually, a process that significantly increases operational expenses and administrative complexity [6]. Additionally, the requirement for a unique network operating system for each vendor introduces substantial software costs beyond hardware expenses [7]. Consequently, conventional networks lack scalability, adaptability, and cost-effectiveness, which are essential to support the dynamic needs of modern technological advancements [8], [9].

Software-Defined Networking (SDN) offers an effective solution to the limitations of conventional networks by introducing a flexible software-driven approach to network management [10], [11]. One of SDN's most compelling advantages of SDN is its decoupling of the control plane from the data plane, which enables centralized control and streamlined oversight across the network [12], [13]. This architecture allows administrators to manage multiple networks through a single unified interface, greatly simplifying network operations and eliminating the need for vendor-specific configurations [14]. SDN flexibility also facilitates the separation of experimental and production environments,

allowing researchers to test and innovate without risking disruptions to the core network functions [15]. Unlike traditional networks, which require decentralized management and individual configurations for each device and vendor, SDN centralizes control, enabling efficient and coordinated management across the network [16]. This centralization alleviates the shortage of skilled network personnel by reducing the time and effort required for network maintenance and troubleshooting [17].

Additionally, SDN significantly reduces costs by minimizing dependence on vendor-specific devices. Its open standards and compatibility with generic hardware allow organizations to lower hardware expenses and eliminate costly software licensing fees, making SDN a more economical option than traditional network models [18]. As a transformative technology, SDN has revolutionized data centers by enabling rapid configuration changes and customized network modifications to meet the demands of dynamic modern environments.

Despite its benefits, SDN faces significant security challenges, with Distributed Denial-of-Service (DDoS) attacks being among the most disruptive ones. DDoS attacks exploit SDN's centralized control structure of SDN, targeting the control plane to overwhelm the network and compromise availability. These attacks are particularly concerning as they threaten both the SDN controller and the broader network infrastructure, presenting substantial challenges in maintaining reliable network performance [19]. Recent incidents have highlighted the growing sophistication and scale of DDoS attacks. For example, Nokia's Threat Intelligence Report (2023) emphasizes the intensifying nature of DDoS attacks, with one of the largest ACK flood attacks targeting major internet service providers, resulting in severe service disruptions [20]. Similarly, botnet-driven DDoS attacks have

recently impacted China and the United States, with compromised devices distributed across countries, such as Japan and Brazil, underscoring the global nature of these threats. Additionally, Yandex experienced an unprecedented DDoS attack, peaking at 21 million Hypertext Transfer Protocol (HTTP) requests per second, a new benchmark in DDoS scale and impact [21]. In light of these developments, strengthening DDoS defenses in SDN environments is crucial for ensuring network resilience and preserving SDN's advantages of SDN in centralized management and flexibility. Securing SDN against these threats is essential for supporting its role as a cornerstone of next-generation networking.

A. SCOPE AND CONTRIBUTIONS OF THE SURVEY

DDoS attacks represent one of the most critical security challenges in SDN and significantly affect the network performance and availability. Numerous survey papers have attempted to address DDoS detection within SDN, each providing valuable insights. However, many of these surveys lack a comprehensive taxonomy to guide readers through the complexities of DDoS attack types and various detection methods tailored to SDN environments. Existing surveys often focus on single detection techniques, such as classification approaches, which limit their scope and applicability for addressing the full range of DDoS threats in SDN. Others have analyzed attacks and defenses without examining the distinct vulnerabilities across SDN layers, namely, the infrastructure, control, and application layers. Additionally, several surveys fail to provide in-depth discussions, actionable insights, and detailed comparisons of detection techniques, leaving readers without a clear understanding of the practical applicability of each method within an SDN context.

To the best of our knowledge, this paper presents the most comprehensive review of DDoS detection in SDN to date, analyzing over 165 original research articles published between 2020 and 2024. Our survey introduced a novel taxonomy of DDoS attacks, categorizing them according to their unique characteristics and impact on different SDN layers. Furthermore, each section offers key insights, detailed discussions on detection methods, and an analysis of techniques specific to each SDN layer, providing both newcomers and experts with valuable resources for navigating DDoS detection strategies within SDN environments. The primary scope and contributions of this study are as follows:

- **Novel Taxonomy of DDoS Attacks in SDN:** We introduce a unique taxonomy that categorizes DDoS attacks based on their mechanisms and impact within the SDN architecture. Covering attack types such as flooding, application-layer, protocol-based, resource exhaustion, botnet-driven, amplification, and reflection attacks, this taxonomy offers a structured framework that clarifies the range of DDoS threats specific to SDN, benefiting both novice and expert readers.
- **Comprehensive Review of Detection Methods for Each Attack Type:** For attack category in the

taxonomy, this survey provides a detailed analysis of the detection techniques, including machine learning, statistical methods, clustering, reinforcement learning, and traditional security frameworks. By systematically organizing and evaluating these methods, the survey highlights each technique's strengths, limitations, and contexts of application, guiding researchers in selecting appropriate detection methods for various DDoS attacks in SDN.

- **Visual Guide to DDoS Detection Techniques:** A dedicated figure summarizes the DDoS detection techniques discussed and categorized by approach (e.g., classification, clustering, statistical analysis, moving target defense). This visual guide serves as a quick reference, enabling researchers and practitioners to efficiently locate detection methods that align with specific attack categories.
- **Summary and Key Insights:** Each major section concludes with a summary and insights that synthesize key points. These summaries reveal effective detection techniques, outstanding challenges, and emerging trends, offering readers clear takeaways and enhancing the survey's value as a reference tool.
- **Comparative Results and Discussion of Detection Techniques:** The dedicated results and discussion section synthesizes findings on the effectiveness and scalability of various detection techniques across SDN attack types. This comparative analysis provides practical guidance for selecting methods suited to specific attack vectors, supporting informed decision making for practitioners and researchers.
- **Identification of Research Gaps and Future Directions:** The survey identifies several underexplored areas in DDoS detection research for SDN, such as hybrid frameworks, clustering techniques, and adaptive detection models. By highlighting these gaps, the survey offers clear directions for future research, encouraging advancements in detection systems that address evolving DDoS challenges in SDN.
- **Valuable Resource for Diverse Audiences:** With its structured taxonomy, organized review of detection methods, section-wise summaries, and in-depth discussions, this study serves as a comprehensive resource. Newcomers gain foundational knowledge of DDoS attacks in SDN, while experienced researchers access recent advancements and specialized techniques, fostering continued innovation in SDN security.

Roadmap: The remainder of this paper is organized as follows. Section II reviews related survey papers on DDoS detection in SDN and highlights their limitations, setting the context for our contributions. Section IV presents an overview of foundational concepts in SDN and DDoS attacks, including SDN architecture, DDoS detection techniques, and challenges specific to integrating DDoS detection into SDN layers. Section V introduces our novel taxonomy of DDoS attacks in SDN, categorizing them by attack type and their

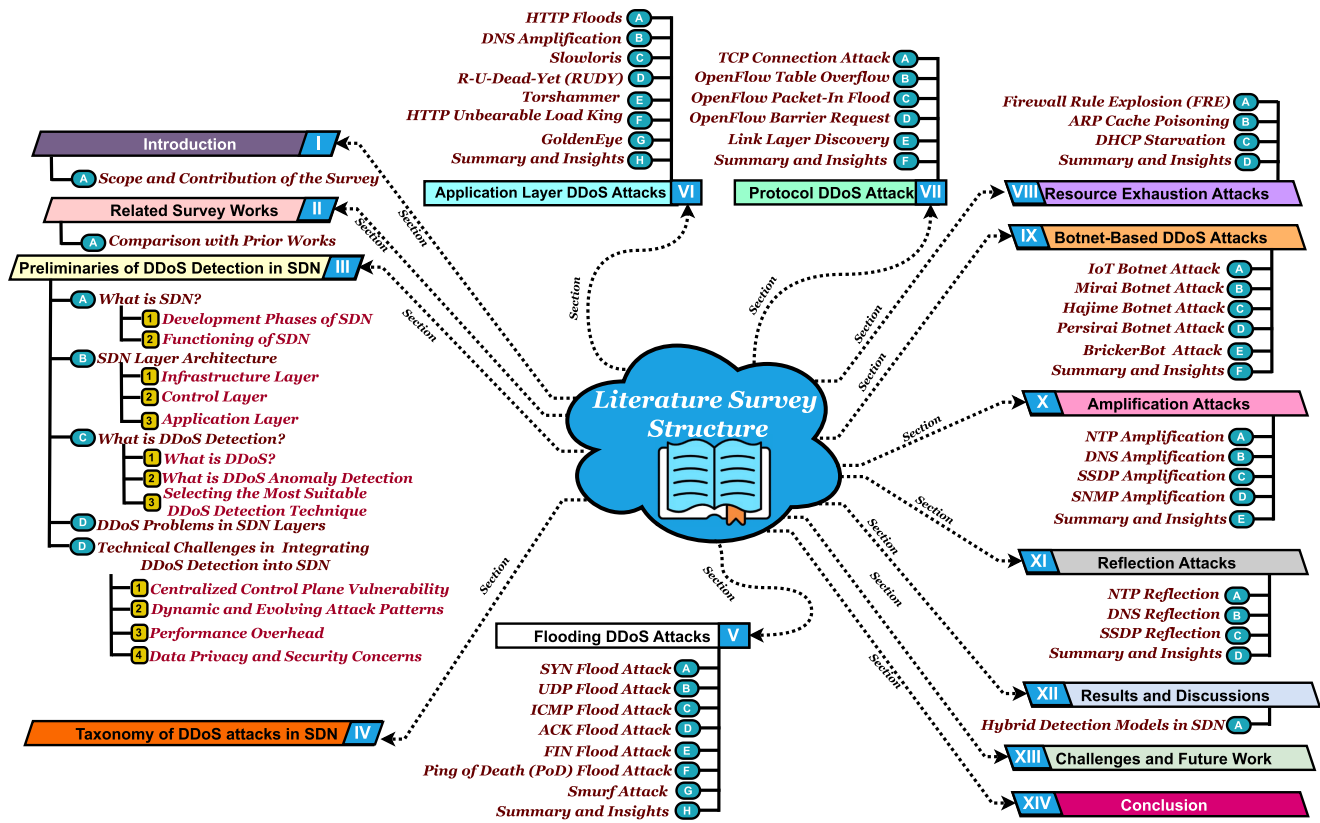


FIGURE 2. Paper organization.

impact across different SDN layers. Sections VI–XI delve into specific categories of DDoS attacks, including flooding, application layer, protocol-based, resource exhaustion, botnet-driven, amplification, and reflection. Each section provides a summary of the detection methods, key insights, and a discussion of the techniques relevant to each SDN layer. Section XII presents a comparative analysis of our Results and Discussion section, evaluating the effectiveness and scalability of the various detection methods. Section XIII outlines the challenges and avenues for future research on DDoS detection in SDN. Finally, we conclude the paper in Section XIV. An outline of the survey is shown in Figure 2.

We employed a rigorous paper selection process to ensure a comprehensive and systematic review, as outlined in Figure 3. The process began with the identification of relevant articles using predefined search protocols across major databases including IEEE Xplore, Scopus, and Web of Science. The search criteria incorporated keywords such as “SDN,” “DDoS detection,” “security,” “hybrid models,” and “adaptive mechanisms,” targeting publications between 2020 to 2024. Initial screening was conducted by manually reviewing the titles and abstracts of 1,450 records to assess their relevance to the study. Articles that met the inclusion criteria, such as peer-reviewed journal papers, conference proceedings, and book chapters published in English, were selected for a full-text review. Quality assessment was then

performed on the full-text articles to ensure methodological rigor and relevance to SDN-based DDoS detection research. Finally, 167 studies were included in this review, providing valuable insights into the taxonomy of DDoS attacks, detection techniques, and future research directions.

II. RELATED SURVEY WORKS

While several surveys have addressed DDoS attacks in SDN, many are limited to specific areas or methodologies and lack a comprehensive view. These studies often fall short in providing a detailed taxonomy of attacks or in holistically covering advanced detection techniques. To address these gaps, our survey introduces a novel taxonomy of DDoS attacks in SDN and examines both traditional and emerging detection methods. Table 1 offers a comparative analysis between this survey and previous studies, evaluating the extent of coverage across several dimensions, including attack taxonomy, SDN layers, evaluation tools, and research challenges. Additionally, it assesses detection approaches, such as classification methods, statistical techniques, clustering, and advanced models, such as reinforcement learning and generative techniques. The table also highlights the integration of novel procedures, such as moving-target defence, which underscores the need for more adaptive and resilient detection mechanisms. By addressing underexplored areas, such as clustering techniques and hybrid detection

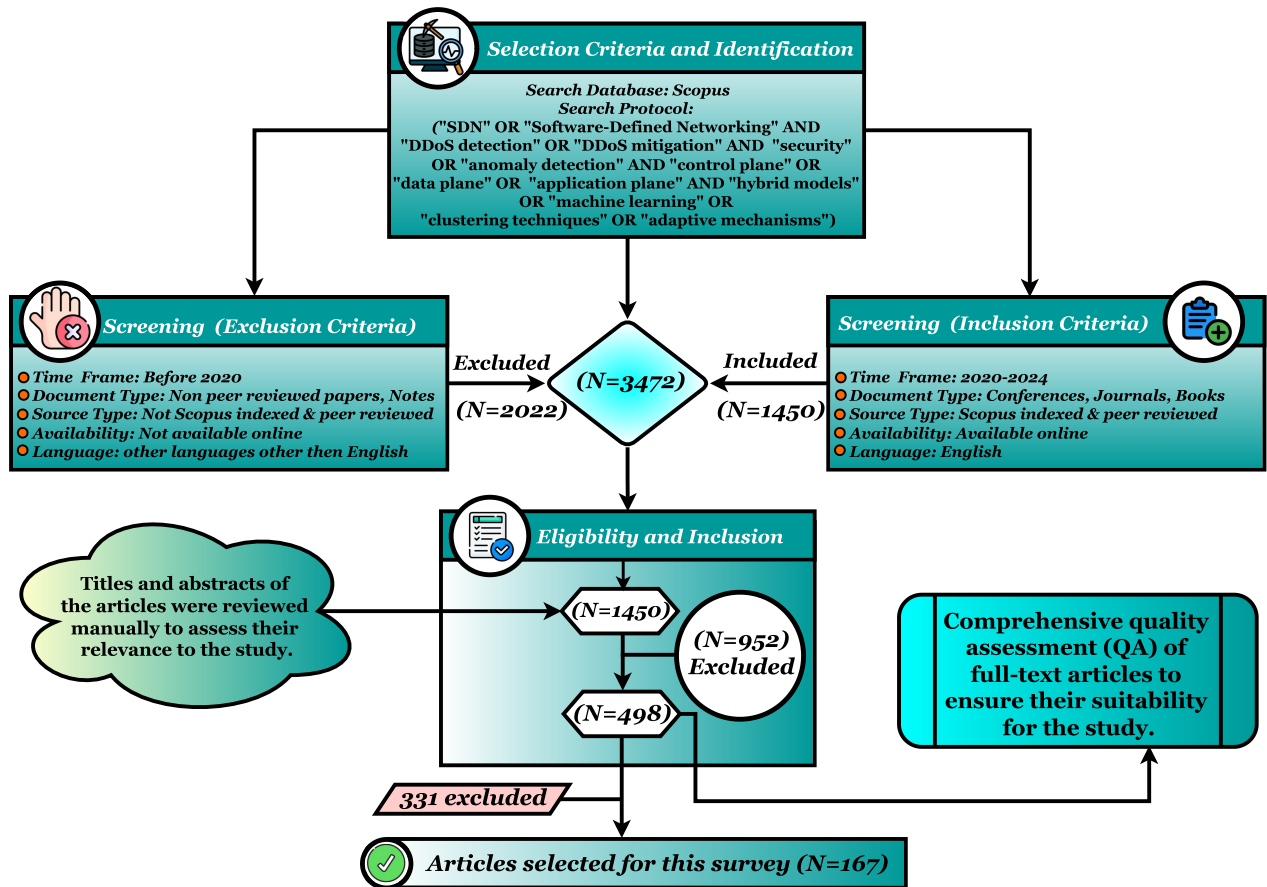


FIGURE 3. Literature source selection process following PRISMA guidelines.

frameworks, often only partially covered or overlooked in prior research, our survey aimed to provide a broader and more comprehensive perspective. The findings from the comparison in Table 1 are elaborated upon in the following paragraphs.

Jain et al. [22] presented a comprehensive survey of DDoS detection, mitigation, and mitigation policies within SDN, emphasizing vulnerabilities across the data, control, and application planes. Their work reviewed mainstream detection techniques, such as ML-based methods, statistical methods, and threshold-based approaches, while also discussing protection solutions, such as blockchain, honeypots, and network slicing. Although their survey identified open challenges related to datasets, tools, and evaluation metrics, it focused primarily on existing methods without delving into more advanced, emerging approaches. In contrast, our survey distinguishes itself by introducing a novel and granular taxonomy of DDoS attacks, classifying them into specific categories, and providing deeper insights into SDN-specific attack dynamics. In addition, while the authors highlight traditional detection techniques, our survey pushes further by evaluating hybrid frameworks and clustering-based approaches that address the limitations of single-method detection.

Kaur et al. [23] focused on DDoS fortification solutions within the SDN framework by employing a Systematic Literature Review (SLR) methodology. It categorizes DDoS attacks based on their targets within the SDN architecture, such as the control plane, data plane, and communication channels. The study emphasizes traditional detection approaches, such as entropy-based techniques, filtering, and ML classifiers, while also identifying key vulnerabilities across SDN layers. Furthermore, it highlights challenges such as the multi-controller architecture, dynamic thresholds, and testing environments. However, the survey primarily focuses on traditional DDoS guard schemes, leaving gaps in areas such as IoT botnets, reflection attacks, and advanced DL-based detection methods, which aim to be addressed through a broader taxonomy and exploration of emerging detection technologies.

In [24], the authors presented DDoS mitigation methods in an SDN environment specifically tailored to the IoT scenarios. The primary motivation is the lack of comprehensive IoT-specific SDN solution mechanisms that can adapt to the unique characteristics and vulnerabilities of IoT ecosystems, such as limited device resources and high granularity. The authors found that although SDN provides programmable capabilities to enhance IoT security, current solutions often

TABLE 1. Comparison with other survey papers. Acronyms: - CP: Covered Paper, PD: Period, DT: Detail Taxonomy of DDoS attacks, ET: Evaluation Tools, SL: SDN Layers, FW: Detail research challenges and future work CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Ref.	CP	PD	Focus of Discussion	E T	S L	D T	F W	DDoS Detection Method							
								C A	S A	T S	C T	N N	G M	R L	M T
[20], 2024	27	2019-23	This paper provides an overview of DDoS attacks in IoT, emphasizing AI-based detection techniques for IoT networks.	✓	*	*	✓	X	X	✓	X	X	X	X	X
[22], 2024	114	2020-24	This survey reviews detection frameworks, including ML, statistical, and blockchain methods, with minimal focus on innovative or hybrid approaches	*	✓	*	✓	✓	✓	✓	X	X	X	X	X
[23], 2021	75	2015-21	This is an SLR review paper identifying vulnerabilities, defence mechanisms, and research gaps within specific layers of the SDN framework.	*	*	*	✓	✓	✓	X	X	X	X	X	X
[24], 2020	25	2015-20	The paper provides a taxonomy of DDoS mitigation approaches in SDN-based IoT, examining SDN control plane and IoT environments.	*	*	*	✓	✓	✓	✓	X	X	X	X	X
[25], 2021	38	2013-20	Provides a taxonomy for DDoS detection and mitigation strategies, covering methods such as statistical, ML, blockchain, NFV, honeynets, and MTD.	*	*	*	✓	✓	✓	✓	✓	X	X	X	✓
[26], 2024	30	2015-24	This survey provides a layered taxonomy of DoS and DDoS attacks in edge computing, specifically targeting vulnerabilities across edge layers.	✓	*	*	✓	✓	✓	X	X	✓	X	X	X
[27], 2022	-	2001-21	This paper presents a taxonomy and mitigation methods for LDoS attacks, covering statistical and ML-based detection with minimal SDN focus.	✓	*	*	✓	X	X	✓	X	X	X	X	X
[28], 2024	80	2018-23	This survey examines DDoS detection in IoT, focusing on IoT constraints and using ML, statistical approaches, and simulation.	✓	X	*	*	✓	X	X	X	X	X	X	X
[29], 2024	-	2020-22	This survey addresses DDoS attacks in IoT, classifying detection methods with a focus on AI-based detection approaches for IoT environments.	X	X	*	*	✓	✓	X	X	X	X	X	X
This paper	167	2020-25	This paper is the first comprehensive review of DDoS anomaly detection in SDN, covering over 165 recent studies and introducing a novel taxonomy that categorizes DDoS attacks by SDN layers. It uniquely presents a structured, layer-by-layer analysis of detection methods, offering fresh insights into their application across the SDN architecture.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
✓ indicates the topic is well covered, X is uncovered, and * is partially covered.															

struggle with scalability and adaptability to different types of DDoS attacks. However, their main focuses were on IoT-centric layers and architectures within SDN (like edge and fog computing), our survey provides a more extensive examination of SDN layers (control, data, and application) affected by DDoS across different scenarios, allowing for a more comprehensive understanding of the impact on SDN architecture as a whole.

Valdovinos et al. [25] conducted a survey on a taxonomy of DDoS detection strategies focusing on various methods such as blockchain and network function virtualization (NFV). The authors categorized strategies into three main types: enhancements to SDN’s inherent security capabilities across the application, control, and data planes; the use of SDN as an enabler for adaptive security measures such as virtualized IDS/IPS; and emerging technologies such as NFV and blockchain, which provide dynamic and decentralized options for DDoS mitigation. Their findings highlight the need for integrated and adaptable solutions that leverage these advanced methods to strengthen SDN resilience against evolving attack vectors. However, while their survey includes emerging techniques, such as honeynets and network slicing, its primary focus is on mitigation strategies. By addressing these underrepresented detection approaches, our survey aims to expand the discussion on evolving detection capabilities in SDN, offering a

more comprehensive perspective on state-of-the-art defence mechanisms.

Chamoli and Mittal [20] presented a review of DDoS attacks in IoT networks, emphasizing how the rapid proliferation of IoT devices has heightened vulnerability to these attacks. Their primary focus was on AI-based detection techniques for IoT-based DDoS attacks. The authors provide a comprehensive overview of AI methods, the IoT botnet lifecycle, various IoT botnet families, the taxonomy of IoT-based DDoS attacks, and a comparative analysis of publicly available IoT datasets. Their taxonomy classifies DDoS attacks into application, transport, volumetric, and zero-day. While this study primarily focuses on IoT-based DDoS attacks, datasets, and AI models, its scope differs significantly from that of the SDN environment. By contrast, our taxonomy categorizes DDoS attacks according to SDN-specific vulnerabilities and layers, such as resource exhaustion and protocol-based attacks, making it more relevant to the SDN context.

Uddin et al. [26] discussed DDoS attacks in the context of edge computing, an environment that bridges IoT devices and cloud systems. Their taxonomy categorizes DoS and DDoS attacks within edge computing into two main types: distributed DoS and non-distributed DoS. The paper also presents an analysis of 30 research-based detection and prevention systems, comparing their effectiveness, limitations,

and applicability across edge layers. While their survey addressed some prevention mechanisms, including federated learning and SDN integration for mitigating DDoS in edge environments, our comprehensive review focuses on a broader range of detection techniques specifically tailored for SDN architectures.

Sudar and Deepalakshmi [27] conducted a survey on Low-Rate Denial-of-Service (LDoS) attacks, a stealthy form of DoS that mimics legitimate traffic patterns, thus making them challenging to detect. The paper summarizes recent detection and mitigation strategies, including statistical methods, ML, and frequency-domain analysis, aimed at accurately identifying LDoS traffic without impacting legitimate user experience. It categorizes LDoS attacks into three main types based on their operational methods: QoS, slow DoS, and service queue attacks. While the authors proposed a taxonomy of LDoS attacks, they did not present a detailed, novel taxonomy of DDoS attacks in SDN. In contrast, our survey offers a comprehensive taxonomy of DDoS attacks specific to SDN, examining how the SDN architecture influences attack vectors and detection methods across various layers (e.g., the control plane and data plane).

Kadri et al. [28] discussed denial-of-service attacks in IoT environments, highlighting the unique security challenges faced by IoT devices due to resource limitations, heterogeneous architecture, and low-power, low-bandwidth constraints. The survey emphasizes the need for more empirical validation of real IoT testbeds to enhance the practical applicability of detection techniques, and notes a lack of holistic approaches that effectively integrate both theoretical and practical validation aspects. Their taxonomy classifies DDoS attacks and detection approaches based on IoT network architecture, covering Layer-Based Attacks and Detection Techniques. In contrast, our survey focused on SDN-specific DDoS attacks, categorizing attacks within SDN environments, such as Resource Exhaustion Attacks (including Firewall Rule Explosion), ARP Cache Poisoning, and DHCP Starvation, along with other SDN-layered vulnerabilities. Our taxonomy centers on SDN's layered architecture and its distinctive detection challenges.

Shukla et al. [29] examine IoT-specific vulnerabilities, emphasizing how the limited resources and weak security of IoT devices make them susceptible to DDoS attacks. The paper reviews AI-based DDoS detection methods for IoT and categorizes attacks according to the IoT layer they target (perception, network, application), detailing vulnerabilities within each. While the authors focused primarily on AI techniques for detection within IoT environments, they did not comprehensively cover all detection methods. In contrast, our survey adopts a broader approach by presenting a taxonomy of DDoS attacks across multiple SDN layers and discussing a wider range of detection techniques beyond AI.

III. COMPARISON WITH PRIOR WORK

Several surveys have explored DDoS detection in SDN environments; however, our work offers unique contributions

that set it apart from prior reviews. Specifically, this survey is based on 167 primary research articles published between 2020 and 2024, offering an extensive and updated analysis of DDoS attacks on SDN. Drawing from a wide range of studies, our work provides valuable insights for both novice and experienced practitioners. Novices benefit from structured taxonomy and foundational overviews, whereas experts gain deeper insights into advanced detection methods and emerging research directions.

This survey is distinguished through its well-structured approach, providing clear summaries, insights, and detailed visualizations for each section. Unlike existing surveys, we integrated figures, tables, and taxonomies to present the information comprehensively and systematically. Each section concludes with a clear summary that synthesizes key findings, enabling readers to quickly grasp the core contributions of the reviewed studies. Furthermore, insights were drawn from the analysis of detection techniques, tools, and datasets, ensuring that the survey serves as a practical resource for researchers.

Our work introduces a novel taxonomy of DDoS attacks, explicitly mapping threats to SDN-specific vulnerabilities across application, control, and data layers. This granular approach contrasts with the generalized classifications seen in previous studies and provides deeper insights into how DDoS attacks exploit SDN architecture. Moreover, we evaluated hybrid frameworks, clustering-based approaches, and adaptive models to address the limitations of single-method detection systems. In contrast to Jain et al. [22], which focuses on mainstream detection methods, such as ML-based and statistical approaches, our work explores advanced frameworks, including reinforcement learning and moving-target defense strategies. Similarly, while Kaur et al. [23] employed a Systematic Literature Review to categorize SDN vulnerabilities and traditional defense schemes, their work lacks the layered taxonomy and detailed evaluation of the emerging detection technologies that we present.

Prior surveys such as Valdovinos et al. [25] and Uddin et al. [26] emphasized mitigation strategies such as blockchain and NFV, but provided limited focus on detection methodologies. Our survey addresses this gap by categorizing detection techniques based on their applicability to SDN layers and emphasizing the practical implementation challenges. Additionally, while Sudar and Deepalakshmi [27] concentrated on LDoS attacks, our survey comprehensively addressed a broader spectrum of DDoS attack categories, including resource exhaustion, botnet-driven attacks, and reflection amplification.

Table 1 highlights the breadth and depth of our survey, comparing it with prior studies in terms of the scope of taxonomy, SDN layer coverage, inclusion of evaluation tools, and detailed discussions of research challenges. By providing a clear structure, integrating comprehensive visualizations, and addressing underexplored areas, this survey serves as a unique and valuable resource for advancing our understanding of DDoS detection in

SDN environments, benefiting both novice and experienced researchers.

IV. PRELIMINARIES OF DDoS DETECTION IN SDN

A. WHAT IS SDN

In this section, we explain what SDN is by outlining its development phases and core functionality.

1) DEVELOPMENT PHASES OF SDN

SDN has evolved through distinct phases, marked by significant technological advancements and shifts in network-management paradigms. This evolution can be broadly categorized into three main eras: the inception, development, and current eras. The first era was the early 2000s, particularly the Ethane project at Stanford University, which aimed to separate control and data planes in networking [30]. This separation is crucial because it enables centralized control, enhanced programmability, and manageability. The introduction of the OpenFlow protocol in 2008 marked a significant milestone, facilitating communication between the control and data planes and allowing dynamic and efficient traffic management [31]. In the same year, the NOX operating system emerged as one of the first SDN controllers to utilize OpenFlow to manage network devices [32]. Nevertheless, the Internet Engineering Task Force (IETF) previously attempted to standardize plane separation with proposals such as Forwarding and Control Element Separation (ForCES) in 2004, which gained traction only with the advent of OpenFlow and the NOX platform, showcasing SDN's practical applications of SDN in academic and research networks [33].

The second era followed the inception era, to increase adoption and innovation. In this era, the establishment of organizations such as the Open Networking Foundation (ONF) in 2011 aimed to promote SDN and OpenFlow technologies [34]. During this era, SDN began to penetrate commercial environments, with companies such as Nicira deploying SDN solutions for data center management [35]. Notably, Google's B4 network, launched in 2012, utilizes SDN principles to enhance its data center interconnectivity [36]. The rise of Network Function Virtualization (NFV) has also played a crucial role during this phase, as it complemented SDN by virtualizing network services that traditionally run on proprietary hardware.

The third era of SDN, often referred to as SDN 3.0, is characterized by a more application-centric approach. This era emphasizes the use of centralized controllers and policy-based automation, allowing for more intelligent and dynamic network management. Technologies such as OpenStack and its Open Virtual Network (OVN) have become integral to SDN, enabling the seamless integration of SDN capabilities with cloud-native applications [37]. The focus has shifted towards creating a more holistic networking environment that supports automation, orchestration, and advanced analytics. Moreover, the integration of emerging technologies, such as AI including ML and DL, into SDN frameworks enhances the

ability to proactively predict network behavior and optimize performance [38]. This era is marked by a strong emphasis on efficiency, scalability, and the ability to meet the demands of modern applications, particularly in the context of cloud computing and the IoT.

2) FUNCTIONING OF SDN

As the name suggests, SDN is a network architecture approach that separates the network control plane from the data plane, allowing for more flexible and dynamic management of network resources through software. SDN guarantees centralized control and management of a network using software applications and enables programmable network behavior through open Application Programming Interfaces (APIs) [39]. In traditional networks, the control and data planes are tightly coupled in individual hardware devices such as routers and switches, making networks complex and difficult to configure and manage [40]. Therefore, SDN functionality improves network agility, as changes can be made via software instead of manual hardware configuration and reduces network complexity by abstracting the control plane [41]. All components of SDNs operate under centralized control for seamless resource allocation, utilizing secure communication protocols that adhere to the principles of programmable networking, thereby enabling the efficient detection and mitigation of DDoS threats.

B. SDN LAYER ARCHITECTURE

As shown in Figure 4, the SDN architecture is typically segmented into three primary layers: the data plane, control plane, and application plane. These layers collaboratively formed a flexible and programmable network framework. Therefore, to comprehensively understand the vulnerabilities inherent in SDN, it is crucial to first describe these layers. This section provides an overview of SDN layers and details their respective roles, components, and functionalities. The subsequent sections examine vulnerabilities specifically in the context of DDoS attacks. However, readers interested in an in-depth analysis of SDN layers, including mitigation strategies, security threats, and developmental aspects, can study the discussions in [1], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], and [20].

1) INFRASTRUCTURE LAYER

This layer, also known as the data plane, forms the foundation of the SDN architecture. It comprises the physical hardware that constitutes the network, including switches, routers, and other networking devices [42]. Therefore, they are responsible for the actual transmission of data packets across networks [43]. Physical switches and routers operating at layer 2 (Data Link) and layer 3 (network) of the OSI model forward packets based on predefined rules, with switches managing local traffic and routers handling inter-segment routing [41], [44]. Network interfaces, in addition to their primary role, act as critical connection points between the

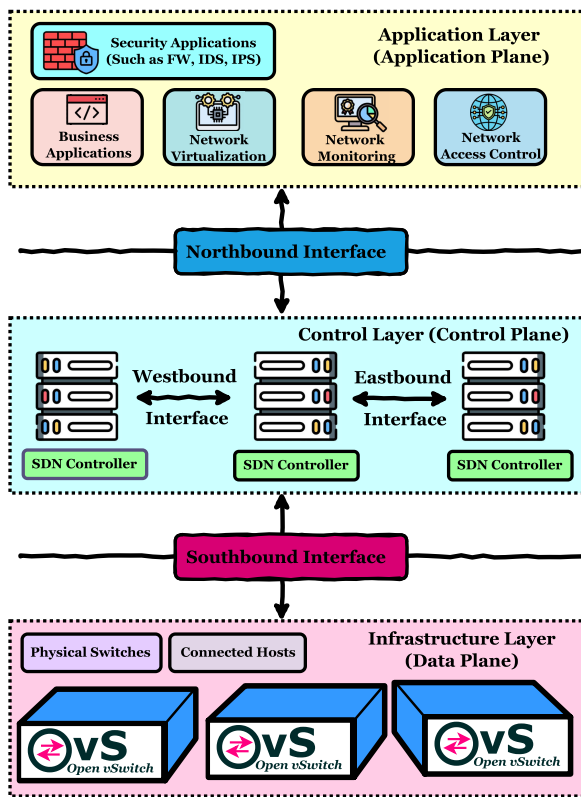


FIGURE 4. Layers of SDN architecture.

hardware and network, enabling seamless communication between devices [45]. Nevertheless, the primary function of the data plane is to forward packets according to the flow entries created by the control layer, which involves examining the packet headers and making forwarding decisions [46]. However, the infrastructure layer also implements Quality of Service (QoS) policies to prioritize certain types of traffic, thereby ensuring efficient bandwidth utilization [47], [48]. Interactions with other layers occur through southbound APIs, with OpenFlow being the most notable protocol [49]. This protocol enables the SDN controller to install, modify, and delete flow entries in the switches.

2) CONTROL LAYER

It is also known as the control plane and serves as the central management point of the SDN architecture. It abstracts the underlying infrastructure and provides a comprehensive view of the network, thereby facilitating dynamic management and configuration [50]. The core component of the control layer is the SDN controller, which is a software entity that operates on a server or cluster of servers [51]. Consequently, it orchestrates the entire network by interfacing infrastructure and application layers. Controllers typically include databases for storing network topology information, flow entries, and other relevant data [52]. Based on its functionalities, the controller manages the configuration and policies of network devices, thereby enabling centralized

control over the network [53]. It can optimize traffic flow by adapting to real-time conditions, such as congestion or link failures, thus dynamically adjusting flow rules [54]. Furthermore, the control layer enforces security policies by managing data flow, implementing firewalls, and monitoring anomalies [55]. Interactions with other layers occur through northbound APIs, allowing applications to request specific network behaviors or configurations [56]. Similarly, the control layer uses southbound APIs such as OpenFlow to push flow rules and receive telemetry data from the infrastructure layer [57].

3) APPLICATION LAYER

As the topmost layer in the SDN architecture, the application layer encompasses various applications that leverage the network resources managed by the control layer [58]. This layer is the epicenter of innovation, in which specialized use-case applications were developed [59]. Based on its components, the Application Layer comprises software applications that utilize the SDN controller capabilities to deliver functionalities such as network monitoring, security, and automation [60]. Consequently, some applications offer graphical user interfaces (GUIs) for network administrators to manage and visualize networks efficiently [61]. In addition, applications within this layer can automate routine tasks including provisioning, configuration, and monitoring, thereby reducing the need for manual intervention. Furthermore, these applications can provide tailored services to meet specific business requirements, such as load balancing, firewall services, and IDS [62]. Interaction with other layers occurs via northbound APIs, which facilitate communication between the application and control layers, enabling applications to request network configurations or obtain network-state information.

C. WHAT IS DDoS DETECTION?

This section presents a concise overview of DDoS detection for the general audience.

1) WHAT IS DDoS?

A DDoS attack is an attempt to render a server or network unavailable by overwhelming it with a massive amount of traffic [44]. Compromise can occur due to the use of multiple compromised computer systems, commonly known as a botnet [63]. This botnet can encompass a range of devices from personal computers to IoT devices, which are leveraged to orchestrate attacks. DDoS attacks in SDN networks can be categorized based on their nature and origin, with control mechanisms targeting different aspects such as saturation, resource exhaustion, and complexity of detection. However, it is crucial to recognize that DDoS attacks are harmful regardless of their nature or origin. For instance, attacks targeting controller saturation can overwhelm an SDN controller with a high volume of traffic, leading to service degradation or failure, as switches send

numerous packet-in messages for unmatched flow entries [9]. Additionally, attackers may exploit the limited resources of the controller by flooding it with requests, thereby exhausting its CPU and memory and impairing its ability to process legitimate traffic. Furthermore, the complexity of detection in SDN environments arises from the difficulty in distinguishing between legitimate and malicious traffic owing to the dynamic nature of the network. This complexity necessitates advanced detection techniques, such as entropy-based methods or machine-learning approaches, because traditional methods may prove insufficient. In contrast, DDoS attacks can stem from various human activities, including malicious intent such as vandalism or corporate sabotage, the creation of botnets through malware infection for covert control, and insufficient security awareness, leading to exploitable vulnerabilities. Regardless of their impact, it is essential to identify anomalies in the network in a timely manner to mitigate the risk of potential DDoS attacks and prevent future disruptions.

2) WHAT IS DDoS ANOMALY DETECTION?

Anomaly detection (AD) refers to the identification of patterns in data that do not conform to expected behavior. Anomaly detection plays a crucial role in identifying unusual traffic patterns that may indicate ongoing DDoS attacks. AD systems analyze network traffic to establish a baseline for normal behavior. When deviations from this baseline occur, such as sudden spikes in traffic or unusual request patterns, the systems flag them as potential threats. Within a practical timeframe for anomaly detection, AD approaches are crucial for evaluating the nature and timing of DDoS attacks. Usually, in SDN, continuous monitoring is essential for several reasons, including real-time detection, adaptive response, and resource allocation. Monitoring can be classified as passive or active. Although there are different definitions in [64] and [65], there are generally two distinct approaches to network monitoring, each with unique functions and applications. Passive monitoring involves observing and analyzing the actual traffic flowing through a network without generating additional traffic. This method is effective for identifying issues that occur intermittently because it captures a comprehensive view of the network performance over time. It is cost effective because it does not require additional resources to generate synthetic traffic. By contrast, active monitoring, also known as synthetic monitoring, involves the generation of simulated traffic to test various performance metrics. The passive approach allows for proactive identification of potential issues before they affect real users. Despite these differences, both types of monitoring systems aim to detect anomalies in the network and determine whether they are harmful or suspicious.

3) SELECTING THE MOST SUITABLE DDoS DETECTION TECHNIQUE

The main challenge in DDoS detection research is determining “*which method is most effective for detecting both*

known and unknown DDoS attacks?” Researchers have found that no single technique works the best in any situation. However, the most suitable model depends on the specific requirements of a problem. Each approach has its strengths and limitations. For instance, when a labeled dataset is available, supervised learning algorithms such as random forests, decision trees, and support vector machines are effective in detecting known DDoS attack patterns in SDN traffic. Similarly, semi-supervised learning methods that combine labeled and unlabeled data can be effective in identifying unknown attacks, although their performance depends significantly on the quality of the labeled data. Conversely, unsupervised algorithms such as clustering and anomaly detection can identify unknown DDoS attacks without labeled data; however, they may have higher rates of false positives. Integrating multiple frameworks into an ensemble can enhance accuracy and reduce false positives. Techniques such as majority voting, stacking, and blending can improve DDoS detection performance beyond what individual models can achieve. Additionally, deep neural networks can automatically learn features from raw SDN traffic data to detect DDoS attacks and capture complex attack patterns but require large, labeled datasets for training. On the other hand, Moving Target Defense (MTD) involves dynamically altering network configurations to hinder attackers, providing an additional layer of defense when combined with detection methods. Despite this, traditional security measures such as firewalls, encryption, and traffic filtering can mitigate some DDoS attacks, although their effectiveness is limited to more sophisticated threats. Therefore, choosing the most effective DDoS detection approach depends on several key factors: availability of labeled data, desired level of detection accuracy, computational resources, and ability to identify both known and unknown types of attacks.

D. DDoS PROBLEMS IN SDN LAYERS

To enhance the efficiency and visibility of SDN, detecting DDoS attacks is essential because of their potential to severely impact network availability and performance. Immediate action is imperative for multiple reasons, including safeguarding network resources, preserving user trust, responding to the dynamic nature of attacks, and minimizing downtime to ensure timely mitigation and continuous service availability. To do so, various detection methods responsible for effectively mitigating DDoS attacks in SDN environments have been discussed. As shown in Figure 5, we categorized these into eight groups, each with unique functions and advantages.

- **Classification Approaches:** This approach leverages machine learning to categorize traffic patterns and detect anomalies associated with DDoS attacks [66], [67]. Such methods excel at real-time detection and can be adapted to new attack methods.
- **Statistical and Analytical Methods:** These techniques examine traffic metrics, such as packet rates and flow

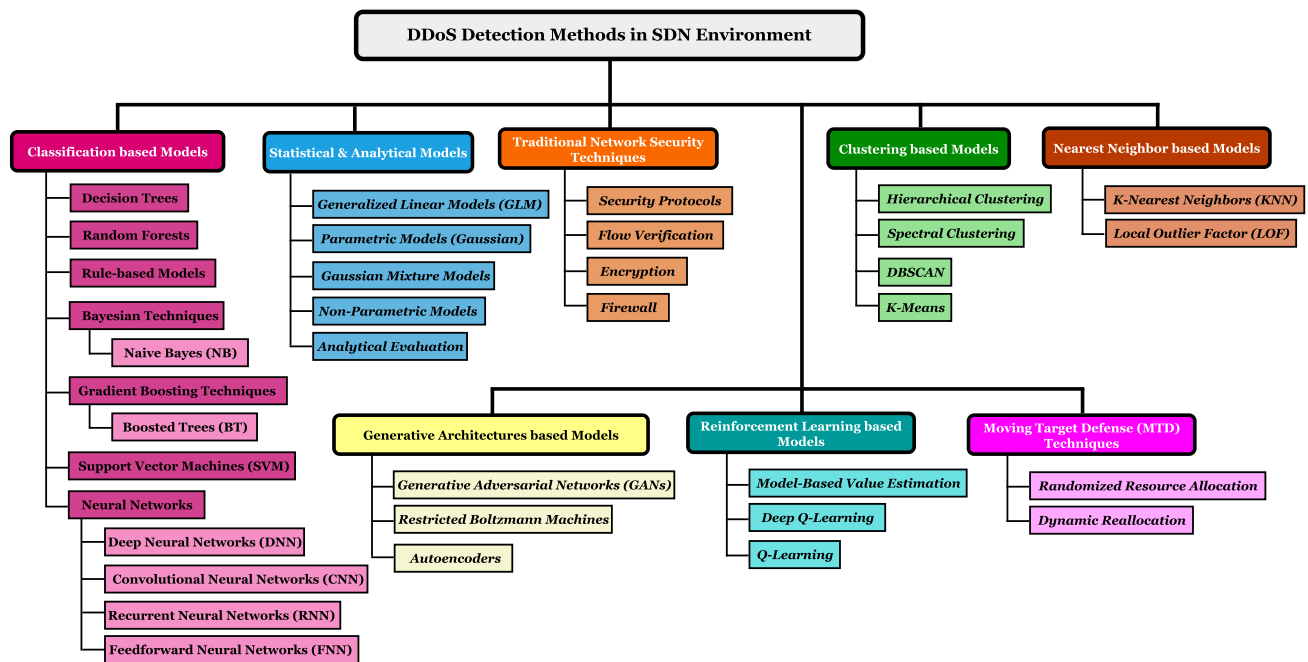


FIGURE 5. Illustrates the diverse range of DDoS detection methods applied in SDN environments, categorized by model type for ease of reference.

counts, to identify deviations from typical behavior [68]. These are crucial for establishing baseline traffic profiles and spotting irregularities.

- **Traditional Security Measures:** These involve established security protocols and practices designed to identify and mitigate DDoS threats [40]. Although straightforward, they may not always adapt well to novel or sophisticated attacks.
- **Clustering Techniques:** Clustering helps detect outliers that may signal attack traffic by grouping similar traffic flows [69]. These methods are effective in recognizing coordinated attack patterns.
- **Nearest Neighbor Approaches:** This technique compares incoming traffic to historical data to identify abnormal patterns [70]. It is particularly useful for detecting less obvious low-volume DDoS attacks.
- **Generative Models:** These create synthetic traffic patterns based on normal behavior, which helps in identifying deviations indicative of an attack [71]. These are beneficial in environments with fluctuating traffic patterns.
- **Reinforcement Learning Methods:** These techniques evolve by learning from traffic patterns to improve detection capabilities over time [72]. They are well-suited for dynamic environments with frequently changing attack strategies.
- **Moving Target Defense Strategies:** These involve dynamically altering network configurations to mislead and mitigate DDoS attacks [73]. They provide a

proactive layer of defense that can reduce the effectiveness of the attack methods.

From the perspective of the SDN layers, each layer employs specific techniques designed to address unique challenges. Therefore, by integrating a combination of classification, statistical analysis, clustering, and artificial intelligence approaches, organizations can significantly bolster their resilience to DDoS attacks.

E. TECHNICAL CHALLENGES IN INTEGRATING DDoS DETECTION INTO SDN

Nevertheless, in a controlled environment, where DDoS detection is implemented using statistical analysis methods, the challenges are relatively minimal. For example, a network administrator may utilize a basic threshold-based detection mechanism to monitor the traffic patterns. If the incoming traffic surpasses a predefined threshold for legitimate requests, the system flags it as a potential DDoS attack. In contrast, integrating ML-based detection methods within SDN introduces additional complexities. Deploying ML techniques to analyze network traffic in real-time for DDoS attack identification involves several challenges. In this section, we address the challenges associated with integrating DDoS detection into the SDN-based scenarios.

1) CENTRALIZED CONTROL PLANE VULNERABILITY

The first challenge in developing DDoS anomaly detection models for SDN networks is to address the issue of a single point of failure [74]. The centralized nature of the SDN

control plane makes it a prime target for DDoS attacks. If the controller is overwhelmed, it can lead to a complete network shutdown, rendering the detection and mitigation efforts ineffective. Another challenge is the exhaustion of resources. Attackers can exploit the limited resources of a control plane by overwhelming it with requests, which complicates the detection of legitimate traffic and may increase the risk of false positive results. Therefore, it is essential to consider the critical role of the centralized control plane in SDN, as it serves as both a potent management tool and substantial vulnerability. The compromise of this component can lead to widespread network disruptions, unauthorized access, and the potential for large-scale attacks, highlighting the necessity for robust security strategies to protect this critical element of the network infrastructure.

2) DYNAMIC AND EVOLVING ATTACK PATTERNS

As SDN technology continues to emerge, DDoS attack techniques evolve rapidly, rendering it difficult for static detection mechanisms to remain effective. Attackers are increasingly employing multi-vector strategies that require detection systems to be adaptable and capable of identifying new patterns in real-time. Furthermore, the dynamic nature of network traffic in SDN environments complicates anomaly detection because traditional methods often struggle to differentiate between legitimate traffic spikes and actual DDoS attacks, particularly in high-traffic scenarios. For example, an online gaming platform hosting a highly anticipated tournament may experience a significant surge in legitimate traffic as thousands of players and spectators log simultaneously. This sudden increase can trigger unnecessary mitigation actions, such as blocking legitimate users, which disrupts the gaming experience and frustrates players. To overcome this, researchers have developed advanced detection systems that can adapt to changing traffic conditions and effectively differentiate between legitimate spikes and actual DDoS attacks in SDN environments [75]. However, they also introduce a range of issues related to complexity, resource demands, adaptability, and security. Addressing these challenges is crucial for the successful implementation and effectiveness of DDoS detection strategies in SDN environments.

3) PERFORMANCE OVERHEAD

Resource allocation and latency are critical factors in the detection and response to DDoS attacks within SDN. The implementation of DDoS detection mechanisms can introduce additional overhead on network resources, potentially compromising overall network performance [76]. Hence, achieving an optimal balance between robust detection and network efficiency is a challenge. Additionally, the time required to detect and respond to DDoS attacks directly impacts the effectiveness of the mitigation strategies. Ensuring timely responses while maintaining high detection accuracy represents a complex and multifaceted issue.

4) DATA PRIVACY AND SECURITY CONCERNS

It is important to recognize that many DDoS attack techniques targeting SDN environments can exploit vulnerabilities related to data privacy and security [77]. Implementing DDoS detection mechanisms often involves analyzing sensitive traffic data, which raises concerns about privacy and regulatory compliance. Ensuring that these detection methods do not compromise the user data is a significant challenge. Moreover, reliance on automated detection systems introduces concerns about their reliability and potential for false positives, which can undermine trust and hinder their adoption. Thus, prioritizing user data privacy remains a crucial challenge for organizations striving to build trust, comply with regulations, and develop advanced detection models that protect SDN integrity and availability.

V. TAXONOMY OF DDoS ATTACKS IN SDN

Existing surveys on DDoS attacks in SDN environments often have limited scope, focusing on specific attack categories or individual detection methods. Many of these studies (see References [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]) concentrate on a single type of DDoS attack or narrow categories, without providing a comprehensive view of the broader landscape of DDoS threats within SDN. This fragmented approach can be challenging for readers, particularly newcomers, who may struggle to understand the full spectrum of DDoS attacks and their impact across different SDN layers.

To address these limitations, we introduce a novel, comprehensive taxonomy that categorizes DDoS attacks in SDN into seven distinct groups, as illustrated in Figure 6. This taxonomy organizes a wide range of DDoS attack types into a unified framework, making it accessible and valuable to both newcomers and experienced researchers in the field. Each category in this taxonomy is further analyzed in terms of its focus within specific SDN layers, the evaluation tools commonly used, and the DDoS detection methods applied (for an in-depth discussion, refer to Section III D). This structured approach provides a holistic perspective on DDoS threats in SDN, serving as a foundation for developing effective detection and mitigation strategies. The seven categories of DDoS attacks are as follows.

- **Flooding DDoS Attacks:** This category includes attacks such as SYN, UDP, and ICMP floods, which overwhelm the target network with excessive requests, deplete resources, and cause service disruptions. For those who are looking for further details, see References [78] and [79].
- **Application Layer DDoS Attacks:** These attacks target the application layer by generating malicious traffic that mimics legitimate requests, such as HTTP floods and DNS amplification attacks, to exhaust server resources [80], [81].
- **Protocol DDoS Attacks:** This category exploits vulnerabilities in network protocols, including TCP connection attacks and OpenFlow-specific attacks such

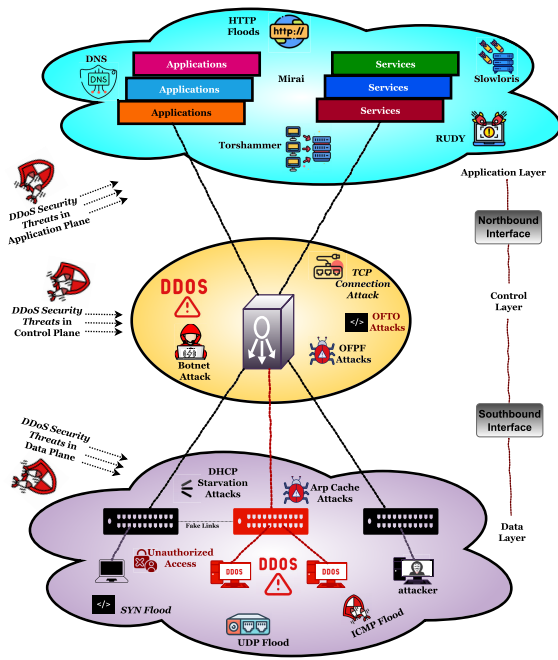


Figure (a)

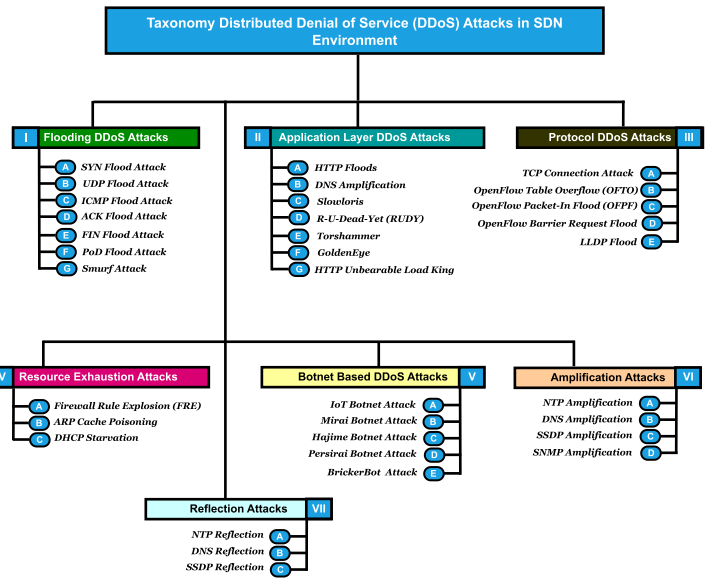


Figure (b)

FIGURE 6. Comprehensive Taxonomy and layered vulnerabilities of DDoS attacks in SDN. Figure (a) illustrates the hierarchical SDN architecture and its vulnerability points, highlighting the application, control, and infrastructure layers. It showcases various types of DDoS attacks targeting each layer: the application layer (e.g., HTTP Floods, DNS Amplification, and botnet attacks), the control layer (e.g., TCP Connection Attacks, OpenFlow Table Overflow), and the data (Infrastructure) Layer (e.g., SYN Flood, UDP Flood, ARP Cache Poisoning). Figure (b) presents a taxonomy of DDoS attack types in SDN environments, categorizing attacks into seven primary groups: Flooding DDoS attacks, application layer DDoS attacks, protocol DDoS attacks, botnet-based DDoS attacks, amplification attacks, and reflection attacks. Together, these visualizations provide a structured view of DDoS attack vectors, demonstrating the multi-layered security challenges and the classification of attack types within SDN.

as OpenFlow Table Overflow (OFTO) and OpenFlow Packet-In Flood (OPPF) [82].

- **Botnet-Based DDoS Attacks:** These attacks are launched through botnets, large networks of compromised devices coordinated to conduct simultaneous attacks. Notable examples include IoT-based botnets such as Mirai, which can simultaneously impact multiple regions. For further details, see References [83], [84], [85], and [86].
- **Amplification Attacks:** Amplification attacks involve sending small requests that trigger large responses, such as NTP and DNS amplifications, resulting in overwhelming traffic directed at the target [22].
- **Reflection Attacks:** Reflection attacks use legitimate servers to send large volumes of traffic back to the target network, and often utilize protocols such as NTP and SSDP to increase the impact of the attack [87].
- **Resource Exhaustion Attacks:** This category includes attacks focused on exhausting specific resources, such as memory or processing power, using techniques such as ARP cache poisoning and DHCP starvation [88].

This taxonomy offers a structured view of DDoS threats specific to SDN, providing readers with a clear and organized reference for understanding and addressing these attacks.

For those seeking deeper insights, relevant references offer extensive coverage of each attack type.

VI. FLOODING ATTACKS

In this section, we provide a detailed review of existing works from the perspective of DDoS flooding attacks in SDN environment. (cf. Section IV-B for details) (cf. Figure 7 and Table. 2)

A. SYNCHRONIZATION (SYN) FLOOD ATTACK

A SYN flood attack exploits the TCP handshake process to overwhelm a server's resources, thereby leading to a potential denial of service. First, the attack works by sending a large number of SYN requests to the target server, preventing it from handling legitimate connection attempts. Second, the TCP handshake involves three essential steps: the client sends a SYN packet to the server, the server responds with a SYN-ACK packet, and the client completes the handshake by sending an ACK packet. However, in a SYN flood, the attacker sends numerous SYN packets, often with spoofed IP addresses, without completing the handshake. Consequently, the server has numerous half-open connections that consume resources. Furthermore, this disruption renders the server incapable of processing legitimate traffic, thereby causing

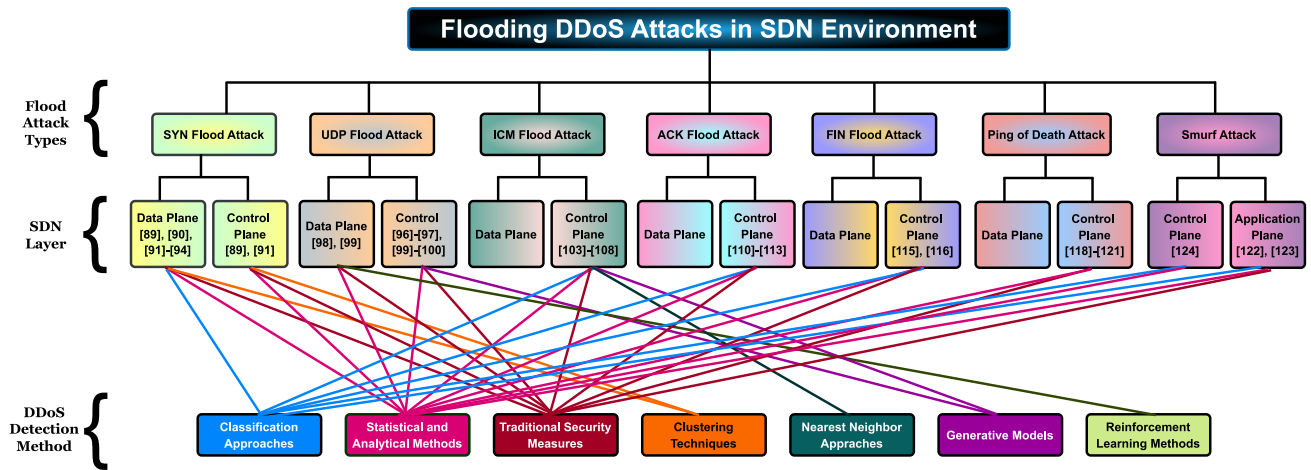


FIGURE 7. Overview of flooding DDoS attacks in SDN, showing attack types, SDN layers, and detection methods from literature.

a denial of service. Numerous studies and methodologies have been proposed to address SYN flooding. For instance, Al-Duwairi et al. [89] introduced the “Intentional SYN Flood Detection and Mitigation in SDN (ISDSN)” mechanism. This approach significantly outperforms existing solutions by reducing the attack detection time, lowering the number of installed forwarding rules, and minimizing CPU utilization. Additionally, the ISDSN mechanism effectively mitigates SYN flood attacks and is resistant to MAC address spoofing, which is a key vulnerability in other systems, such as SLICOTS. Goldschmidt and Kucera [90] examined network-based techniques for mitigating SYN Flood denial attacks. This study proposed three specific methods that employ SYN authentication mechanisms to enhance defense against such attacks. Modified network-based mitigation techniques, including RST Cookies, SYN Drop, and TCP Handshaker, effectively protect against SYN Flood attacks, even sophisticated variants involving spoofed IP addresses. These methods introduce only minor delays in connection establishment, typically less than 0.2 milliseconds for subsequent connections, despite the initial delay caused by the mitigation process. Furthermore, these methods are compatible with various operating systems and browser applications, although some configurations such as newer Fedora distributions may interfere with RST Cookies.

From the perspective of attacks that consume server resources by sending numerous SYN packets, Nashat and Hussain [91] have introduced a unique method for detecting SYN flooding attacks. Their approach utilized Multifractal Detrended Fluctuation Analysis (MFDFA) to exploit the multifractal nature of SYN flooding traffic, allowing it to be distinguished from normal traffic. This method achieves higher detection rates with lower computational overhead than traditional techniques. In addition, it employs an adaptive threshold to identify abnormal behaviors in network traffic, thereby improving the accuracy of attack detection. Another study by Yang et al. [92] addressed the

issue of overwhelming server performance caused by SYN flooding attacks. Their solution employs updated techniques to block attackers more effectively than traditional methods, by focusing on MAC addresses rather than IP addresses. The detection model uses an evolved cuckoo hashing approach that combines whitelists and blocklists. Overall, this method demonstrates superior effectiveness in handling SYN flooding attacks compared to conventional TCP reset methods, achieving a 2% improvement in detection accuracy over existing techniques. AlArnaout et al. [93] proposed a “Robust Attack Path Tracing Algorithm” (RAPT) to mitigate SYN-Flood DDoS cyberattacks. This technique involves edge routers generating IP traceback packets to record the attack path and identify the source of the malicious traffic. This approach mitigates the impact of SYN-flood attacks by discarding SYN packets from the identified attackers. The RAPT algorithm can be deployed on both the edge and core routers to detect and trace SYN-flood attacks in real time, thereby protecting the network infrastructure from being overwhelmed by malicious traffic. Despite its effectiveness, the algorithm is designed with low computational and message overhead, ensuring efficient detection and tracing capabilities. Studies have employed machine-learning techniques for traffic classification and programmable data plane filtering to mitigate SYN flood attacks. For example, Dimolianis et al. [94] proposed a schema that achieved high detection accuracy for SYN Flood attacks using ML models, significantly outperforming traditional SYN cookies mechanisms. Their approach includes a signature reduction process that effectively decreases the number of filtering rules required, thereby enhancing performance and minimizing collateral damage to legitimate traffic. This method notably improves packet filtering efficiency, resulting in higher rates of blocked malicious traffic and reduced backscatter traffic compared to SYN cookies.

Key Insight: Traditional techniques and statistical analyses have been widely employed to detect and mitigate SYN flood

attacks in SDN [89], [90], [91], [92], [93], [94]. However, as attackers become increasingly sophisticated, there is a pressing need for more advanced methods to address these threats effectively. These traditional approaches often suffer from high false positive rates and struggle to detect low-rate DDoS attacks. Consequently, they may be inadequate to keep pace with the speed and scale of modern DDoS attacks. Therefore, researchers should focus on leveraging ML, DL, and hybrid techniques to develop robust and scalable defence systems.

B. USER DATAGRAM PROTOCOL (UDP) FLOOD ATTACK

UDP flood attacks are another significant type of DDoS attack that exploit the User Datagram Protocol (UDP), a connectionless networking protocol. In this attack, an adversary sends a large number of UDP packets to random ports on the target server. Consequently, the server checks for applications that listen to ports. When no applications are found, the server responds with ICMP (Internet Control Message Protocol) “Destination Unreachable” messages, which can overwhelm the server and render it unresponsive to legitimate traffic. Although it is challenging to pinpoint the exact first occurrence of UDP flood attacks, they have been recognized as a major threat since the early 2000s, particularly as internet usage and reliance on UDP for various applications have increased [95]. A broader approach for developing UDP flood DDoS detection tools for general SDN networks was presented in [96]. The authors found that the entropy method was effective in detecting UDP flooding packets within an SDN environment, offering a more efficient solution for mitigating such attacks. Their solution was implemented using a POX controller, which is a well-known open-source SDN controller.

An improved method for traffic regeneration using scene reconstruction to enhance the detection and defense against network attacks was presented by Wang et al. [97]. The authors integrated network topology reconstruction, probability models, and packet sequence models to regenerate the network traffic data from Pcap files. Subsequently, the SDN environment was simulated using Mininet with a Ryu controller. This method involves preprocessing the Pcap data to extract key information, construct network topologies, and automatically generate attack commands for scenario replication. The proposed framework successfully regenerates network traffic data with a higher similarity to real network conditions than the traditional methods. From the perspective of UDP attacks on the IoT SDN environment, Javanmardi et al. [98] proposed “Mobility-aware Rate Limiting (M-RL)” to address vulnerabilities in traditional detection methods, particularly in environments where adversaries employ spoofing and mobility. To elaborate it further, authors demonstrated that the new technique significantly improves the detection of UDP flooding attacks in IoT networks, achieving an accuracy of over 99%. The performance of traditional methods, such as rate-limiting,

was lower, with maximum accuracies of 77%, 48.9%, and 64.3%, respectively.

Some studies have focused on addressing multiple flood attacks simultaneously. For example, [99] presented a novel contribution to the development of a versatile defense system within a P4-based SDN environment, capable of mitigating both SYN flooding and UDP flooding attacks. Unlike many existing defense systems, which are often limited to specific protocols such as TCP or UDP, the proposed system is adaptable and robust, effectively addressing multiple types of flooding attacks within a single framework. Furthermore, by leveraging P4, the authors implemented much of the detection logic directly on the switch, reducing the communication overhead between the switch and the controller, and enabling faster real-time detection and response. A similar study using a different detection method was presented by Jaszcz and Połap [100]. They introduced a new framework called “Artificial Intelligence Merged Methods (AIMMs)” for detecting UDP and TCP attacks. Their proposed scheme achieves a high detection accuracy of 99.5% when tested on the BOUN dataset. The results show that combining AI methods, specifically neural networks and k-nearest neighbors (k-NN), with soft set inference and weighted averaging techniques offers significant advantages for identifying flood DDoS attacks.

Key insights: The algorithms proposed in [96], [97], [98], and [99] do not fully leverage advanced analytical techniques for enhancing network security. Only [100] has introduced the AIMM technique, which uses AI to protect against UDP attacks. However, this study was limited by the use of a single dataset and scalability issues, highlighting the need for further exploration of different AI techniques. Given the evolving tactics of attackers, there is a need for continuous adaptation of detection and mitigation strategies. This requires ongoing research and development to keep up with the new attack vectors and techniques. Therefore, research in this area should focus on developing various mitigation strategies, including rate limiting, deep packet inspection, AI techniques, and cloud-based DDoS protection. Further studies are required to enhance these strategies to effectively address the dynamic nature of UDP flood attacks.

C. INTERNET CONTROL MESSAGE PROTOCOL (ICMP) FLOOD ATTACK

An ICMP flood attack, often referred to as a ping flood, is a denial-of-service attack that targets devices or networks with an overwhelming volume of ICMP echo request packets [101]. This form of attack is relatively easy to execute due to the inherent simplicity of the ICMP protocol, the widespread availability of tools for generating such packets, and the potential to amplify the attack using botnets. The ICMP protocol, originally designed for network diagnostics, is straightforward to manipulate, and attackers can easily generate ICMP echo requests using basic command-line tools or scripts. Additionally, attackers often employ IP spoofing

techniques to conceal their identity, complicating efforts to trace the source of the attack and allowing them to operate with a relative impunity. An ICMP flood attack involves sending a high volume of ICMP echo request packets to a target device, which then responds with echo replies [102]. The process begins with the attacker generating numerous ICMP echo requests using tools, such as hping or custom scripts. These packets are directed to the IP address of the target. Therefore, if a botnet is employed, requests can come from multiple sources, amplifying the intensity of the attack. Consequently, the target device is overwhelmed by requests that exhaust the CPU, memory, and bandwidth. Consequently, the device struggles to manage legitimate traffic. Furthermore, the high volume of responses further depletes system resources, exacerbating the impact of the attack.

ICMP flood attacks can significantly affect various technologies, including network devices, cloud services, IoT devices, and business operations [103]. Several strategies can be employed to mitigate these attacks, such as traffic filtering, rate limiting, anomaly detection, and disabling the ICMP. Recent studies have investigated the use of DL techniques to enhance the detection and mitigation of ICMP flood attacks. For instance, a study by [104] introduced an advanced mitigation system utilizing a dynamic learning system (DLS) that integrates a Complete Autoencoder (CA) with ensemble learning techniques. This system aims to improve the accuracy and efficiency of DDoS attack detection and mitigation in an ISP environment. Additionally, research conducted by Gadze et al. [105] explored the application of DL techniques, specifically Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN), to mitigate ICMP attacks. The study found that LSTM achieved an accuracy of 89.63%, surpassing traditional ML models, such as Support Vector Machines (SVM) and Naive Bayes. However, it did not outperform K-Nearest Neighbors (KNN), which achieved an accuracy of 99.4%. Similar work involving CNN models was presented by the authors of [106]. Their proposed method, which integrates edge computing with a 2D Convolutional Neural Network (CNN), achieved high accuracy rates of up to 99.5% for packet traffic and 99.8% for packet features in detecting ICMP attacks. However, while the study effectively combined edge computing with a CNN, it lacked detailed information on the implementation and management of edge computing components.

From the perspective of ML techniques, Tuan et al. [107] found that KNN and XGBoost were highly effective in mitigating ICMP attacks, achieving an efficiency rate exceeding 98%. Their mitigation scheme effectively reduced the impact of DDoS attacks while maintaining the integrity of benign traffic. In terms of traditional detection methods, Onyema et al. [108] discussed the impact of ICMP protocol-based attacks on SDN environments, focusing particularly on flooding attacks such as ping floods and Smurf attacks. They proposed a “Security Policy Protocol (SPP)”

that effectively defends against these flooding attacks. Moreover, the model demonstrated the capability to address attacks originating from unprotected legacy equipment.

Key Insight: Research has shown that SDN architectures, owing to their centralized control planes, are particularly vulnerable to ICMP flood attacks, underscoring the critical need for effective mitigation mechanisms. Various techniques, including classification and statistical methods, have proven valuable in enhancing the identification of traffic anomalies associated with ICMP floods, as evidenced by studies, such as those by [104] and [105]. However, the current study had notable limitations. Many studies [100] to [107] relied on single or generated datasets, which may not cover the full spectrum of network conditions or attack types, impacting the generalizability of the models. In addition, the lack of standardized evaluation methodologies makes it difficult to compare the effectiveness of different approaches, and some solutions may struggle with scalability in large SDN environments. To address these issues, future research should explore a broader range of ML and DL methods using diverse datasets to improve accuracy and robustness. Emphasis should be placed on developing real-time analysis systems for rapid response to ICMP floods, refining feature selection to enhance algorithm performance, and using benchmark datasets to reduce biases and improve the effectiveness of AI models.

D. ACKNOWLEDGMENT (ACK) FLOOD ATTACKS

An ACK flood attack is a type of DDoS attack that specifically targets a Transmission Control Protocol (TCP) by inundating a server with a large volume of ACK packets. This attack exploits the TCP handshake process, where ACK packets are used to confirm the receipt of data between the two endpoints [109]. By bypassing the initial connection request (SYN packet) and directly flooding the target with ACK packets, the attacker causes severe resource exhaustion and network congestion, leading to a significant disruption of services. In SDN environments, ACK flood attacks can disrupt operations significantly. Resource exhaustion occurs because the processing capacity of the target server is overwhelmed by a high volume of incoming ACK packets, thereby hindering its ability to manage legitimate traffic. Furthermore, an excessive influx of packets can result in network congestion, leading to delays and packet loss for legitimate users. In addition, the server's struggle to cope with the flood may cause users to experience slow responses or complete service outages. This raises an important question within the research community “*Is there a universally accepted technique for mitigating ACK floods?*” Although various strategies have been proposed, no single method has achieved universal acceptance. Therefore, researchers have often advocated a combination of techniques tailored to specific network architectures. For instance, Sudar et al. [110] presented a

research work focusing on classification approaches through the C4.5 algorithm and statistical/analytical methods through proxy-based mechanisms. This technique aims to remove half-open connections to maintain the availability of TCP connections for legitimate users. Another study in [111] addressed ACK flood attacks by utilizing a system framework that integrates the “*Extended Berkeley Packet Filter (eBPF)*” and the “*eXpress Data Path (XDP)*”. This framework combines eBPF for monitoring and data extraction with XDP to provide an efficient defense against abnormal IP addresses identified by the detection algorithm. Apart from merging methods, there are studies that fall under the category of single-method detection, as demonstrated by Al-Naeem [112]. The system predicts ACK attacks by analyzing the variability in transmission behavior with a high degree of accuracy. More specifically, the study identified that the most effective delay between transmissions to prevent overwhelming the receiving end is 23 ms. Shah et al. [113] proposed a framework for mitigating ACK flood attacks in cloud computing environments using an SDN-based statistical anomaly detection model. Their approach demonstrated superior performance in detecting both source-based and spoofing attacks compared with existing solutions.

Subsequently, developing a unified model that integrates both detection and mitigation strategies is crucial to effectively address ACK flood attacks. Such a model can streamline response efforts by employing a centralized approach to manage traffic flows and dynamically adjust mitigation tactics based on real-time analysis. Future research should focus on designing models that encompass multiple layers of the network stack (application, transport, and network layers) to provide a comprehensive view of traffic patterns and enhance the overall effectiveness of the mitigation strategies. Additionally, implementing segmented network architectures can help isolate critical services from potential attack vectors, thereby limiting the impact of ACK floods by confining attacks to less critical segments of the network.

E. FIN FLOOD ATTACKS

An FIN flood DDoS attack is a type of DDoS attack that overwhelms network resources by inundating a target server with excessive finish (FIN) packets. This attack exploits TCP, thereby disrupting normal operations and causing denial of service for legitimate users [114]. In a typical TCP connection, an FIN packet signals the termination of the communication. When a client sends an FIN packet, it indicates that the data transmission is complete and requests closing of the connection. The server responds with an ACK and may send its own FIN packet to finalize the connection closure. Upon reviewing research databases, we found that few studies have specifically investigated FIN flood DDoS attacks. Although FIN attacks pose a genuine threat, their relatively lower resource requirements, less disruptive nature, and overlap with defenses against SYN flood attacks may explain their limited research focus compared with

other DDoS attack types. Nedyalkov and Georgiev [115] conducted research utilizing IP network modeling platforms to simulate and assess the impact of various TCP flood attacks, particularly FIN attacks, on VoIP connections. Their study evaluated how the asterisk-free PBX responds to different TCP flood attacks, identifying the type (TCP SYN, TCP FIN, or TCP RST) that has the most severe impact on VoIP connections and traffic. Furthermore, the authors introduced a traffic characterization and monitoring tool to classify and mitigate TCP attacks including FIN flooding. Conversely, another study [116] examined seven types of DDoS attacks, including FIN flood attacks, to develop a more efficient detection method using a Poisson distribution framework. This approach requires less training time than traditional ML techniques. Their research demonstrated that this technique outperformed existing detection systems and was validated across various scenarios using virtual machines to simulate DDoS attacks. However, although the Poisson distribution method proved effective in detecting multiple DDoS attacks, the study did not extensively address the nuances and potential evasion techniques specific to FIN flood attacks.

Thereafter, key insights from these studies underscore the importance of understanding the distinctive traffic patterns associated with FIN flood attacks and enhancing detection methods to address them more effectively. Moreover, future research should prioritize the development of specialized countermeasures and refinement of intrusion detection systems to better address the unique characteristics of FIN flood attacks. Additionally, integrating advanced analytics, such as machine learning, can further enhance the accuracy of identifying subtle patterns in these attacks.

F. PING OF DEATH (PoD) FLOOD ATTACKS

PoD attacks exploit vulnerabilities in the Internet Protocol (IP) by sending oversized packets to a target machine [117]. These data fragments exceed the maximum allowable size of 65,535 bytes, potentially causing the target system to freeze or crash because of its inability to process such large data. The assault typically involves fragmenting the oversized packet into smaller segments, bypassing size restrictions before being reassembled at the destination, leading to system failures. Unlike certain DDoS attacks that utilize reflection or amplification techniques to magnify traffic, Ping of Death relies on an attacker directly transmitting oversized packets. Consequently, the attacker must possess sufficient bandwidth and resources to independently overwhelm the target. PoD attacks can significantly degrade the network performance by exhausting resources, leading to device failure and increased latency. SDN is vulnerable to PoD results. If the central controller is overwhelmed, it may serve as a single point of failure, rendering the entire network infrastructure susceptible to disruption. These attacks may target the controller itself or the switches under its management, thereby compromising overall stability.

The authors of [118] explored the deployment of an IDS to monitor network communication for suspicious activities. These systems can automatically block malicious data and alert administrators to potential threats, thereby enabling a proactive response to unusual traffic patterns that may indicate ping-of-death attacks. Abdollahi and Fathi [119] emphasized that using specialized DDoS protection services adds an additional defence layer. Such services filter network traffic and block malformed data before they reach the target infrastructure, thereby effectively mitigating the risks posed by the PoD assaults. Regarding ML, Almorabea et al. [120] presented several algorithms capable of analyzing network communications in detail to recognize malformed or oversized ICMP packets that exceeded the maximum allowed size. This study compared three ML models: logistic regression, k-nearest neighbor (KNN), and support vector machine (SVM). Among these, the KNN model demonstrated superior performance, achieving a 99.67% detection accuracy, an error rate of 0.33%, and an F1-score of 99.67%. However, the utilized dataset was generated from an IoT testbed that emulated both normal and malicious flood attacks, raising questions regarding the generalizability of these results to real-world scenarios. Furthermore, SDN enables the implementation of more sophisticated security policies tailored to the specific needs of platforms such as WhatsApp and Telegram. Hakim et al. [121] demonstrated that using Snort as an IDS can safeguard against PoD attacks targeting social media platforms. The system successfully integrated Snort (NIDS) with WhatsApp and Telegram to provide real-time notifications to administrators.

Conversely, legacy systems may lack the advanced capabilities of modern architectures, making them particularly vulnerable to PoD attacks. Therefore, future studies should conduct thorough assessments of both legacy and modern systems to identify the vulnerabilities that can be exploited by PoD attacks. This includes an examination of the impacts of different configurations and network topologies. Moreover, incorporating real-world case studies of PoD incidents can provide valuable insights into the impact and effectiveness of various mitigation strategies. Additionally, developing and refining advanced detection techniques that utilize DL and anomaly detection to recognize PoD attack patterns in real time can enhance response capabilities. The implementation of multilayered defense strategies that combine technical measures (such as firewalls and IDS/IPS) with organizational policies and user education is essential for creating a robust security posture against PoD attacks. Furthermore, developing frameworks for the continuous monitoring and adaptation of defenses based on emerging threats and evolving attack methodologies is critical for ensuring long-term security.

G. SMURF ATTACKS

Smurf attacks pose a significant threat due to their ability to amplify traffic directed at a target without the need for a

large botnet, allowing attackers to generate up to 100 times more traffic than initially sent [122]. This amplified traffic can result in network congestion, where the flood of ICMP replies overwhelms the victim's bandwidth, delaying or dropping legitimate traffic and causing service disruptions that may lead to financial losses and reputational damage. Recent research has focused on improving the detection and mitigation strategies for Smurf attacks, particularly in SDN environments. For instance, Shadrooh and Norvag [123] introduced "*SMoTeF*," a framework designed to enhance smurfing detection in financial networks by using a new algorithm based on maximum temporal flow, which reduces false positives by considering the temporal order of events. Their study, which evaluated "*SMoTeF*" across three real-world datasets, demonstrated that it outperformed existing baselines, delivering superior fraud detection with minimal runtime overhead. In addition, Vijayakumar et al. [124] presented a deep neural network-based detection system capable of identifying anomalies such as Smurf attacks by recognizing abnormal traffic patterns. This system achieved a detection accuracy of 99.85% for identifying the excessive ICMP traffic characteristics of Smurf DDoS attacks. However, despite its high accuracy, the detection and response times may not meet the requirements for real-time mitigation in critical environments such as healthcare, where any delay could have serious consequences for IoT-based health systems, underscoring the importance of timely interventions.

H. SUMMARY AND INSIGHTS

Figure 7 provides a detailed overview of various DDoS flooding assaults within the SDN environment, mapping their impact across different SDN layers (data, control, and application) and showcasing the mitigation techniques employed. Attacks such as SYN Flood, UDP Flood, ICMP Flood, and Smurf assaults target specific layers of the architecture, with defensive strategies like CA, SA, TS, CT, NN, GM, and RL being utilized to counter them. For instance, SYN Floods primarily affect both the data and control planes, which are mitigated through classification and statistical analysis [89], [90], [91], [92], [93], [94]. Conversely, UDP and ICMP floods are typically identified using advanced statistical techniques and AI-driven approaches in the control plane [96], [97], [98], [99], [100]. More complex incidents, such as Smurf attacks, target the application layer, and are handled using reinforcement learning and generative models [122], [123], [124]. These approaches reflect a broader shift towards AI-powered adaptive security mechanisms, providing proactive, multilayered defense across the SDN ecosystem.

The literature on SDN security reveals a strong trend towards integrating ML and AI with traditional defenses. Algorithms such as LSTM, CNN, and k-NN are key tools for anomaly detection in network traffic, aiding the identification of DDoS threats. Reinforcement learning and generative models, particularly in frameworks such as M-RL, further drive automated countermeasures such as the dynamic

TABLE 2. Detection techniques for flood DDoS attacks on SDN. This table summarizes research efforts on flood attacks in SDN, listing attack types, detection methods, evaluation tools, and the SDN layers targeted by each solution. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Reference	Flood Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[89], 2020	SYN	Data, Control	ISDSDN installs forwarding rules only for validated SYN packets, reducing exhaustion and processing.	POX, Mininet, Scapy, Curl	✓									
[90], 2021		Data	Enhanced SYN mitigation with modified technique.	Not specified		✓								
[91], 2021		Data, Control	MFDFA on TCP traffic detects shifts from normal to attack patterns.	CICDDoS2019, Python		✓	✓	✓						
[92], 2023		Data	Cuckoo hashing integrates whitelists and blacklists.	Tools: P4 switches		✓	✓							
[93], 2023		Data	RAPT traces IP packets, identifies attacks.	Tools: Wireshark		✓	✓							
[94], 2021		Data	Signature-based ML detects SYN flood attack.	Dataset: generated	✓	✓	✓							
[96], 2022	UDP	Control	Entropy method detects UDP flooding anomalies.	Tools: POX		✓								
[97], 2023		Control	Solution regenerates traffic for accurate attack.	CIC-DDoS2019		✓	✓			✓				
[98], 2024		Data	M-RL disables malicious IoT, detects UDP accurately.	Tools: PyTorch		✓	✓				✓			
[99], 2021		Data, Control	Authors presented P4-based SDN defense system that can detect and mitigate UDP flooding attacks.	Tools: Mininet and P4		✓								
[100], 2022		Control	AIMM detects UDP attacks in real-time.	Dataset: BOUN	✓				✓					
[104], 2020	ICMP	Control	DLS provides real-time ICMP mitigation.	Dataset: Generated	✓	✓				✓				
[105], 2021		Control	LSTM enhances ICMP flood attack detection.	Dataset: Generated	✓	✓								
[106], 2022		Control	CNN model detects abnormal DDoS traffic.	Dataset: Generated	✓									
[107], 2020		Control	KNN and XGBoost to detect ICMP attacks.	CAIDA2007		✓		✓						
[108], 2022		Control	Authors suggested SPP can mitigate ICMP attacks.	Mininet, RYU		✓	✓							
[110], 2023	ACK	Control	Proposed hybrid approach to identify ACK attack.	CAIDA2007	✓	✓								
[111], 2022		Control	Developed automatic detection and defense system.	Tools: NID, FW		✓	✓							
[112], 2021		Control	Method predicts ACK attacks within 23 ms.	Tools: GUI		✓								
[113], 2022		Control	Statistical detection model for mitigating attacks.	Tools: Statistical		✓								
[115], 2024	FIN	Control	Traffic monitoring tool that analyzes network.	Tools: GNS3		✓								
[116], 2023		Control	IDS system to identify abnormal traffic patterns.	Tools: Snort	✓	✓	✓							
[118], 2020	PoD	Control	IDS system filtering out over-length packets.	Tools: IDS		✓								
[119], 2021		Control	Snort along with alert can detect PoD attacks.	Tools: Snort			✓							
[120], 2023		Control	Presented KNN, LR, and SVM to detect PoD attacks.	Tools: Zeek		✓	✓							
[121], 2020		Control	IDS leveraging media to guard against PoD attacks.	Tools: Snort			✓							
[122], 2023	Smurf	App	Smurf attack identification using COOJA simulator.	Tools: COOJA	✓		✓							
[123], 2024		App.	Method mitigates smurfing, reduces false positives.	Tools: IDS	✓	✓								
[124], 2023		Control	Developed a DNN-based detection of Smurf attacks.	Dataset:ECU-IoHT	✓									

deactivation of malicious IoT devices [94], [98], [105], [107]. Statistical approaches, such as entropy-based detection and MFDFA, remain valuable in recognizing subtle shifts in network behavior [91], [96], [113]. Clustering methodologies and nearest neighbor classifiers, including k-means clustering and cuckoo hashing, enable efficient sorting and isolation of malicious traffic, especially in large-scale environments [92], [93], [120]. Simulation tools such as Mininet, Python, and Wireshark are frequently used to create realistic network scenarios to test these security measures, whereas datasets such as CICDDoS2019 and CAIDA2007 provide essential benchmarks for evaluating their accuracy [91], [97], [110]. Advances in adaptive security frameworks driven by RL and MTD strategies highlight a shift towards proactive, self-learning defenses that adapt to evolving threats in real time [98], [104]. These integrated systems combine AI-based

techniques, traditional security measures, and sophisticated statistical analysis to deliver comprehensive protection within the SDN landscape [98], [100], [111]. However, significant challenges persist, particularly regarding the availability of diverse representative datasets that can capture the full range of potential attack vectors. There remains a critical need for systems capable of adjusting to novel threats in real time without relying solely on predefined signatures. This is especially crucial for application-layer security, as demonstrated in studies on Smurf attacks, which emphasize the necessity of a holistic approach to SDN defenses [122], [123], [124]. Future research on flooding DDoS attacks in SDN should prioritize the development of hybrid models that combine ML techniques with traditional statistical methods, thereby enhancing the detection accuracy by leveraging the complementary strengths of both approaches. A key area

for advancement is feature engineering, with a focus on creating tailored methods for specific attack types to improve the precision and efficiency of model training. In addition, investigating distributed SDN controller architectures holds promise for increasing system resilience and improving load balancing under DDoS attack scenarios. Addressing these research areas will contribute to a deeper understanding of flooding DDoS attacks and lead to more effective mitigation strategies in SDN environments.

Insights: Table 2 provides a comprehensive overview of detection techniques for Flood DDoS attacks within SDN environments. It categorizes recent research by attack types, detection methods, and evaluation tools while also specifying the SDN layers targeted by each solution. Detection methods are systematically organized into categories such as Classification, Statistical Analysis, and Machine Learning techniques. This structured presentation offers a clear academic insight into current approaches and tools in SDN-based DDoS mitigation research.

VII. APPLICATION LAYER ATTACKS

In this section, we provide an in-depth discussion of application-layer attacks with a specific focus on how these attacks interact with various layers of the SDN architecture. We explore the unique vulnerabilities present at the application layer and examine the role of SDN centralized control and application plane separation in mitigating such threats. In addition, we analyze existing defense mechanisms and their effectiveness in addressing application-layer DDoS attacks, emphasizing the integration of network programmability and dynamic traffic management in SDN environments (cf. Section IV for details) (cf. Figure 8 and Table. 3).

A. HTTP FLOODS

HTTP flood attacks are a type of denial attack that specifically target web applications by overwhelming them with HTTP requests [125]. In SDN, these attacks pose unique challenges and vulnerabilities owing to the reliance of the architecture on centralized control [126]. HTTP flood attacks can exhaust the resources of the SDN controller by bombarding it with a high volume of legitimate looking GET or POST requests [127]. This can lead to a denial of service, where the controller becomes unable to process legitimate traffic, thereby disrupting the functionality of the entire network [128]. When it comes to the flow table, the influx of requests can cause the flow tables in SDN switches to become overloaded. Each request can create new flow entries, leading to saturation and potential failure in handling legitimate traffic [129]. As the controller and switches become overwhelmed, the latency in processing requests increases, which can degrade the performance of applications that rely on timely data transmission. Recent research has focused on developing effective mitigation strategies for HTTP flood attacks in SDN environments. A study proposed a proxy-based solution that filters out malicious HTTP requests before they reach the target

server [130]. This approach is designed to be lightweight and minimizes the load on the server by using a mathematical model to simulate client-server interactions and detect invalid requests. It offers a practical and resource-efficient method for detecting and mitigating HTTP flood attacks. Another recent study introduced a time-based solution for detecting and mitigating high-rate HTTP flood attacks [131]. The authors used entropy-based techniques to detect the abnormal traffic patterns. The dynamic threshold adapts over time, making it more flexible for dealing with varying network traffic volumes. This time-based analysis approach is lightweight because it relies primarily on statistical measures (entropy) rather than resource-intensive ML methods, making it suitable for real-time DDoS detection in environments with limited computational power.

Several studies have explored the use of AI algorithms to detect HTTP attacks. “HTTPScout” focuses on using supervised technique to detect and mitigate HTTP flood attacks within an SDN environment [132]. By leveraging the SDN architecture, “HTTPScout” can dynamically monitor and analyze traffic patterns by applying ML to distinguish between legitimate and malicious HTTP traffic. This integrated approach provides both detection and mitigation using an SDN controller to block malicious sources. The authors of [133] used DL and optimization techniques (Rider Optimization Algorithm) to detect and prevent HELLO flood attacks in IoT environments. Although its focus is on HELLO floods, the use of ML for anomaly detection and route optimization can be adapted to HTTP flood attacks. The combination of deep learning with optimization makes this approach more computationally demanding and highly accurate in complex networks. Research has also focused on integrated architectures that combine various detection techniques, such as Time-based Analysis & Integrated Defense Architecture methods, to enhance the resilience of SDNs to HTTP flooding attacks [134]. The paper proposes a solution that uses an SDN controller to detect and mitigate HTTP GET flood attacks. The SDN controller monitors traffic over time and triggers blocking mechanisms when request rates exceed predefined thresholds. Time-based monitoring allows for dynamic detection, whereas the integration of the SDN controller provides a centralized defense architecture that optimizes network-wide resource management and attack response.

B. DOMAIN NAME SYSTEM (DNS) AMPLIFICATION

DNS amplification attacks pose significant risks to SDN environments, particularly in the application layer [135]. These attacks exploit the inherent features of DNS, allowing attackers to send small queries that generate large responses directed at a target, overwhelming it with traffic. This can lead to service disruption, degraded performance, and even complete denial of service. Amplification attacks can exhaust the bandwidth and processing resources of network devices, including switches and routers, within the SDN architecture.

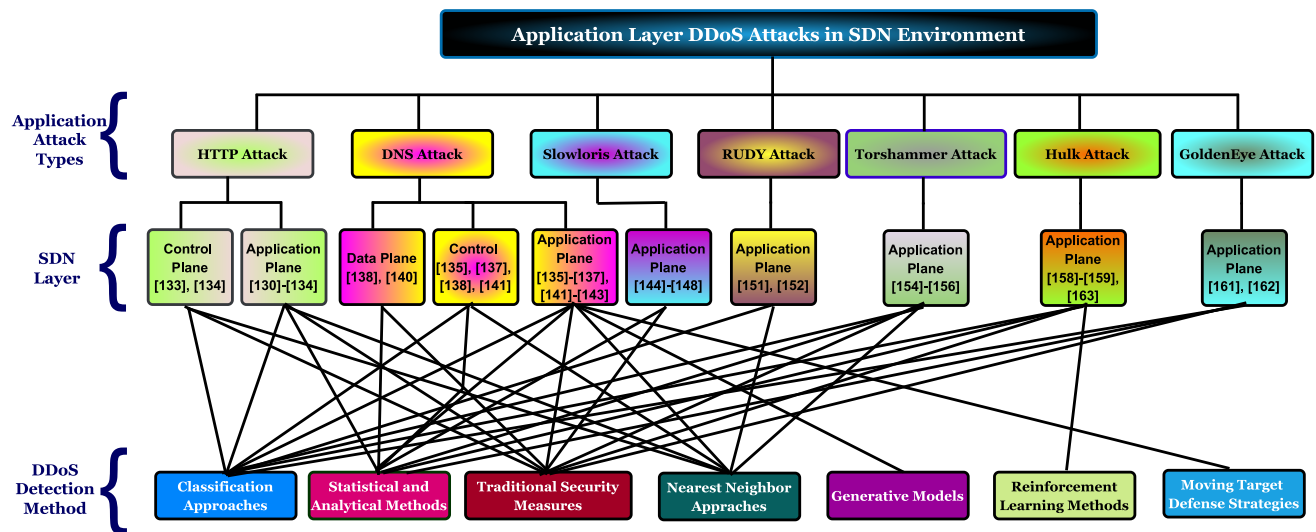


FIGURE 8. Detection methods for DDoS attacks in the application layer of SDN. This figure categorizes the detection methods applied across SDN layers (application, control, and data planes) for DDoS attacks targeting the application layer, such as Hulk, Slowloris, RUDY, DNS amplification, and HTTP floods.

This can lead to an increased latency and reduced availability of services for legitimate users. In other hand, once an attacker gains control over the network resources through DNS amplification, they may launch additional attacks (e.g., data exfiltration or further DDoS attacks) against other targets.

Several studies have focused on mitigating DNS amplification attacks in SDN environments. Some studies have integrated flow-control mechanisms into SDN controllers to effectively manage and block unsolicited DNS responses. Gupta et al. [136] proposed two prevention techniques, “Port-Mapping” and “Port-Merge-IP” which have proven successful in preventing DNS amplification attacks and protecting the victim’s network from harm. These algorithms also introduce 30% less delay compared with existing DNS prevention methods that rely on authentication mechanisms. Fouladi et al. [137] integrated Discrete Wavelet Transform (DWT) and auto-encoder neural networks for DNS anomaly detection. The proposed system reduces computational complexity by dynamically activating detection based on the hit rate of the flow table. The evaluation demonstrated that the system outperforms traditional time-domain approaches in differentiating between DNS attacks and normal traffic, particularly in scenarios where flash crowds or advanced DDoS tools are involved.

Abou El Houda et al. [138] developed a method called “WisdomSDN” to improve the mitigation of DNS amplification attacks. Their approach uses stateful DNS mapping, enabling OpenFlow switches to manage traffic autonomously, thereby reducing the load on the SDN controller. The entropy-based detection method effectively identifies suspicious traffic and achieves a high detection rate with minimal overhead. The experimental results show that WisdomSDN mitigates DNS amplification attacks while

maintaining low latency and protecting both the control and data planes of the network. Savenko et al. [139] presented detecting botnets that use DNS tunneling for command-and-control communication. The article highlights the challenges in detecting such attacks, owing to their complexity and subtlety. authors proposed a novel approach that leverages a comprehensive analysis of botnet features and architectural characteristics to effectively identify DNS tunneling-based botnet activities. The key finding is that the proposed method is highly efficient in detecting these types of attacks.

Bloom filters are used as a defense mechanism to quickly detect abnormal traffic patterns associated with amplification attacks, allowing proactive measures before significant damage occurs. For instance, Dai et al. [140] introduced the “DAmPADF” framework, which reduces the false positive rate in filtering DNS responses and achieves a higher filtration ratio of illegitimate traffic compared with traditional methods that rely solely on bloom filters. Systems that monitor the amplification factors and Time-to-Live (TTL) headers have been proposed [141]. These systems generate alerts and initiate mitigation strategies when suspicious activities are detected. Even low-rate attacks, often missed by traditional methods, are effectively identified through the combination of ML classifiers and DNNs. The experimental results demonstrate that this approach successfully mitigates DDoS attacks while minimizing both false positives and false negatives. Authors of [142] analyzed open DNS resolvers for DDoS attack risks, focusing on infrastructure diversity. They found over 2.6 million resolvers, with 59% lacking DNSSEC support. A selective model targeting problematic resolvers was proposed, aiming to reduce DNS amplification by prioritizing high-risk resolvers. Furthermore, Nesary and Aydeger [143] proposed a vDNS solution that enhances DNS security through dynamic server configuration changes,

making it more difficult for attackers to exploit static vulnerabilities. Utilizing MTDD strategies, the system effectively reduced the success rate of DNS amplification attacks. The study highlights the potential of vDNS as a scalable and agile solution that is capable of adapting to evolving attack vectors while maintaining DNS functionality.

C. SLOWLORIS

An Slowloris (SW) attack is an application layer (layer 7) attack that strategically sends partial HTTP requests to a target server, forcing it to open multiple TCP connections [144]. The attacker maintains these connections by sending small amounts of data at regular intervals, thus preventing the server from timing them out [145]. Consequently, as the number of active connections increases, the server's resources become strained, eventually reaching their maximum connection limit, which results in denial of service for legitimate users. Furthermore, in the context of SDN, the centralized controller responsible for managing network flows can be exploited to exacerbate attacks. Conversely, attackers may utilize virtualized resources or distributed botnets to launch multiple slowloris attacks from diverse sources, thus complicating both detection and mitigation efforts [146]. On the other hand, one of the key characteristics of Slowloris is its low bandwidth requirement, which allows attackers to execute the attack with minimal resources. This stealthy nature often enables an attack to evade traditional intrusion-detection systems that primarily focus on high-volume traffic patterns.

Politeknik Negeri Jakarta investigated the effectiveness of configuring IP tables to enhance security against Slowloris DDoS attacks [147]. The study involved testing web servers under simulated Slowloris attacks in a Kali Linux environment, with performance metrics, such as connection status and response times, closely monitored. After applying the IP table configurations, the server successfully maintained the service availability and showed a significant reduction in malicious requests. The findings suggest that, while HTTPS offers some security, it is insufficient on its own against Slowloris attacks, making robust IP table-based firewall rules a necessary and effective defense mechanism. A work by Faria et al. [148] developed a tool to detect and block slow-loris attacks. This method efficiently blocks attacks without restricting the number of parallel connections, thereby reducing false positives. The findings indicate that the new system uses significantly less processing power and memory than other tools, such as Snort, while effectively mitigating Slowloris attacks by analyzing traffic behavior and blocking attackers at their first network hop.

D. R-U-DEAD-YET (RUDY)

RUDY attacks represent a significant form of application-layer DoS attack, specifically targeting web applications by exploiting how servers process HTTP requests [149]. These attacks function by sending extremely slow requests that maintain open connections for extended periods, ultimately

leading to resource exhaustion because the server depletes its memory and processing power [150]. Moreover, because RUDY attacks require fewer resources than traditional DDoS attacks, attackers can execute them without the need for large botnets or an extensive infrastructure. Consequently, these attacks are difficult to detect and mitigate. One of the works by Rani et al. [151] developed an efficient IDS that uses ML techniques to detect both known and previously unknown RUDY attacks. Their hierarchical ML model achieved high accuracy (99.07%) in identifying these attacks. The system outperformed models such as Recurrent Neural Networks (RNN) and DNN in terms of the training time, accuracy, and complexity. Furthermore, it effectively identifies a wide range of DDoS attacks, including those launched by the RUDY tool. The authors suggest investigating the integration of Federated Learning (FL) techniques into their approach. This allows the model to learn and classify unknown attacks incrementally across distributed devices without centralizing data, potentially enhancing privacy and security in D2D networks. Another work carried out by Bocu and Iavich in [152] focuses on the development and implementation of an advanced FL Detection system specifically designed to identify Low-Rate DDoS (LDDoS) attacks, including RUDY attacks. Notably, their study demonstrated that the proposed model, which integrates a bidirectional LSTM network with an attention mechanism, significantly enhances the detection of RUDY and other slow-rate DoS attacks. Furthermore, the system processes real-time corporate data traffic and effectively distinguishes between legitimate traffic and slow attacks, such as RUDY, by analyzing the data packet patterns over time.

E. TORSHAMMER

Every Torshammer (THA) attack executed against a web server carries its own nuances and can impact the server's performance in distinct ways [153]. Once launched, slow-rate attacks continuously deplete server resources, making it increasingly difficult to halt attacks without targeted intervention. Adversaries often exploit this persistence using Torshammer's gradual approach to evade conventional detection systems [154]. These attacks present significant threats to web servers, leading to serious service disruptions and potential server failures. Therefore, the timely detection and mitigation of THA attacks are imperative to ensure network security and maintain server integrity against adversarial actions. One such work towards the identification of THA-based DDoS attacks was carried out by the authors in [155]. The authors developed a tool to simulate and analyze the impact of HTTP-based LDDoS THA attacks. Torshammer sends slow but legitimate HTTP POST requests to a target web server to exhaust its resources over time by keeping connections open through multiple incomplete requests. As the server waits for these requests to be completed, its resources become tied up, eventually leading to a denial of service for legitimate users. Owing to its stealthy

nature, which mimics regular traffic behavior, victims are often unaware of the attack. The authors conducted extensive experiments and demonstrated that potential damage to THA can cause web services. They also proposed a methodology for detecting and mitigating slow-rate DoS attacks in modern network environments, particularly within 5G infrastructures, and performed a comprehensive analysis of Torshammer's behavior, examining its key components and highlighting the need for more advanced detection mechanisms in next-generation networks.

Another significant study aimed at identifying critical patterns in low-rate DDoS attacks, particularly Torshammer, was conducted by Dantas et al. [156]. The authors framed the motivation for their study by discussing that certain low-rate DDoS attacks, such as Torshammer, are stealthy and can cause severe disruptions to web services by overwhelming server resources without raising immediate alarms. The study identified the unique characteristics of Torshammer attacks by analyzing hardware performance counters (HPCs), network traffic patterns, and system-level indicators. By employing ML models and feature selection techniques, the authors were able to pinpoint specific HPCs that are highly sensitive for detecting THA attacks. Ultimately, they successfully identified three key HPCs that can significantly improve detection performance in real-time environments. Furthermore, the authors categorized the attacks into various dimensions, including security impact, performance degradation, availability issues, and computational overhead, highlighting the wide-ranging effects of such low-rate attacks on the network infrastructure.

Until now, previous works have primarily relied on network traffic analysis and heuristic approaches to identify potential threats posed by LDDoS attacks, such as Torshammer. However, a more advanced model utilizing ML techniques has been proposed in a recent study. The authors in [157] implemented an IDS to detect slow HTTP POST attacks such as Torshammer. In their approach, the authors first identified the distinctive traffic patterns generated by THA attacks and then modeled these patterns using anomaly detection methods. After defining a baseline for normal network traffic, the IDS was trained to recognize deviations indicative of ongoing attacks. Through their experiments, the authors achieved high detection rates with an accuracy of over 98%, successfully identifying THA attacks in a simulated network environment.

F. HTTP UNBEARABLE LOAD KING (HULK)

A HULK attack is a type of DDoS attack that targets web servers by overwhelming them with a high volume of seemingly legitimate HTTP requests. This attack is particularly effective against web applications and can significantly disrupt services, making it a concern in SDN environments. The basic command to execute a HULK attack using Python is `"python2 hulk.py http://example.com"`,

where `"python2"` indicates the version of Python, `"hulk.py"` is the attack script, and `"http://example.com"` represents the target URL. Usually, SDNs rely heavily on APIs for communication between components. A successful HULK attack can exploit vulnerabilities in these APIs, allowing attackers to manipulate network behavior or gain unauthorized access. The open-source nature of the HULK tool has allowed researchers to create forks and enhancements. One such study addressing DDoS attacks in SDN environments was carried out by Chafika et al. [15]. The authors proposed a robust self-protection framework that utilizes DL and SDN architectures to detect and mitigate hull attacks at the application layer. The framework continuously monitors network traffic and analyzes flow patterns using a deep learning model that is resilient to adversarial examples. Once suspicious traffic patterns indicative of Hulk attacks are detected, the system automatically issues mitigation policies through the SDN controller to block or redirect the malicious traffic. The framework also employs adversarial training to ensure that the detection model remains effective against evasion tactics and continuously updates itself based on new attack data to maintain the robustness and security of the network.

Cai et al. [158] conducted a study on detecting anomalous traffic caused by Hulk-based DDoS attacks in SDN layers. Their project aimed to identify malicious traffic patterns generated by the Hulk DDoS tool, which targets vulnerable SDN networks to overcome them and cause service disruptions. The study simulated high-rate HTTP flooding attacks that degrade network performance. Using a Random Forest classifier to analyze key traffic features, they optimized the detection by selecting a small set of important network flow features, achieving an accuracy of 99.95%. While the results are promising, further research is needed to improve detection across various network environments by incorporating methods such as reinforcement learning and generative models for more robust detection mechanisms. Another significant study addressing the use of simulation systems to defend against Hulk attacks in SDN environments was conducted by Aliyev [159]. The primary goal of this work is to simulate layer 7 hull attacks to better understand their impact and develop mitigation strategies while minimizing the overhead on system resources. The authors aimed to ensure that even systems with limited computational power could effectively simulate these attacks without a significant overhead. From the empirical analysis, the authors demonstrated that their simulation tool could accurately simulate hull attacks and help organizations identify vulnerabilities in their systems. The authors further evaluated the proposed simulation model to determine its practical application. The experimental results indicated that although the Hulk simulation caused some additional overhead, the tool remained effective for vulnerability identification. After reviewing the study, it is evident that, while simulation tools are increasingly being used to understand the impact of Hulk attacks, the frequency

and complexity of these attacks continue to grow. Therefore, there is a growing need to develop machine-and deep learning-based models that can detect and mitigate such attacks in real time within an actionable time frame.

G. GOLDENEYE

GoldenEye (G-Eye) is a tool primarily used for executing denial-of-service attacks, particularly against web servers. G-Eye can significantly impact SDN environments owing to its ability to generate a large number of connections and requests, potentially overwhelming the network's resources. In an SDN environment where traffic is managed centrally, a G-Eye attack can disrupt established policies and lead to unexpected behaviors in network management [160]. To execute an attack using Goldeneye (strictly for educational purposes, as unauthorized use is illegal), ensure that Python is installed in the system. The command to initiate a basic attack is: `"goldeneye.py http://www.example.com"`. The attack can be customized by adjusting parameters, such as `"w"` for the number of workers, `"s"` for the number of sockets, and `"m"` to specify the HTTP method. One such study from the perspective of detecting and mitigating GoldenEye-based attacks in SDN layers was conducted by Mahmoud et al. [161]. in their recent study. The proposed framework has two major aims: detection of application-layer DDoS attacks, such as GoldenEye, and real-time mitigation of these attacks within the SDN environment. From the viewpoint of attack detection, the authors analyzed several SDN-specific metrics such as packet flow rates, controller response times, and the number of active flows in the network. Similarly, to identify malicious activity, the authors used detailed flow statistics alongside attack simulation data, which helped visualize and identify malicious traffic patterns generated by G-Eye. By utilizing various SDN-related factors such as flow duration and packet inter-arrival times, the framework was able to enhance the accuracy of anomaly detection and provide a robust defense mechanism against G-eye attacks. A more generalized approach towards the development of anomaly detection tools for GoldenEye-based attacks in SDN environments was presented in [162]. The authors first discussed the architecture of SDN and then highlighted specific attack patterns generated by tools such as GoldenEye, which cause abnormal HTTP traffic flows and connection issues. Subsequently, the authors implemented and validated their framework using LSTM-CNN networks to detect these attacks. The framework was tested on the CICIDS 2017 dataset, which includes various attack types such as GoldenEye, and employed different configurations of LSTM-CNN networks, analyzing the impact of hyperparameters such as the number of layers and learning rates. The results showed that the optimized LSTM-CNN models significantly improved detection performance. Despite these findings, the authors emphasize that there is a critical need to develop real-time detection frameworks capable of promptly identifying and mitigating such attacks in dynamic SDN environments.

H. SUMMARY AND INSIGHTS

Section VII reviews how DDoS application layer attacks in SDN environments have been addressed through various research efforts. Existing studies have primarily focused on mitigating attacks such as Hulk, Slowloris, RUDY, DNS amplification, and HTTP floods through a variety of ML, DL, and other advanced detection methods. For instance, Random Forest classifiers have demonstrated detection accuracy rates as high as 99.95% for Hulk attacks [158]. Similarly, entropy-based detection with dynamic thresholds has been effectively employed to manage fluctuating traffic patterns caused by application-layer attacks [131]. Moreover, DNNs and autoencoder models have been utilized to further enhance the detection precision for abnormal traffic in SDN environments [137]. In contrast, targeted approaches to DNS-based attacks have yielded promising results. Techniques such as proactive mapping and Bayesian filtering have significantly improved the detection and mitigation of DNS amplification attacks [138]. Furthermore, programmable switches and SDN modules have enhanced traffic monitoring and attack detection at the data plane level, leveraging SDN's inherent flexibility of SDN to deploy adaptive security measures [139], [142]. Nevertheless, significant gaps remain in the literature. Although many solutions are effective, they tend to focus on specific attack types, thereby limiting their broader applicability in different network environments. Additionally, although ML and DL models demonstrate high accuracy, their adaptability to evolving attack patterns in real time continues to pose challenges. Therefore, future research should prioritize the enhancement of scalability and robustness of these detection systems. The integration of RL and GM offers promising avenues for developing more adaptive and proactive security mechanisms [156], [163].

Figure 8 categorizes the detection methods for DDoS application layer attacks in SDN environments, organized by SDN layers (application, control, and data planes) and specific attack types. Most research efforts have focused on the application plane, targeting attacks such as HTTP floods, DNS amplification, Slowloris, RUDY, Hulk, and GoldenEye [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159]. Commonly applied techniques include clustering [130], [138], nearest neighbor approaches [141], and statistical and analytical methods [131], [136], all of which have been deployed across multiple attack types. Moreover, machine and deep learning approaches dominate this landscape, with RF classifiers proving particularly effective in detecting Hulk attacks [158]. However, advanced detection techniques such as RL and GM remain underexplored, with relatively few researchers incorporating them in their studies [156], [163]. For example, RL and generative methods have been selectively applied to attacks such as RUDY and Hulk, revealing a gap in their broader application across various DDoS attack types.

TABLE 3. This table summarizes the research efforts on DDoS attacks in the application layer of SDN, listing attack types, detection methods, evaluation tools, and SDN layers targeted by each solution. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning Methods, MT: Moving Target defense strategies.

Reference	App Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[130], 2020	HTTP	Application	Proxy-based filter for detecting invalid requests	Emulation		✓								
[131], 2023		Application	Entropy-based detection with dynamic thresholds	CICIDS2017		✓	✓							
[132], 2023		Application	ML-based mitigation integrated with SDN	Mininet and RYU	✓				✓					
[133], 2020		Cont, App.	Deep learning-based detection and route optimization	MATLAB 2018a	✓				✓					
[134], 2020		Cont, App.	Time-based monitoring and blocking.	OpenFlow			✓							
[135], 2023	DNS	Cont, App.	Proposed layered defense for DNS servers.	Wireshark		✓	✓							
[136], 2019		Application	Introduces two techniques to prevent DNS attacks	Mininet	✓	✓								
[137], 2022		Cont, App.	DWT extracts; auto-encoder detects anomalies.	Mininet,GNS3		✓					✓			
[138], 2022		Data,Control	Proactive mapping DNS; Bayesian detects anomalies.	Mininet, OVS	✓	✓								
[139], 2021		Application	Proposed a model detects botnet DNS attacks	Linux utilities		✓								
[140], 2024		Data Plane	Bloom filters to detect DNS amplification.	C++,MAWI		✓	✓							
[141], 2020		Cont, App.	ML-based classifiers for DNS traffics.	CICIDS2017	✓				✓					
[142], 2020		Application	Selective model targets high-risk resolvers	Zmap		✓								
[143], 2022		Application	MTD for DNS, implemented in NFV frameworks	Mininet,Scapy		✓							✓	
[144], 2020	SW	Application	An advanced tool was designed to protect web servers	Apache2, Nginx			✓							
[147], 2022		Application	Firewall tools to counter Slowloris attacks.	Kali Linux			✓							
[148], 2020		Application	Presented SD-WoW to block Slowloris attacks	Open-WRT		✓	✓							
[151], 2023	RUDY	Application	Proposed DHMLM system to detect RUDY attack	Wireshark	✓				✓					
[152], 2024		Application	Developed FLD-LRDDOS to prevent RUDY attacks	RUDY, HULK	✓				✓					
[155], 2021	THA	Application	Proposed smart IDS that leverages ML to detect THA	CICFlowMeter	✓				✓					
[156], 2023		Application	ML-based detection uses selected HPCs.	HULK, LOIC	✓	✓			✓					
[157], 2020		Application	IDS detects traffic deviations from baseline.	LOIC, Xerxes	✓		✓		✓					
[163], 2020	Hulk	Application	Proposed DL-based technique to protect Hulk attacks	ONOS, Apache	✓	✓						✓		
[158], 2023		Application	Proposed Random Forest to detect Hulk attacks	Scikit-Learn	✓									
[159], 2023		Application	Suggested simulation tool identifying Hulk attacks	Portal: Online	✓	✓	✓							
[161], 2022	G-Eye	Application	Proposed DNN to detect errors of GoldenEye	PyTorchFI	✓									
[162], 2022		Application	Presented LSTM-CNN to detect GoldenEye attacks	CICIDS 2017	✓									

Consequently, this indicates a need for further research to explore these underutilized techniques in response to evolving threats in SDN environments.

Furthermore, although methods such as statistical analysis and clustering are widely adopted, they may not be sufficiently adaptive to the increasingly sophisticated attack patterns observed in today's networks. This underscores the need for an improved real-time detection and scalability. Therefore, future research should seek to integrate more dynamic approaches, such as reinforcement learning and generative models, to strengthen defense mechanisms against DDoS attacks across all SDN layers [159], [163].

Insight: Table 3 presents a summary of DDoS detection techniques focused on application-layer attacks in SDN. It categorizes research by attack types, detection methods, evaluation tools, and SDN layers impacted. Detection approaches include Classification, Statistical Analysis, and Reinforcement Learning, providing an organized view of current methodologies in mitigating application-layer DDoS threats within SDN.

VIII. PROTOCOL ATTACKS

In this section, we categorize the studies that have identified protocol DDoS attacks and concentrate on their efficient prediction from the perspective of all layers within the SDN architecture (cf. Figure 9 shows the visual representation of the attack categories and layers, and Table 4 presents a comparative analysis of the detection methods and their effectiveness).

A. TCP CONNECTION ATTACKS

A TCP connection attack exploits vulnerabilities in the Transmission Control Protocol (TCP), particularly during a handshake process, to disrupt services or gain unauthorized access [164]. Notable examples include SYN Flood, TCP Hijacking, and TCP Veto attacks. In an SYN Flood attack, the attacker overwhelms the server by sending a large volume of SYN packets, preventing it from completing connections and depleting its resources. TCP Hijacking, on the other hand, involves predicting sequence numbers to seize control

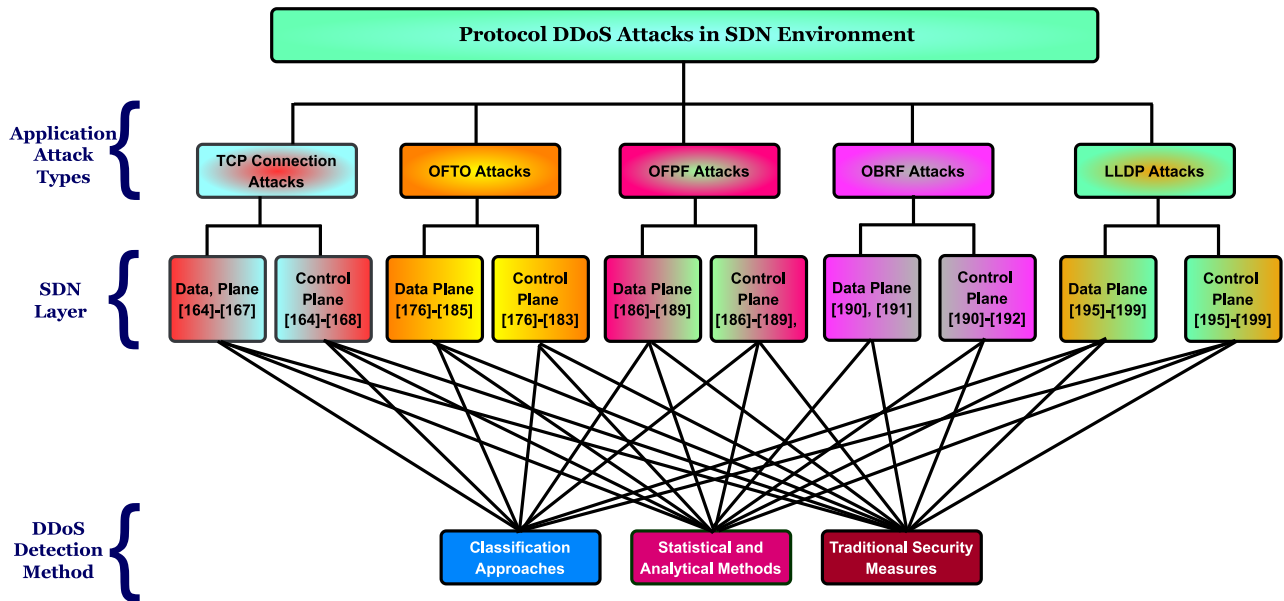


FIGURE 9. Taxonomy of DDoS protocol attacks in SDN environment. This figure categorizes different protocol-based DDoS attacks in SDN, maps them to their corresponding layers (Data Plane, Control Plane), and highlights the detection methods used (Classification Approaches, Statistical and Analytical Methods, Traditional Security Measures).

of an active session and inject malicious data. A TCP Veto attack occurs when an attacker sends a counterfeit packet with the correct sequence number, causing the legitimate packet to be discarded without interrupting the connection. Recent research has explored various mitigation strategies to address these vulnerabilities, focusing on improving TCP security and preventing resource exhaustion.

From the perspective of security analysis in SDN, a noteworthy work has been presented by Kumar et al. [165]. The authors primarily focused on identifying and analyzing vulnerabilities in the Multipath TCP (MPTCP) protocol across different layers of SDN. By exploiting the manipulation of the Data Sequence Signal (DSS) in MPTCP, the authors demonstrated how attackers could induce nonresponsive traffic, leading to DoS attacks. Through detailed experimentation and analysis, they proposed a novel detection model based on data sequence map skipping, which effectively mitigated these attacks by validating the integrity of the transmitted signals. This work highlights the critical security challenges inherent in the interaction of MPTCP with SDN architectures. Rahman et al. [166] introduced an approach to analyze vulnerabilities in TCP connection attacks within a software-defined industrial network. This research focuses on identifying and exploiting weaknesses in TCP connections to demonstrate how they can be manipulated to perform DoS attacks. The system designed for this analysis allows researchers to simulate real-world SDN environments and to track the impact of these attacks on network performance. One of the main objectives of this system is to understand the behavior of TCP connections under attack, providing insights into how these vulnerabilities

can degrade network stability and availability. The authors explored various attack scenarios in depth, demonstrating that their model can effectively highlight the weaknesses in TCP connections and offer pathways for further development of mitigation strategies. Although the proposed approach primarily addresses TCP in SDN, it can potentially be adapted to other protocols with similar functionality, extending the scope of the analysis to other network infrastructures.

An advancement in the detection of TCP attacks was presented by Hartpence and Kwasinski [167], who utilized sequential neural networks to analyze network traffic patterns instead of relying on predefined signatures. Their model processes real-time TCP packet data to detect abnormal behaviors, such as port scans, achieving over 99% accuracy while significantly reducing false positives compared with traditional systems such as Snort. This classification-based approach allows for a detailed examination of packet types and adapts to new attack patterns, although the detection of entirely novel attacks may require retraining to ensure ongoing accuracy. Another significant study on TCP attacks, conducted by Feng et al. [168], revealed a new type of off-path TCP hijacking attack that exploits weaknesses in the identification field of the IP protocol (IPID) assignment method used in recent versions of the Linux kernel (4.18 and later). The primary goals of this study are to elucidate the mechanics of an attack and evaluate its real-world implications for network security. The authors outlined the three-step attack process. Initially, an off-path attacker downgrades the IPID assignment from a more secure per-socket method to a less secure hash-based policy, thereby resulting in a shared IPID counter that creates a side

channel. In the second step, the attacker monitors the shared counter to detect the active TCP connections. Finally, the attacker infers the sequence and acknowledgment numbers of these connections, enabling them to hijack the connection, which may involve terminating the connection or injecting false data. The research findings indicate that over 20% of Alexa's top 100,000 websites are vulnerable to this attack, highlighting significant security risks. Through various case studies, the authors demonstrated the effectiveness of the attack in scenarios such as SSH DoS, manipulation of web traffic, and BGP routing table poisoning. To mitigate these vulnerabilities, the authors propose a novel IPID assignment method aimed at strengthening defense against such attacks. They developed a prototype of this defense mechanism in Linux 4.18, confirming its effectiveness in preventing TCP hijacking.

B. OPENFLOW TABLE OVERFLOW (OFTO)

OpenFlow is a pioneering protocol in the field of SDN that serves as the first standard communication interface between the control and forwarding layers [169]. This enables network controllers to manage how data packets are routed through network devices, thereby allowing for greater flexibility and efficiency in network management. OpenFlow offers numerous advantages that enhance network performance and management.

- **Centralized Control:** OpenFlow allows a logically centralized controller to manage multiple switches and simplifies the network configuration and management.
- **Flexibility and Customization:** Network operators can create custom features and services without relying on vendor-specific hardware or software, thereby enabling rapid service deployment.
- **Reduced Operational Costs:** By automating the network configurations, OpenFlow minimizes manual errors and downtime, leading to lower operational expenses.
- **Network Virtualization:** OpenFlow supports the virtualization of networks, allowing seamless integration with computing and storage resources, which is essential for cloud environments.
- **Interoperability:** As an open standard, OpenFlow fosters a multivendor ecosystem that allows different devices to be managed through a unified protocol.
- **Enhanced Traffic Management:** OpenFlow enables sophisticated traffic management strategies, such as dynamic path adjustments based on real-time network conditions.

Figure 10 illustrates the evolution of OpenFlow using its key versions. OpenFlow started with *Version 1.0* in December 2009, laying the foundation for the protocol [170]. *Version 1.1*, released in February 2011, introduced several enhancements such as support for multiple flow tables [171]. In February 2012, *Version 1.2* added further improvements and refinements [172]. *Version 1.3*, launched in October

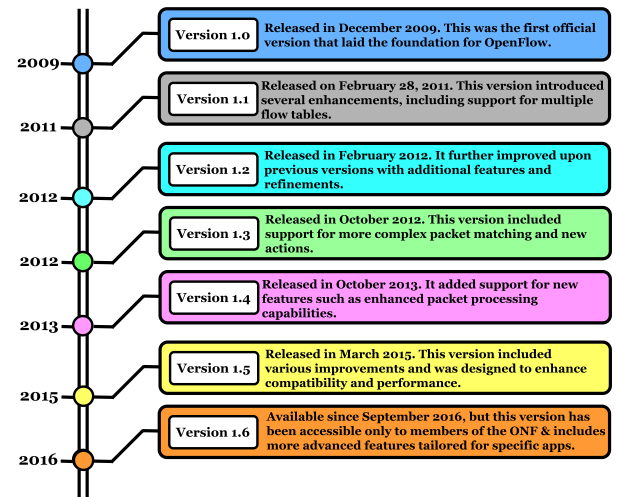


FIGURE 10. Evolution of OpenFlow versions.

2012, expands the protocol with more complex packet matching and new actions [170]. In October 2013, *Version 1.4* focused on enhancing packet processing capabilities [173], [174]. *Version 1.5*, released in March 2015, introduced improvements designed to enhance the compatibility and performance [175]. Finally, *Version 1.6*, available since September 2016, introduced advanced features tailored for specific applications, although it has only been accessible to members of the ONF [172].

OpenFlow table overflow is a specific threat, in which an attacker floods an OpenFlow switch with excessive flow entries. This can exhaust the switch's resources, causing performance degradation or even crashes. Mitigating the OFT overflow requires a multifaceted approach that combines various strategies tailored to specific network conditions and requirements. The authors in [176] presented a significant contribution to understanding OFT Overflow attacks in SDN, emphasizing emerging vulnerabilities that can disrupt the data plane. They introduced the LOFT attack, which efficiently targets the flow tables in SDN switches at a minimal attack packet rate. The LOFT attack methodology involves manipulating flow rule settings to create a shared side channel, detecting active TCP connections by observing network behavior, and inferring critical sequences and acknowledgment numbers. Experimental evaluations in a real-world SDN testbed revealed that over 20% of the top 100,000 websites are vulnerable to these attacks, affecting applications such as SSH DoS, web traffic manipulation, and BGP routing table poisoning. In response, the authors proposed LOFTGuard, a defense mechanism that enhances SDN security by enabling collaboration between data and control planes, effectively mitigating LOFT attacks while maintaining low overhead.

POAGuard [177] was introduced as a notable defence mechanism against OFTO attacks. It addresses the vulnerabilities exploited by a Preemptive Overflow Attack (POA),

which disrupts normal applications by taking advantage of the flow entry eviction mechanism. This research highlights how POA can significantly degrade network performance by using the widely deployed Least Frequently Used (LFU) eviction algorithm, which requires fewer flow entries compared to traditional rate-based overflow attacks (ROA). The authors further discussed the challenges in detecting POA owing to the absence of distinctive flow features, underscoring the need for robust detection methods. In response, they proposed POAGuard, a defense mechanism that encompasses a table segmentation method, score-based eviction algorithm to remove suspicious flow entries, and concept drift-based detection method to identify and defend against POA attacks. Extensive evaluations demonstrate the effectiveness of POAGuard in mitigating POA, revealing that, while both POA and ROA are significant threats to SDN, the lack of attention on preemptive strategies presents substantial research potential for future work in securing SDN architectures against such vulnerabilities.

FTODefender [178], *FTMaster* [179], and *FTOP* [180], all proposed by Tang et al., focus on mitigating Low-rate OFTO attacks through advanced ML techniques and flow eviction strategies. Furthermore, *FTODefender* utilizes a CRITIC-based detection mechanism combined with a LightGBM-LR classification model to differentiate between normal and malicious flows, thereby effectively evicting harmful rules and blocking their sources [178]. In contrast, *FTMaster* enhances this approach by enabling the real-time monitoring of flow tables and employing machine learning for LFTO attack detection, followed by targeted mitigation through flow eviction [179]. Similarly, *FTOP* integrates several modules for prediction, detection, mitigation, and prevention, aimed at addressing both LFTO attacks and Flash Crowds [180]. This system utilizes random forest classifiers to efficiently detect and evict low-priority or malicious flows, demonstrating substantial effectiveness in simulations, underscoring the importance of real-time defense mechanisms.

On the other hand, *FloRa* [181], *SFTO-Guard* [182], and *LtRFT* [183] introduce innovative ML-driven solutions specifically designed for slow-rate OFTO attacks. *FloRa* employs CatBoost along with advanced flow analysis techniques to detect and blacklist malicious flows, achieving impressive detection accuracy with minimal overhead [181]. Moreover, *SFTO-Guard*, also developed by Tang et al., enhances the mitigation process through predictive analytics and adaptive eviction strategies, maintaining a low operational overhead while facilitating swift responses to threats [182]. Conversely, *LtRFT* applies learning-to-Rank algorithms to model the flow-entry eviction process, prioritizing the removal of malicious entries [183]. This approach has shown robust results in detecting malicious flows, significantly reducing the attack duration, and adding minimal latency, thereby illustrating its practicality within SDN environments.

Other notable solutions include *NFV-GUARD* [184], *HSDT* [185], and *CEOF* [186], each contributing a unique methodology to combat OFTO. *NFV-GUARD* employs NFV to dynamically filter out attackers and mitigate overflow attacks without relying on the underlying network infrastructure [184]. This approach is particularly advantageous for small-to medium-sized networks, where traditional distribution methods may fail. Moreover, the “*History-Based Dynamic Timeout (HSDT)*” scheme adapts flow entry timeouts based on statistical data and flow history, allowing for the quick removal of malicious flows while maintaining normal traffic performance [185]. This dynamic adjustment ensures that the flow table occupancy remains low, thereby enhancing resilience against overflow attacks. Conversely, the *CEOF* employs a two-phase approach, where it first estimates the number of flow entries using HyperLog for overflow detection, followed by the eviction of unnecessary entries using Hierarchical Agglomerative Clustering [186]. This innovative methodology significantly reduces redundant entries and improves network throughput, as evidenced by the simulation results. Overall, to advance research on detecting OFTO attacks, future studies should focus on developing hybrid approaches that combine DL techniques with real-time monitoring and adaptive mitigation strategies. Additionally, exploring the integration of predictive analytics and dynamic flow management can enhance the resilience and efficiency of SDN systems against evolving attack vectors.

C. OPENFLOW PACKET-IN FLOOD (OPPF)

OPPF is a type of DDoS attack that exploits communication between OpenFlow switches and controllers. This attack can significantly impact the southbound interface, leading to DoS conditions. Attackers commonly use techniques, such as packet injection, table miss exploitation, and flooding with various payloads. Tools such as Scapy and Hping are often employed to create and send custom packets designed to initiate these attacks. For simulation purposes, platforms such as *OMNeT++* enable testing of packet flooding against various network configurations. OPF attacks on southbound interfaces often target resource exhaustion, network congestion, and controller overload. Researchers have explored various mitigation strategies, including traffic monitoring, filtering, rate limiting, table miss optimization, and decoupling mechanisms. Asgari and Akbari [187] focused primarily on traffic monitoring and flow analysis using entropy-based detection. Their method involves monitoring packet distributions and analyzing flow table statistics to detect anomalies, such as DDoS attacks, specifically OPF attacks. It combines aspects of traffic monitoring with an advanced analytical technique (entropy) rather than filtering or rate-limiting. Although their approach does not directly involve rate limiting or table miss optimization, it uses flow table statistics to optimize detection accuracy

and controller performance, indirectly contributing to table miss optimization by detecting floods early and managing flows efficiently. Another study utilizing traffic monitoring and classification to mitigate OFPF attacks was conducted by Hu et al. [188]. The major focus of this study is to address the vulnerabilities of the SDN southbound interface by employing entropy-based methods to detect abnormal traffic patterns. The authors implemented a classification system using supervised learning techniques, specifically an SVM classifier, to identify the anomalies caused by OFPF attacks. Furthermore, a whitelist-based mitigation mechanism is introduced to distinguish between legitimate and malicious traffic. The proposed method demonstrated high accuracy in detecting DDoS attacks in an experimental setup. However, while the mitigation system effectively reduces the impact of OFPF attacks, further improvements in scalability and handling of more complex attack scenarios can enhance the overall system performance. Another comprehensive study on traffic monitoring and intentional dropping as a mitigation strategy was conducted by Kareem et al. [189]. The research highlights the security challenges of SDN, particularly the vulnerabilities in the data and control planes that expose the network to various attack vectors. The study focuses on DDoS attack detection by analyzing the entropy variation in the destination IP addresses. The authors utilized the flexibility of the OpenFlow protocol and POX controller to implement their detection method. The proposed approach calculates the entropy to identify OpenFlow packet flood attacks and demonstrates its ability to detect such attacks within 0.445 s of occurrence, showcasing its potential for real-time attack detection in SDN environments.

D. OPENFLOW BARRIER REQUEST FLOOD

OBRF refers to a scenario in which an OpenFlow controller sends multiple barrier request messages to a switch, potentially causing congestion and the delayed processing of subsequent messages [190]. This situation arises when the controller needs to ensure that all previously issued commands are processed before executing any new commands. However, improper management can inadvertently lead to a flood of requests. The OpenFlow protocol includes a Barrier Request message “(OFPT_BARRIER_REQUEST),” which is sent by the controller to instruct the switch to complete all pending messages before processing new ones. The switch acknowledges this by queuing any subsequent messages until it sends back a Barrier Reply “(OFPT_BARRIER_REPLY)” after completing the earlier operations. If the controller sends multiple Barrier Requests in quick succession, it can overwhelm the switch because each request forces the switch to pause the processing of new commands, creating a backlog. This is particularly likely in high-throughput environments, where rapid state changes occur frequently. In addition, if the controller fails to manage its request rate properly, the situation can worsen. However, continuous flooding with Barrier Requests can lead to resource exhaustion during

the switch. As it struggles to handle queued requests, the switch may eventually run out of the memory or processing capacity. To address this issue, several strategies have been proposed and implemented to mitigate the effects of excessive Barrier Requests in OpenFlow. Mishra et al. conducted a study focusing on detecting abnormal behavior in SDN, particularly related to OBRF attacks [191]. The article primarily addresses the detection of anomalous traffic patterns by monitoring entropy variations in network flows, especially focusing on identifying sudden drops in entropy as an indicator of potential OBRF attacks. The authors further discussed various attack scenarios and their effects on the SDN controller and flow tables. Subsequently, an entropy-based defensive mechanism was introduced to mitigate these attacks by monitoring traffic flows and identifying unusual traffic patterns. The method was evaluated in an SDN-Cloud environment using simulation tools such as Mininet and a POX controller, demonstrating the effectiveness of entropy variation monitoring in mitigating OBRF attacks and protecting the SDN southbound interface from overload and disruption. A final study addressing the mitigation of OBRF attacks in the SDN southbound interface was conducted by Shoaib et al. [192]. The authors analyzed the vulnerabilities of SDN control planes by evaluating how timing-based side-channel attacks can be leveraged to exploit the system. To address this, the authors developed a novel defense mechanism called *Netkasi*, which introduces randomized delays to controller responses to obscure timing information from attackers. The solution effectively prevents adversaries from distinguishing between switch-served and controller-served packets, thereby protecting SDN from control plane exhaustion caused by OBRF attacks. Another key contribution of this study is the use of randomized response times to evaluate the system’s effectiveness in obfuscating the controller behavior. The authors further compared their solutions to traditional techniques to demonstrate their effectiveness in mitigating timing-based attacks. Although OBRF attacks remain a critical challenge in SDN, this study presents an innovative approach, suggesting that future improvements could involve integrating modern ML techniques to further enhance detection and mitigation capabilities.

E. LINK LAYER DISCOVERY PROTOCOL (LLDP) FLOODS

LLDP attacks in SDN are a type of security vulnerability that exploits the topology discovery mechanisms in SDN architectures [193]. These attacks can alter the network topology as perceived by the SDN controller, potentially leading to data breaches, service disruptions, or other malicious outcomes [194]. For instance, an attacker may use compromised hosts to flood a network with excessive traffic (e.g., ARP requests), thereby increasing the round-trip time (RTT) of legitimate LLDP packets. The attacker can then relay LLDP packets through out-of-band channels (channels that bypass the usual data paths), falsify the links between switches, and create a false network topology. Undoubtedly,

LLDP floods can overwhelm the network controller by sending a high volume of LLDP packets, exhausting resources, resulting in incorrect topology mapping, and causing potential network outages. Popic et al. [195] highlight the vulnerability of LLDP in SDN environments, particularly the susceptibility of the control plane to attacks exploiting link discovery protocols. They proposed the use of a “*Network Configuration Protocol (NETCONF)*” in combination with “*YANG*” modeling to mitigate the risks posed by LLDP floods. By enhancing the controller’s ability to differentiate between legitimate and malicious LLDP messages, their solution strengthens the security and integrity of SDN topology-discovery services. The *LINK-GUARD* framework proposed by Al Salti and Zhang [196] is an effective method for preventing LLDP-based DDoS attacks. By introducing packet authentication and the real-time verification of LLDP messages, “*LINK-GUARD*” ensures that only legitimate packets are processed by the SDN controller. This approach significantly reduces the likelihood of topology poisoning attacks caused by LLDP floods, and the framework has been demonstrated to detect fabricated links in milliseconds, thereby providing real-time protection with minimal overhead.

Soltani et al. [197] focus on the broader scope of topology poisoning attacks in SDN, including LLDP floods. They proposed an MLLG that used real-time latency data to detect abnormal link creation caused by LLDP floods. Their system is particularly adept at identifying false links and mitigating negative impacts on the controller’s resource management by continuously analyzing link latency data for anomalies. Joseph et al. [198] present a mitigation approach named *LiFAMA*, which integrates hashed message authentication codes (HMAC) to authenticate LLDP packets. This solution actively prevents the addition of fake links during the discovery process, addressing one of the primary weaknesses of the LLDP in SDN environments. Their approach minimizes the link discovery time while ensuring that the controller only processes verified legitimate links, which are crucial in preventing DDoS attacks based on LLDP floods. Hussain et al. [199] extended conventional LLDP discovery mechanisms by proposing an optimized protocol that reduces the number of LLDP packets exchanged between controllers and switches. Their Extended Indirect Controller Legacy Forwarding (E-ICLF) scheme helps mitigate the impact of LLDP floods by limiting unnecessary packet exchanges and reducing the risk of controller overload in multicontroller hybrid SDN environments. Overall, future studies should focus on the development of robust authentication methods for LLDP packets to prevent spoofing. Implementing cryptographic techniques such as *HMAC* or digital signatures can significantly enhance the integrity of LLDP communications. Finally, implementing real-time monitoring systems using AI techniques, such as ML and DL, can help promptly detect and respond to LLDP attacks as they occur. This could involve deploying lightweight models at edge devices for immediate threat assessment.

F. SUMMARY AND INSIGHTS

Various DDoS attacks exploit vulnerabilities in SDN protocols, including TCP, OFTO, OFPF, OBRF, and LLDP, to compromise the network. Moreover, the literature reveals numerous strategies and techniques proposed to mitigate these threats. TCP-based DDoS attacks are significant threats to both the data and control layers of SDNs, and often exploit flaws in connection management. In addition, many frameworks focus on improving detection through ML approaches such as neural networks and sequence mapping [164], [165]. Furthermore, statistical methods such as *IPID* assignment [168] and PLC-based detection [166] address specific attack scenarios. However, most approaches have only been tested in controlled environments, thereby limiting their real-world effectiveness. Despite this, ML offers promising detection capabilities; however, scalability in large SDNs remains a challenge. Therefore, future efforts should prioritize real-world applicability and scalability while exploring dynamic adaptations, potentially using RL to counter evolving threats.

OFTO attacks exhaust flow tables in SDN switches, leading to resource depletion and network disruptions. Various recent solutions, such as *LOFTGuard* and *FTODefender*, tackle low-rate table overflow attacks [176], [178], whereas ML methods such as *FLORA* and *CEOF* have been shown to enhance detection accuracy [181], [186]. Likewise, these methods, which span statistical and clustering approaches, demonstrate potential. Nevertheless, they often require fine-tuning to reduce detection errors. Thus, future research should aim to develop adaptive mechanisms that can autonomously adjust to varying flow table sizes and network configurations. Therefore, techniques such as generative models and RL can offer more dynamic and effective responses to this challenge.

Packet-in floods overwhelm the SDN controller with excessive requests, disrupting both data and control planes. For example, detection strategies include entropy-based methods [187], [191] and SVM classifiers [188]. “*OpenFlow Entropy-Based Detection Model*” [189] explores DDoS attack detection in SDN using entropy variation in destination IP addresses. It leverages the OpenFlow protocol and POX controller to identify UDP flood attacks, achieving detection within 0.445 seconds of attack onset. The proposed model focuses on real-time monitoring of traffic behavior to improve security in SDN environments.

OBRF attacks focus on the control plane by overwhelming it with barrier requests and disrupting normal operations. In contrast to other more widely studied attacks, this vector remains relatively underexplored. Nonetheless, frameworks such as *Netkasi* [192] and *OFaaS* [190] have introduced initial defenses that utilize entropy-based detection mechanisms. However, the literature offers limited comprehensive solutions for mitigating OBRF attacks. Consequently, future research could explore advanced traffic analysis techniques as well as strategies such as generative models and moving target defense, which have the potential to enhance resilience.

TABLE 4. This table summarizes the protocol attacks of SDN environment. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Reference	Proto. Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[164], 2022	TCP	Data,Control	Framework detects low-rate TCP denial.	Mininet, RYU		✓	✓							
[165], 2021		Data,Control	Data sequence map skipping to detect TCP attacks	Iperf, Dummynet			✓							
[166], 2022		Data,Control	Traditional based techniques to detect TCP attacks	OpenPLC			✓							
[167], 2020		Data,Control	Sequential neural networks to analyze traffics	Snort, NMAP	✓	✓	✓							
[168], 2022		Control	IPID assignment method to detect TCP hijacking	Ubuntu 18.04		✓	✓							
[176], 2023	OFTO	Data,Control	LOFTGuard mechanism to protect OFTO attacks	Floodlight		✓	✓							
[177], 2023		Data,Control	Proposed POAGuard to protect POA attacks	Mininet, RYU		✓	✓							
[178], 2024		Data,Control	Proposed ETODefender to detect OFTO attacks	Mininet, RYU		✓	✓							
[179], 2023		Data,Control	Proposed FIMaster to monitor low rate OFTO attacks	Mininet, RYU		✓	✓							
[180], 2024		Data,Control	Presented FTOP to protect low rate OFTO attacks	Mininet, RYU		✓	✓							
[181], 2024		Data,Control	Proposed FLORA, ML-based to detect OFT attacks.	Mininet, RYU	✓		✓							
[182], 2023		Data,Control	Proposed SFTO-Guard to protect OFTO attacks	Mininet, OV's		✓	✓							
[184], 2021		Data,Control	Presented NFV-GUARD to protect overflow attacks	Mininet, RYU		✓	✓							
[183], 2023		Data,Control	Presented LFTT to track flow table attacks in SDN	Mininet		✓	✓							
[185], 2023		Data	Proposed HSDT to limit memory of OFTO attacks	VM, RYU			✓							
[186], 2022	OFFP	Data,Control	Presented CEOF ML-based to detect OFTO attacks	Simulated	✓		✓							
[187], 2022		Data,Control	Entropy-based detection method for OFFT attacks	Mininet, POX		✓	✓							
[188], 2017		Data,Control	Proposed SVM classifier to mitigate OFFP attack	Mininet,TFN2K	✓	✓								
[189], 2022		Data,Control	Approach calculates entropy to identify OFFP attacks	Mininet,POX		✓	✓							
[190], 2021	OBRF	Data,Control	Presented OFaaS to mitigate OBRF attacks	Wireshark, OV's		✓	✓							
[191], 2022		Data,Control	Entropy-Based Detection method for PBRF attacks	Mininet, POX		✓								
[192], 2021		Control	Proposed NetKas to mitigate OBRF attacks	Mininet, RYU		✓								
[195], 2020	LLDP	Data,Control	Proposed NETCONF to mitigate LLDP floods	YANG Model		✓	✓							
[196], 2022		Data,Control	Proposed LINK-GUARD to detect LLDP floods	Mininet, RYU		✓	✓							
[197], 2023		Data,Control	Proposed MLLG to detect abnormal LLDP attacks.	Mininet,Floodlight	✓		✓							
[198], 2022		Data,Control	Presented LiFAMA to authenticate LLDP packets.	Mininet,POX			✓							
[199], 2022		Data,Control	Proposed E-ICLF to mitigate LLDP floods.	Mininet,POX			✓							

LLDP flood attacks target network topology discovery by overwhelming the control plane with the discovery packets. To address this, *NETCONF* and *LINK-GUARD* [196], [197] employ mitigation and detection techniques. Additionally, *LiFAMA* [198] focuses on authenticating LLDP packets to prevent such attacks. Furthermore, ML-based methods such as *MLLG* [197] enable real-time detection by analyzing network traffic. However, the key challenge is to reduce the computational cost of these detection mechanisms, particularly for large networks. Ultimately, future work should explore clustering techniques and deep learning to automate detection threshold tuning and minimize both false positives and network overhead.

Insight: Table 4 focuses on protocol attacks in SDN and highlights that most studies rely on three main DDoS detection techniques: CA, SA, and TS. While these methods have been valuable for detecting and mitigating protocol-specific attacks, such as TCP connection issues and OpenFlow-based vulnerabilities, their reliance on these techniques limits the exploration of other potentially more advanced and adaptive approaches. Techniques such as RL, CT, and GM can provide better detection mechanisms, particularly

for dynamic, real-time protocol attacks. Investigating these alternatives is crucial for addressing the evolving complexity and frequency of attacks, enabling more resilient defenses in SDN environments.

IX. RESOURCE EXHAUSTION ATTACKS

In this section, we classify the studies that have addressed resource-exhaustion attacks and examine their effective detection methods across various SDN layers, including the application, control, and data planes. These attacks are categorized into three types: firewall rule explosion, ARP cache poisoning, and DHCP starvation DDoS attacks, each affecting different SDN planes. The detection methods applied include classification approaches, statistical and analytical methods, traditional security measures, and moving target defense strategies, as shown in Figure 11 (cf. Table 5 for details).

A. FIREWALL RULE EXPLOSION (FRE)

FRE is a specific type of DDoS attack that exploits the resource management capabilities of firewalls [200], [201]. This attack overwhelms firewalls by generating an excessive

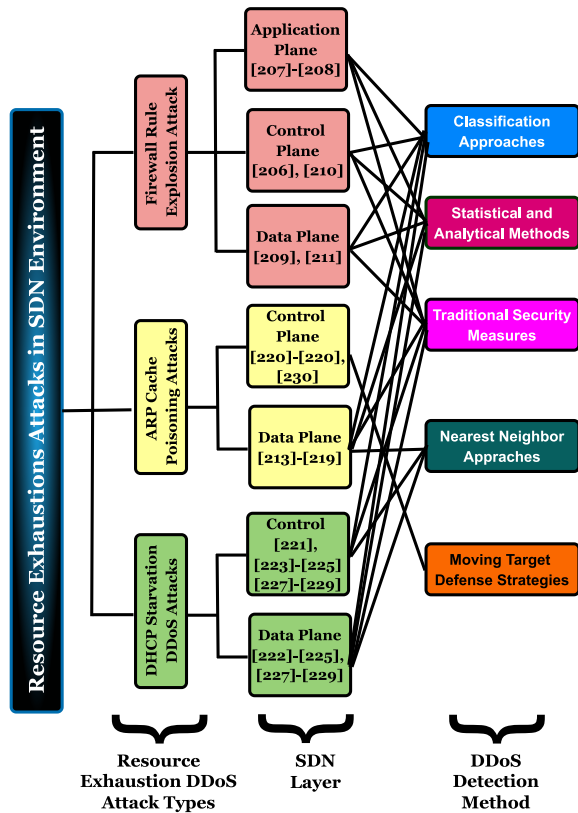


FIGURE 11. Classification of resource-exhaustion attacks in SDN environments across different SDN layers, highlighting the corresponding detection methods and affected planes for each attack type.

number of rule creation requests, thereby leading to resource exhaustion and service disruption across multiple layers of SDN architecture [202]. In the control plane, attackers can flood the controller with requests to create or modify firewall rules, thereby consuming CPU and memory resources [203], [204]. As a result, this overload can lead to significant slowdowns or even crashes, which disrupt the network's control and management capabilities. Moreover, in the data plane, the attack causes flow-table overflows in switches, preventing them from storing new rules [204]. Consequently, this results in packet drops and degraded network performance, as legitimate traffic cannot be effectively processed. Furthermore, the application layer is vulnerable because malicious API calls can generate excessive requests for rule updates, further draining resources, and potentially causing application crashes [205]. Ultimately, this affects the services that are dependent on the SDN infrastructure. Thus, Firewall Rule Explosion, which targets multiple layers of SDN architecture, poses a substantial threat to the stability and performance of these networks.

Pioneering work using ML for FRE prevention in SDN environments was carried out by *Prabakaran et al.* [206], who addressed the significant challenges posed by firewall rule explosions, especially in the context of decoupled control

and data planes of SDN. Their solution involved leveraging ML algorithms, such as BayesNet, naïve Bayes, and decision trees, within a virtualized stateful firewall to detect and mitigate rule explosions. BayesNet achieved the highest prediction accuracy of 92.87%, minimizing the overload caused by excessive rule generation, and enhancing SDN scalability and security during DDoS attacks. This innovative approach offers a proactive solution to rule explosions, with future work aimed at further optimization.

Kavach is an approach to enhance firewall detection capabilities in dynamic networks, presented by Aswal et al. [207]. It addresses the issue of FRE, where the rapid growth of rules leads to inefficiency and security vulnerabilities. The authors introduced an AI-based method to update firewall rules dynamically, enabling the system to learn from past attacks and adapt to evolving threats without manual intervention. Using ModSecurity and ML frameworks trained on the NSL-KDD dataset, the system detects and blocks novel attack vectors, improving accuracy and mitigating FRE attacks, and ultimately strengthening network security in SDN environments.

The authors of [208] addressed the issue of FRE in SDN by implementing a Web Application Firewall (WAF) using SDN to detect and mitigate SQL injection attacks. They focused on reducing the overhead of firewall rule expansion, particularly in dynamic environments where traffic complexity increases. The solution involves dynamically generating firewall rules based on traffic patterns and attack signatures, which helps manage the scalability challenges faced by traditional firewalls. By leveraging the SDN's centralized control plane and regular expression-based detection techniques, their approach ensures efficient rule management and avoids unnecessary rule accumulation. A comparison with ModSecurity highlights the superior scalability and performance of SDN-based WAF, significantly reducing latency and controller overhead, even under heavy traffic conditions.

In NFV, the firewall rule management is vital for network security. The authors in [209] investigated vulnerabilities from firewall rule explosions, where complex rule sets grow with network size. They explored how cache side-channel attacks, using the “Flush + Reload” technique, could infer firewall rules by analyzing cache traces during packet processing. This method bypasses traditional probe-and-response approaches, proving to be more efficient as the rule sets expand. Simulations using VyOS demonstrated that their approach can infer rules with over 90% accuracy using a few probe packets. To mitigate these risks, the authors suggested improved virtual machine isolation and better cache management in NFV environments. Another notable study that addresses the problem of firewall policy anomalies is carried out by Togay et al. in [210], where they propose a “Firewall Policy Anomaly Detection (FPAD)” framework. As network devices multiply, firewall rulesets grow, leading to rule explosions of overlapping, redundant, and conflicting rules that undermine security and efficiency. The “FPAD”

framework, which is based on logic programming and Prolog, simulates packet flows and detects four main anomalies: shadowing, redundancy, generalization, and correlation. This automated system reduces the rule complexity and enhances the performance. Tested on simulated policies, it demonstrated high accuracy, particularly benefiting SDN, where efficient firewall management is critical for maintaining security and performance.

Nevertheless, Amal and Venkadesh [211] introduced a hybrid honeynet system, “*H-DOCTOR*”, designed to enhance firewall configurations using honeypots. This system aims to detect real-time attacks, such as DDoS and ransomware, within IoT and cloud environments. By employing both low- and high-interaction honeypots, the system lures attackers, analyzes their behavior, and dynamically updates firewall rules to improve security. The “*H-DOCTOR*” system outperformed traditional methods, achieving 86% accuracy in ransomware detection compared to 73.25% in conventional systems. The study also explored the potential for integrating ML models to further improve anomaly detection, particularly against zero-day vulnerabilities. Future work should focus on enhancing these techniques to better safeguard against advanced threats.

B. ARP CACHE POISONING

Address Resolution Protocol (ARP) cache poisoning is a type of exhaustion attack in which an attacker sends falsified ARP messages over a local network [212]. This results in the association of the attacker’s MAC address with the IP address of a legitimate device, thereby allowing the attacker to intercept or manipulate the traffic intended for that device. In the context of SDN, this attack poses significant risks owing to the centralized nature of control in SDN architectures. ARP cache poisoning can exhaust resources across all three planes of SDN, and the data plane, which is responsible for forwarding packets based on the rules set by the control plane, can become overwhelmed by a flood of malicious ARP packets. The control plane that manages flow entries and network policies can be directly targeted. An attacker can send continuous ARP replies that appear legitimate, causing the SDN controller to incorrectly update its flow tables. Applications that rely on accurate network state information may be misled by poisoned ARP cache. This can result in incorrect monitoring and management decisions, further complicating the network’s operational integrity.

Researchers have focused on various methods for detecting and mitigating ARP cache-poisoning attacks in SDN environments. One study addressing the persistent issue of ARP spoofing was conducted by Ibrahim et al. [213], who proposed an enhanced SDN controller equipped with a module to monitor and intercept ARP packets. Their solution incorporated a proxy ARP functionality, enabling the controller to respond to ARP requests, thereby reducing

broadcast traffic and preventing ARP spoofing an attack vector that can lead to DoS and Man-in-the-Middle (MITM) attacks. Through experiments in a simulated SDN environment, this study demonstrated the effectiveness of this approach in preventing ARP spoofing and mitigating ARP broadcast traffic, leading to improved response times and reduced CPU load. However, the authors identified scalability as a potential challenge, as larger networks may impose higher CPU loads on the controller, thereby affecting the overall performance.

“*D-ARP*” is a robust mechanism proposed by Morsy and Nashat [214] to defend against ARP spoofing attacks. The proposed scheme enhances network security by integrating cryptographic keys with ARP packets and signing each ARP request and response. This ensures that any mismatch between the MAC and IP addresses is detected, preventing unauthorized access and protecting the network from spoofing attempts. A key advantage of *D-ARP* is its compatibility with existing ARP protocols, making it easy to implement without requiring significant changes to the network infrastructure. The method utilizes DHCP servers and network scanning tools, such as Nmap, to accurately identify legitimate MAC addresses, ensuring that any unauthorized devices are swiftly detected and blocked. The results of the study show that *D-ARP* achieves a high detection rate with zero false positives or negatives, and imposes minimal performance overhead. The authors also recommend future improvements to expand the capabilities of “*D-ARP*”, particularly by addressing more complex ARP-related vulnerabilities and testing them in larger, more dynamic network environments.

Several studies have been conducted on the efficient detection and prevention of ARP poisoning in SDN environments. One such work was presented by Zhao et al. [215], where the authors focused on the vulnerability posed by ARP spoofing, which can result in attacks such as DDoS through the manipulation of MAC-IP mappings in ARP tables. In this approach, both manual and automated detection techniques were explored. Tools such as Wireshark are used to detect duplicate IP addresses, whereas Arpwatch monitors MAC-IP pair changes in real time. The authors highlighted that while manual methods such as static MAC entries can mitigate small-scale attacks, they are not feasible for larger, dynamic networks. Automated approaches, which involve monitoring ARP tables for inconsistencies, show more promise, but still face scalability challenges.

The detection latency of ARP attacks was determined by Majumdar and Chintala [216]. The authors proposed a novel OS-based monitoring system that continuously observes ARP tables for anomalies, enabling the immediate identification of ARP poisoning incidents. This automated detection system offers significant performance improvements over traditional manual approaches, as it operates in real time to detect inconsistencies in the ARP cache, thereby drastically reducing the response time. Although the evaluation was

conducted in a controlled, simulated environment, this study demonstrated the effectiveness of this automated technique in enhancing network security by addressing ARP-related vulnerabilities more efficiently.

Regarding AI-based detection techniques, Atmojo et al. [217] presented a k-NN classification method to analyze network traffic and identify anomalies that indicate ARP poisoning. The motivation behind this study is the significant threat posed by ARP poisoning, which can result in communication deadlocks, identity fraud, and data theft in networks. The contribution of this study is to provide an accurate and efficient mechanism for detecting ARP poisoning attacks that minimizes false positives. This methodology involves the use of network traffic analysis to extract key features from network flows, which are then fed into a k-NN model to classify normal versus malicious traffic. The dataset used for training and testing the model was generated from simulated network traffic, which included both normal operation and ARP poisoning attacks. The experimental results demonstrated the effectiveness of the model, with a true positive rate of 97.67% and an overall detection accuracy of 98.7%. The study also highlights the limitations of previous detection models that rely on complex configurations and fail to distinguish ARP poisoning from normal ARP activity. Another ML-based work done by Ahuja et al. [218], introduced a hybrid framework combining “*Convolutional Neural Networks (CNN)* and *Long Short-Term Memory (LSTM)*”, to classify benign and malicious ARP traffic. The methodology involved creating a custom dataset using Mininet, a network emulator, to simulate ARP attacks. The generated dataset includes various network features such as MAC/IP addresses and packet types. The study evaluated multiple ML approaches, with the CNN-LSTM hybrid model achieving an accuracy of 99.73%, outperforming other methods in detecting ARP-based attacks. The experimental tools included Python for model implementation, Mininet for network simulation, and a traffic-logging system for data collection.

Thomas et al. [219] propose a detection algorithm implemented using Scapy and Python that identifies malicious ARP packets by systematically analyzing MAC and IP address inconsistencies. Their method not only detects ARP poisoning attempts, but also suggests using static ARP entries as a preemptive measure to mitigate the risk of such attacks. The experimental setup was conducted in a Kali Linux environment using a VMware workstation simulating ARP spoofing to test the detection algorithm. The study found that their method is effective in identifying and preventing ARP attacks, especially in scenarios prone to DDoS attacks. A similar study suggested a scheme called “*GaTeBaSep*” [220]. The authors propose a game-theoretic approach that leverages SDNs to predict optimal defensive strategies using the Nash equilibrium. The model frames the interaction between the attacker and defender as a non-cooperative game, where the attacker aims to poison ARP caches, and the defender seeks to prevent it. The

solution uses a mixed-strategy Nash equilibrium to identify the best defense tactics supported by a player utility-based algorithm to detect, block, or redirect malicious users to a honeypot. Simulations demonstrate that this approach is more effective in preventing, detecting, and recovering from ARP spoofing attacks than existing methods in the literature.

C. DHCP STARVATION

DHCP starvation is one of the most severe attacks targeting a Dynamic Host Configuration Protocol (DHCP) in SDN environments. In this type of attack, the attacker sends multiple DHCP requests with either fake or spoofed MAC addresses to exhaust the pool of available IP addresses on the server. This results in legitimate users denying access to the network, leading to a DoS. The threat becomes even more dangerous in the context of SDN, where the central controller plays a pivotal role in managing network traffic [221], [222]. The flexibility of SDN architecture, characterized by its centralized control plane and programmable nature, offers advantages in network management but also introduces new vulnerabilities. As traffic flows through the SDN controller, DHCP starvation attacks can lead to a significant increase in the controller load, higher latency, and reduced network throughput. For instance, in a study involving RYU, OpenDaylight, and Floodlight controllers, researchers found that DHCP starvation attacks cause severe disruptions, including CPU overhead and throughput reduction [223]. Additionally, the lack of native authentication mechanisms in DHCP further exacerbates the impact of the attack, allowing attackers to send unlimited DHCP discovery messages, overwhelming the server, and blocking legitimate clients from obtaining IP addresses [223], [224].

Several methods have been proposed to mitigate these attacks in SDN. One commonly discussed technique is port security, which limits the number of MAC addresses allowed per switch port [225]. This method is effective in preventing certain types of DHCP starvation attacks by immediately disabling the attacker’s port when multiple MAC addresses are detected. However, this approach is less effective against more sophisticated attacks, where the attacker uses their real MAC address in the Ethernet header but a spoofed address in the DHCP request, bypassing port security mechanisms [221], [226]. Another solution involves rate-limiting, where the DHCP request rate is monitored and restricted to prevent the attacker from overwhelming the server [222], [227]. Despite its simplicity, rate-limiting can lead to high false-positive rates and requires careful tuning of thresholds, especially in dynamic environments where legitimate traffic fluctuates [228]. Advanced solutions such as “*DHCPguard*” have also been developed to specifically protect SDN controllers from DHCP-related attacks. This method involves snooping DHCP messages and implementing rate-limiting combined with IP pool-recovery mechanisms. “*DHCPguard*” has been shown to improve

TABLE 5. Detection techniques for resource Exhaustion Attacks (REA) on SDN. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Reference	REA Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[206], 2022	FRE	Control	ML-based prediction using stateful firewall VNF	Python, Mininet	✓	✓	✓							
[207], 2021		Application	AI for dynamic firewall rule updates	Python	✓	✓	✓							
[208], 2023		Application	SDN-based WAF for SQL injection detection	Python	✓	✓	✓							
[209], 2020		Data Plane	Cache side-channel analysis for firewall rule inference	C, KVM, VyOS		✓	✓							
[210], 2022		Control	Anomaly detection in firewall policies	Prolog	✓	✓	✓							
[211], 2023		Data Plane	Honeypot-based firewall for ransomware detection	Python, Docker	✓	✓	✓							
[213], 2020	ARP	Data Plane	Proposes an SDN module to prevent ARP spoofing.	Mininet, POX			✓							
[214], 2022		Data Plane	Cryptographic key-based detection of ARP spoofing	DHCP, Nmap		✓								
[215], 2020		Data Plane	Detecting ARP spoofing using manual techniques	Arpwatch			✓							
[216], 2023		Data Plane	OS-based ARP poisoning detection	Python/Bash			✓							
[217], 2021		Data Plane	K-NN model for detecting ARP poisoning	Python					✓					
[218], 2022		Data Plane	CNN-LSTM model to detect ARP Poisoning	Python, Mininet	✓				✓					
[219], 2024		Data Plane	Combines static ARP with Scapy-based detection	Python, Scapy			✓							
[230], 2024		Control	Game-theory to optimize detection	Mininet, Python									✓	
[220], 2024		Control	Game-theoretic approach to defend ARP	Ovs, Python									✓	
[221], 2018	DHCP	Control	Proposed detection for DHCP starvation	ONOS/Java		✓								
[222], 2024		Data Plane	Analysis of vulnerabilities and countermeasures	RADIUS Server			✓							
[223], 2023		Data,Control	Proposes detection and neutralization of rogue DHCP	Python, Scapy	✓									
[224], 2023		Data,Control	Detects DHCP starvation using port scanning	Python		✓								
[225], 2023		Data,Control	Mitigates DHCPv6 starvation using DUID and MAC	Python, DUID			✓							
[226], 2024		Control	Analyzes and mitigates starvation in SDN controllers	RYU, Floodlight		✓								
[227], 2023		Data,Control	Detects rogue attacks with Python scripts	Python, Scapy	✓									
[228], 2021		Data,Control	Uses machine learning to classify DHCP DoS attacks	Python	✓				✓					
[229], 2021		Data,Control	Security analysis of DHCP in SDN with mitigation	POX		✓	✓							

network resilience by restoring the IP address pool and preventing malicious DHCP messages from affecting the server's performance. For instance, experimental results with “*DHCPguard*” demonstrated a throughput increase of up to 94% and a 92% decrease in CPU usage under attack scenarios on a POX controller [229].

To detect DHCP starvation in real time, researchers have adopted an ML approach. Classifiers, such as random forests, have been trained on custom DHCP datasets to identify anomalous traffic patterns indicative of starvation attacks. This method offers high detection accuracy but requires extensive computational resources and well-curated training datasets, as misconfigurations in the training phase can lead to biased results [228]. Additionally, these ML models have demonstrated significant potential for detecting other network-layer attacks alongside DHCP-related threats [228]. Despite these advancements, it is clear that traditional mitigation techniques such as port security and rate limiting are insufficient to fully secure SDN environments from DHCP starvation attacks. More comprehensive, multilayered security architectures that integrate both server-centric and network-centric protections are required to enhance network robustness. For example, port security mechanisms can be reinforced with DHCP snooping and machine-learning-based

detection systems to create a more adaptive and responsive security framework [222], [223].

D. SUMMARY AND INSIGHTS

Exhaustion DDoS attacks in SDN exploit vulnerabilities in resource allocation, leading to the depletion of critical network resources, which can severely impair SDN functionality. These attacks target various components of SDN, including the control, application, and data planes. The most notable subtypes of resource-exhaustion attacks are FRE, ARP Cache Poisoning, and DHCP Starvation, as summarized in Table 5. FRE involves overloading firewall systems with excessive or redundant rules, exhausting control plane resources, and significantly reducing the ability of the system to manage and secure network traffic. Moreover, ML approaches [206], [208] have been proposed to predict attack patterns and dynamically optimize firewall rule updates, thereby enhancing the system resilience. In addition, challenges such as cache side-channel attacks [209] and firewall misconfigurations [210] highlight the need for more robust and adaptive firewall mechanisms. Furthermore, these solutions must be scalable to manage the increasing complexity of modern networks. ARP Cache Poisoning

targets the data plane by manipulating the ARP tables, which disrupts legitimate communication between hosts. For instance, cryptographic detection methods [214] and AI-based models [217], [218] have been developed to mitigate such attacks effectively. Similarly, game-theoretic approaches [220] provide novel methods for predicting attacker behavior and improving the efficiency of defense mechanisms. Additionally, these approaches help optimize responses by anticipating potential decisions that attackers may make, thereby strengthening the overall security of SDN environments. DHCP Starvation depletes the available IP addresses by flooding DHCP servers with fake requests, leaving legitimate clients unable to connect. Consequently, detection and mitigation techniques such as manual port scanning [224] and ML methods such as Random Forest [228] have been employed to combat these attacks. However, ongoing challenges, including rogue DHCP servers [223], [229], indicate that further refinement of detection strategies is required to address these evolving threats. To summarize, early detection and automated mitigation are recurring themes in the literature on resource-exhaustion attacks. Therefore, by integrating AI-based solutions with other detection methods, SDN systems can become more resilient to these types of attacks. Nonetheless, as SDN scale in size and complexity, it is crucial that defense mechanisms continue to evolve in response to increasingly sophisticated threats.

X. BOTNET-BASED ATTACKS

In this section, we provide an in-depth discussion of the functionalities, limitations, and comparative analysis of detection models from the perspective of botnet-based DDoS attacks on SDN. The analysis focuses on how these detection methods operate across different SDN layers, including the control, data, and application planes, to address key challenges, such as botnet denial of service. This section further examines how existing works utilized either a single detection method or multiple methods in combination, depending on the specific categories of botnet-based DDoS attacks that they aimed to mitigate (cf. Section IV for details) (cf., for Figure 12 and Table 6).

A. IoT BOTNET ATTACKS

IoT botnets have introduced significant security challenges, particularly in SDN environments. These attacks exploit the interconnected nature of IoT devices and programmability of SDN controllers, enabling large-scale DDoS attacks, reconnaissance, and other malicious activities [231]. Botnets formed from compromised IoT devices can severely affect SDN performance, requiring advanced detection techniques to mitigate their effects. Nevertheless, IoT botnets such as Mirai and Echobot take advantage of poorly secured IoT devices using default credentials, hardcoded passwords, and insecure configurations [232], [233], [234]. These botnets connect the infected devices to a command and control (C&C) server to coordinate large-scale attacks.

Mirai, for instance, launched massive DDoS attacks by leveraging infected devices to overwhelm SDN controllers with malicious traffic, thereby disrupting legitimate communications [234], [235]. The programmable nature of SDN controllers makes them a prime target for these attacks because attackers can dynamically inject malicious flow rules into the network [236], [237]. In SDN environments, attackers flood the control plane, leading to congestion and packet drops by exploiting OpenFlow protocol vulnerabilities [236], [237]. This weakens the SDN network's ability to manage resources and disrupts the normal communication between the data and control planes. Machine and deep learning are commonly employed to enhance botnet detection in SDN. Traditional security solutions face limitations owing to the dynamic and real-time nature of IoT-botnet attacks. Researchers have proposed feature selection and adaptive models to improve detection accuracy and efficiency.

- **Feature Selection for Efficiency:** Given the resource constraints in IoT devices and SDN controllers, an optimized feature selection is essential. Alqahtani et al. [238] propose a genetic algorithm-based feature selection method, reducing the number of features while maintaining high detection accuracy. This technique streamlines the detection process, making it suitable for SDN environments in which the computational resources are limited.
- **Adaptive Learning for Evolving Threats:** IoT botnet traffic is subject to concept drift where attack patterns change over time. Shao et al. [239] address this by integrating adaptive online learning and ensemble models. These techniques dynamically update detection models in real time to accommodate evolving threats, thereby ensuring the relevance of detection systems.
- **GAN-based Solutions for Imbalanced Data:** Another key challenge is dealing with imbalanced datasets, because attack data are often underrepresented. Habibi et al. [240] employ CTGAN models to generate synthetic data, enhancing model robustness by simulating underrepresented attack scenarios. These methods ensure better detection accuracy across diverse attack patterns.

Several studies have developed advanced frameworks for IoT botnet detection using both network flow analysis and deep-learning architectures. Sriram et al. [241] propose a DL-based network flow detection framework, which extracts optimal features from traffic flows and categorizes attacks based on traffic patterns. This approach improves the precision of SDN controllers in identifying and mitigating botnet attacks. Similarly, “IoTEnsemble,” which was proposed by Li et al. [242], uses clustering techniques to group network activities and profile legitimate traffic patterns. This ensemble framework significantly reduces false positives by focusing on activity-specific behaviors that are particularly relevant in SDN networks with dynamic traffic. Another promising approach was proposed by Hussain et al. [243]

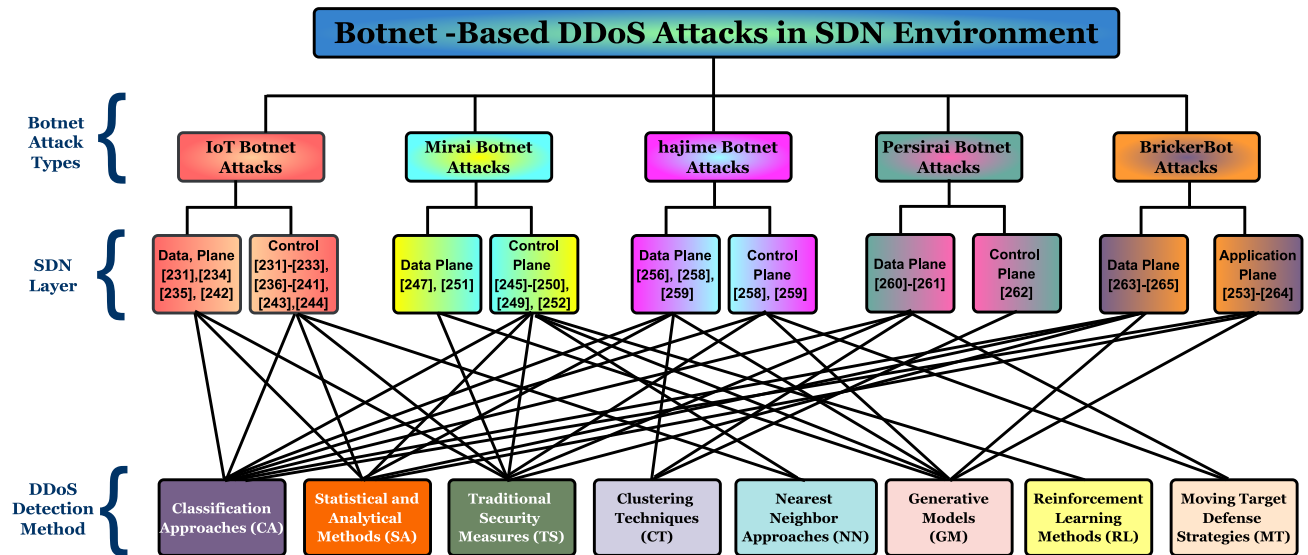


FIGURE 12. Classification of botnet-based DDoS attacks in SDN Layer and detection method. This figure categorizes botnet-based DDoS attacks in SDN, including IoT botnets, Mirai, Hajime, Persirai, and BrickerBot, across different SDN layers: the Data Plane, Control Plane, and application plane. The detection methods applied in these studies were CA, SA, TS, CT, NN, GM, RL, and MT. The references associated with each attack type indicate studies focusing on specific botnets, SDN layers, and corresponding detection methods.

using a two-fold ML framework. This model combines DL for early stage detection (e.g., scanning attacks) with traditional ML algorithms for the comprehensive detection of DDoS attacks, ensuring that the SDN controller remains protected from both the initial probe and full-scale attacks. Although many models demonstrate high detection accuracy on benchmark datasets such as Bot-IoT and N-BaIoT, practical deployments face additional challenges. The primary issues include the following.

- **Heterogeneous Traffic Patterns:** IoT devices generate diverse traffic, complicating the modeling of normal behavior. The proposed framework addresses this problem by employing multiclass classifiers and embedding techniques, as suggested by Ge et al. [244].
- **Real-Time Detection Requirements:** Because SDN networks operate in real time, detection frameworks must process and analyze traffic without delay. Adaptive online models, such as those proposed by Shao et al. [239], ensure that detection models update dynamically without interrupting the network operations.
- **Encrypted Traffic:** In modern IoT systems, some data streams are encrypted, complicating the traffic analysis. Researchers have addressed this issue by focusing on flow-level analysis rather than on deep packet inspection, as demonstrated by Sriram et al. [241].

B. MIRAI BOTNET

With the increasing adoption of IoT devices in SDN environments, Mirai botnets have become a serious security concern. Originally designed to exploit weak IoT device

credentials, Mirai evolved to launch DDoS attacks that threaten SDN infrastructure. Thus, it is crucial not only to develop detection models for Mirai attacks, but also to implement proactive defenses to prevent infections and protect SDN-based systems. Several studies have addressed these issues by proposing advanced detection frameworks, forensic methods and preventive strategies. Affinito et al. [245] analysed the evolution of Mirai's scanning behavior over six years, revealing how the botnet continues to adapt by targeting new ports, making traditional defenses ineffective against its evolving attack patterns. Zhang et al. [246] took a forensic approach by analyzing Mirai's command-and-control infrastructure, identifying key artifacts left on servers. This study provides practical insights for remotely acquiring evidence, which is critical for securing distributed SDN networks. In terms of detection, Nakip and Gelenbe [247] introduced an Auto-Associative Dense random neural network, demonstrating superior detection accuracy with minimal false alarms, which is ideal for SDN environments where false positives can disrupt operations. Cruz et al. [248] compared multiple ML techniques, including KNN, SVM, and Logistic Regression, achieving 99% accuracy in detecting Mirai attacks. These results highlight the effectiveness of tailored ML solutions for different types of attacks, such as SYN and UDP flooding, which directly affect the SDN controllers. Vajrobol et al. [249] applied adversarial learning by integrating LSTM and XGBoost models. Their approach achieved 97.7% detection accuracy, enhancing the robustness of the detection mechanism against adversarial threats, which is a vital aspect of the dynamic nature of SDN environments. Similarly, Azhari et al. [250] achieved 92.91% detection accuracy using an SVM model, further validating the utility

of lightweight ML techniques for resource-constrained SDN deployments. On the preventive side, Kelly et al. [251] assessed the vulnerability of IoT devices to Mirai attacks and recommended specific configurations to reduce infection risks. These hardening techniques are particularly relevant for securing the interconnected nature of the SDN. Borys et al. [252] emphasized the importance of adaptive security frameworks to address emerging Mirai variants that engage in both DDoS and crypto-jacking attacks. Finally, Snehi and Bhandari [253] proposed a conceptual defense model for IoT-DDoS mitigation using neural networks, suggesting that similar models can be integrated within SDN control layers. They emphasized the importance of continuously adapting security mechanisms to counter evolving threats, such as the Mirai. Overall, existing detection and mitigation frameworks offer promising solutions, and the continuous emergence of new Mirai variants underscores the need for real-time anomaly detection and adaptive machine-learning models. Future work should focus on seamlessly integrating these approaches into SDN infrastructure to ensure robust, long-term security.

C. HAJIME BOTNET

Hajime Botnet is a notable example of a peer-to-peer (P2P) IoT botnet that emerged as a competitor to the well-known Mirai botnet. It is characterized by its unique operational structure and capabilities, which differentiate it from traditional botnets. Unlike centralized botnets, which rely on a single command and control server, Hajime operates on a decentralized P2P model. This means that each infected device (or “bot”) can communicate directly with other bots, making it more difficult to disrupt the network because there is no single point of failure. Although Hajime itself has not been observed to execute attacks, it has the potential to be harnessed for DoS attacks. Its ability to control large numbers of IoT devices could theoretically be leveraged to flood networks with traffic, overwhelming services, and disruptions. For instance, if an attacker gains control over Hajime-infected devices within an SDN setup, they can manipulate data flows or hijack control over network resources by injecting malicious flow rules into the SDN controller. Hajime’s sophistication lies in its decentralized architecture, advanced propagation techniques, ambiguous purpose, and adaptability compared with Mirai’s straightforward approach focused primarily on DDoS attacks through a centralized model [254]. The Hajime botnet was first identified in October 2016 by researchers in Rapidity Networks [255]. Unlike other botnets such as Mirai, Hajime initially stood out for its unusual behavior, refraining from launching attacks immediately. Since its discovery, Hajime has undergone continuous evolution, incorporated new features, and refined its strategies. However, the relatively short period since its emergence has limited the availability of extensive historical data for comprehensive analysis.

Anh and Nakamura [256] provide foundational insights into the dynamics of Hajime botnets in IoT ecosystems. Their study highlights how Hajime establishes persistence in infected devices while preventing reinfections from other botnets, such as Mirai, adding complexity to its impact. Moreover, the research underscores Hajime’s potential influence on SDN networks, as its ability to secure compromised devices reduces the risk of malicious intrusions, particularly those leading to DDoS attacks. Building on this, Makiyara and Yamaguchi [257] introduced Hajime as a white-hat botnet, positioning it as an active measure within a “*botnet defence system (BDS)*”. They suggested that Hajime’s patrol-like operations could be adapted to SDN environments, suppressing botnet threats and reducing vulnerabilities. However, their work also noted the risks of unauthorized control over devices, emphasizing the need to balance security benefits with potential drawbacks. This study provides a conceptual framework for using Hajime-inspired mechanisms to strengthen the SDN security. Furthermore, Kobayashi [258] work extends these ideas through Markov chain-based modelling to simulate the propagation and containment of botnets, such as Hajime, in SDN environments. The model demonstrates how deploying Hajime-like bots can effectively reduce infection rates and limit the spread of malicious agents, such as Mirai. In addition, this mathematical framework offers practical insights into the implementation of patrol systems within SDN networks, showing how theoretical models can inform real-world defence strategies. Empirical research by Pour et al. [259] complemented these systems with real-world evidence derived from a Darknet data analysis. Their study utilized deception techniques to collect malware artifacts, thereby shedding light on Hajime’s evolving strategies. Through their “*HoneyComb system*,” the team captures botnet activities at scale, validating the effectiveness of white-hat botnets such as Hajime. Moreover, these findings reinforce the potential of the models proposed by Makiyara and Yamaguchi [257] and Kobayashi [258] and demonstrate the practical value of deploying such strategies in SDN environments.

D. PERSIRAI BOTNET

The Persirai botnet is a sophisticated malware family that primarily targets IoT devices, particularly Internet Protocol (IP) cameras. It exploits vulnerabilities in over 1,000 different camera models, positioning itself as a major threat to the IoT landscape. As IoT devices become integral to SDN networks, the attack surface for threats, such as Persirai, expands significantly. These environments inherit the security weaknesses of connected IoT devices, making them increasingly vulnerable to botnet attacks that exploit both default credentials and software vulnerability. Recent studies have shown that Persirai has even overtaken Mirai as the most common malware affecting IP cameras, highlighting the urgency to develop robust security frameworks. A key study by Kim et al. [260] examined the behavior of Persirai

and other Mirai variants using a “*graphical security model (GSM)*”. Their model captured propagation phase scanning, exploitation, and infection, demonstrating how Persirai leverages vulnerabilities, such as “*CVE-2014-8361*”, to transform IoT devices into propagating nodes for large-scale DDoS attacks. Although this study did not focus extensively on mitigation strategies, it emphasized the importance of security evaluations to understand and manage the spread of these botnet attacks effectively in SDN environments. A complementary approach by Chandavarkar [261] focused on hardcoded credentials, another key vector exploited by Persirai. This study highlights how the botnet targets IP cameras by deploying a malicious shell script that runs in memory to create a botnet while blocking other attackers from reinfected compromised devices. Although a system reboot can remove malware from memory, changing default passwords is critical to prevent reinfection, as insecure configurations remain a persistent vulnerability in IoT and SDN deployment. Another perspective on mitigating Persirai attacks is provided by Susilo and Sari [262], who explored the use of DL models to detect botnet-driven DDoS attacks in SDN. Their work employed algorithms, such as CNNs and RF, to identify abnormal traffic patterns, demonstrating promising results in intrusion detection. They suggested that ML-based solutions can significantly improve detection accuracy, although further enhancements in training and testing accuracy are needed to ensure that these models perform effectively in real-world SDN scenarios.

E. BRICKERBOT

SDN allows for rapid reconfiguration of network resources and policies in response to threats. This adaptability is vital when facing evolving malware, such as BrickerBot, which may change its tactics based on the network environment [263]. One of the studies focusing on the detection and mitigation of evolving botnet-related attacks in IoT networks, including BrickerBot, was presented by Baz in [264]. The authors highlighted the importance of adaptive intrusion detection systems in coping with botnets that modify attack patterns over time. BrickerBot stands out by conducting PDos attacks, in which it corrupts firmware and disables storage, rendering devices permanently inoperable. This destructive strategy differentiates BrickerBot from other botnets, which typically rely on network disruption through traffic flooding. To address such threats, the authors propose a “*Self-Evolving Host-Based Intrusion Detection System (SEHIDS)*”. This framework evolves dynamically based on real-time traffic analysis using lightweight ANNs to monitor devices and adapt their architecture as needed. This adaptability enables “*SEHIDS*” to detect anomalies such as BrickerBot’s behavior early, thereby preventing severe damage. Unlike static detection systems, “*SEHIDS*” uses anomaly-based detection to identify new or evolving attacks beyond predefined patterns.

A significant contribution to the study of destructive IoT botnets was presented by the authors in [265], focusing on BrickerBot. Unlike Mirai or Hajime, which exploit devices for DDoS attacks, BrickerBot’s goal is to permanently disable IoT devices by corrupting their firmware. It uses stateless random scanning and balanced dictionary attacks to target vulnerable devices running embedded Linux systems using BusyBox. A key insight from this study is BrickerBot’s non-replicative nature. Rather than expanding into a large botnet, BrickerBot aims to disable devices, creating detection challenges. Its attacks bypass typical signs of botnet activity, such as persistent infections or C&C server communications, thus complicating detection. The authors highlighted how restricted-range scanning reflects BrickerBot’s focus on disrupting devices within specific local networks, allowing rapid, localized attacks. Although this study offers valuable insights into BrickerBot’s strategies, it emphasizes random-scanning and dictionary-based attacks. Future studies could explore AI-based methods for early detection and mitigation. SDN frameworks also present opportunities to dynamically detect and block destructive botnets using automated real-time responses.

F. SUMMARY AND INSIGHTS

The research summarized in Table 6 covers 33 studies focusing on detecting and mitigating botnet-based DDoS attacks within SDN environments. Among these, 20 studies employed CA approaches [231], [232], [233], [244], [250], [253], [259], [260], and [262], [263], [264], [265]. SA methods were used in thirteen studies to detect traffic anomalies and patterns for identifying threats [231], [232], [234], [237], [239], [240], [244], [245], [252], [256], [263], [264], [265]. In addition to CA and SA, other approaches include TS-based solutions, used in seven studies [235], [236], [246], [251], [257], [261], and [265], and CT techniques, which appeared in three studies [233], [243], [259]. NN methods have been employed in two studies [248] and [262]. Four studies used GM models to address data imbalance using synthetic data [247], [258], [263], and [264], whereas only one study applied RL frameworks to develop an adaptive mitigation [249]. Finally, four studies explored MT strategies to add unpredictability to network configurations [257], and [260].

The studies in Table 6 reveal the trends and challenges in the detection of botnet-based DDoS attacks in SDN. Most efforts have focused on CA-based approaches to detect and mitigate attacks, especially in the control plane, preventing botnet propagation within SDN [231], [234], [237], [239]. Data-plane approaches play a complementary role in managing real-time traffic variations and improving anomaly detection [241], [242]. Although CA and SA dominate the field, clustering techniques and moving target defense present promising opportunities for improving detection adaptability against evolving threats [243], [257]. Additionally, multi-agent patrol systems and graph-based models

TABLE 6. Summary of botnet-based DDoS attack detection techniques in SDN. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Reference	Botnet Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[231], 2023	IoT	Data,Control	Proposed genetic algorithm to detect IoT botnet.	Python	✓	✓								
[232], 2022		Control	Collaborative method to stop botnet attacks	Python	✓	✓								
[233], 2022		Control	Ensemble learning for botnet detection	Python	✓			✓						
[234], 2022		Data,Control	Detects IoT botnets early using collaborative learning	Python	✓	✓								
[235], 2022		Data	Feature selection using mutual information	Python	✓		✓							
[236], 2021		Control	Efficient feature engineering for detection	Python	✓		✓							
[237], 2020		Control	Compared multiple ML models for botnet detection	Python	✓	✓								
[238], 2020		Control	Optimizes XGBoost for botnet detection	Python	✓									
[239], 2021		Control	Proposed a sequential detection architecture	Python	✓	✓								
[240], 2023		Control	Uses GANs for data augmentation	Python	✓	✓								
[241], 2020		Control	Proposes network flow detection for botnet	Python	✓	✓								
[242], 2022		Data plane	Ensemble clustering to detect botnet	Python	✓									
[243], 2021		Control	Introduced online learning for botnet threats	Python	✓			✓						
[244], 2021		Control	Develops a universal feature set for detection	Python	✓	✓								
[245], 2023	Mirai	Control	Analyzed evolution of Mirai’s scanning behavior	Python		✓								
[246], 2020		Control	IoT-DDoS defense using neural networks	Python			✓							
[247], 2021		Data,Control	Dense Random Neural Networks for botnet detection	Python						✓				
[248], 2021		Control	Evaluates ML models for Mirai detection	Python	✓				✓					
[249], 2024		Control	LSTM and XGBoost for detection	Python						✓				
[250], 2022		Control	SVM fro lightweight mirai detection	Python	✓									
[251], 2020		Data plane	Suggested hardening strategies for IoT devices	Python			✓							
[252], 2020		Control	Evaluates DDoS attack of IoT	Python		✓								
[249], 2024		Control	Proposed neural network to defend IoT systems	Python	✓									
[256], 2024	Hajime	Data plane	Tracks infection patterns	Python		✓								
[257], 2021		Control	Patrol function using white-hat botnet for BDS	Petri nets		✓						✓		
[258], 2023		Data,Control	Markov chain-based modeling of Hajime’s spread	Python						✓				
[259], 2020		Data,Control	Darknet-based data analysis	Python	✓			✓						
[260], 2020	Persirai	Data	GSM model to capture Persirai attack	Mininet	✓							✓		
[261], 2020		Data	Analysis of hardcoded credentials in IP cameras	Not mentioned			✓							
[262], 2020		Data,Control	DL learning models for intrusion detection in SDN	Python	✓				✓					
[263], 2022	Bricker -Bot	Data,App.	Comprehensive IoT malware detection methods	iOS	✓	✓				✓				
[264], 2022		Data,App.	SEHIDS: Adaptive intrusion detection for BrickerBot	Not mentioned	✓	✓				✓				
[265], 2021		Data	Testing threat prevention in IoT networks	QEMU	✓	✓	✓							

are gaining traction, especially in tackling Hajime botnets, reflecting a trend toward more intelligent and dynamic defenses [258], [259]. GAN-based data augmentation offers practical solutions for handling imbalanced datasets, whereas RL introduces real-time adaptability to counter changing attack patterns [240], [249]. However, challenges remain, such as ensuring latency and availability when deploying MT strategies and optimizing the scalability of clustering techniques in large networks.

The proposed approaches are technically feasible but require careful execution. Clustering techniques, although promising, require optimization to reduce false positives and efficiently manage high-dimensional traffic. Reinforcement learning can enhance adaptability but requires extensive training and continuous feedback to operate effectively in dynamic

environments [249]. MT approaches are viable because of SDN's programmability; however, precise management is required to prevent performance degradation [257]. Likewise, GANs can improve detection by generating realistic synthetic traffic, but they require constant monitoring to avoid introducing bias. Future research should focus on cross-layer solutions that integrate both control and data planes for a more comprehensive defense framework. Hybrid models that combine CA, CT, RL, and MT strategies provide more robust and adaptive solutions for SDN security. Additionally, multi-agent patrol systems are essential for maintaining dynamic and continuous monitoring across the network. These strategies ensure that SDN environments remain secure against both current and future botnet-based DDoS attacks.

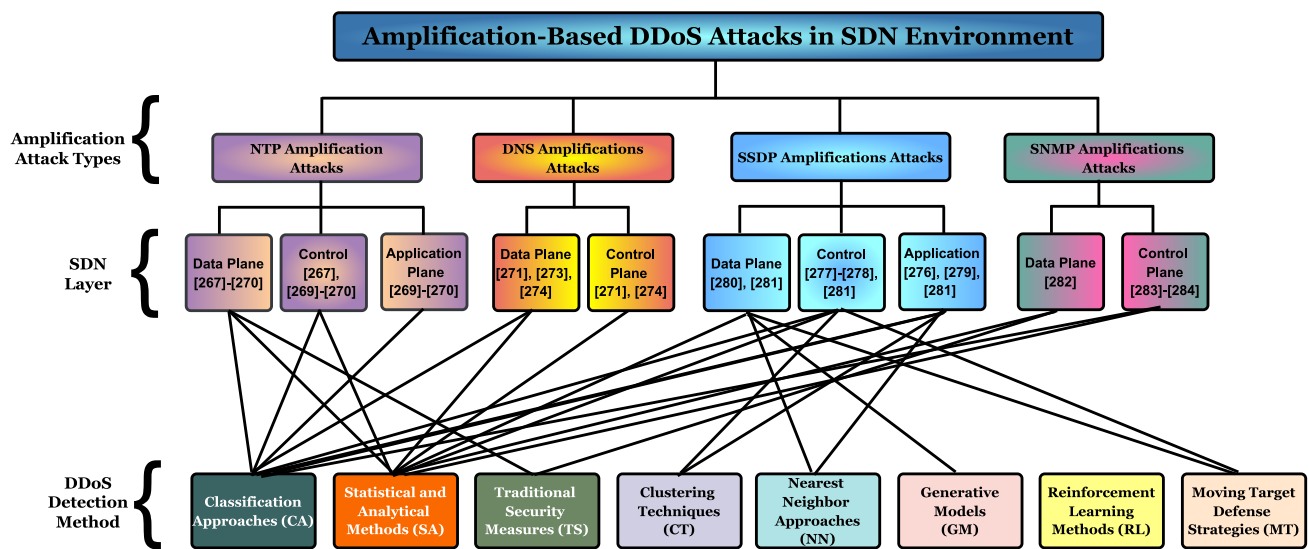


FIGURE 13. Framework of amplification-based DDoS attacks in SDN environments, categorized by attack types (NTP, DNS, SSDP, and SNMP) and their corresponding SDN layers (data, control, and application planes). The figure illustrates the multi-layered nature of these attacks and the need for targeted defense mechanisms across different SDN planes.

XI. AMPLIFICATION ATTACKS

In this section, we provide a detailed review of existing studies on amplification-based denial-of-service attacks within SDN environments. Figure 13 presents a structured overview of these attacks, categorized by attack types (NTP, DNS, SSDP, and SNMP) and their corresponding SDN layers, including the data, control, and application planes. The figure also highlights the variety of detection methods employed across these studies, such as the CA, SA, CT, RL, and MT strategies (cf. Section IV provides details). This visual framework offers valuable insight into how different attack vectors interact with SDN layers and emphasizes the diversity of the defense mechanisms implemented. For a comprehensive analysis of the referenced studies, along with the specific mitigation strategies and methodologies applied, we encourage readers to refer to Table 7.

A. NTP AMPLIFICATION

NTP amplification is another type of denial-of-service attack that exploits a Network Time Protocol (NTP) to flood a target with overwhelming traffic. Attackers send small requests to publicly accessible NTP servers using spoofed IP addresses that correspond to the IP of the target. The most commonly exploited command is “*monlist*,” which returns a list of the last 600 clients that queried the server [266]. As the response can be up to 206 times larger than the request, this results in significant amplification. The detection of NTP-amplification attacks involves several strategies. Honeypots mimic vulnerable services to log malicious traffic patterns, whereas network telescopes monitor unused IP spaces to capture backscatter and scans indicative of spoofed attacks. Flow-based detection methods focus on traffic

thresholds such as packet volume and idle timeouts to identify suspicious flows. In addition, hop count and timestamp verification ensure that incoming requests are authentic and not part of a spoofing attempt.

First, Ismail et al. [267] proposed a system that utilizes ML algorithms such as RF and XGBoost to detect and classify general DDoS attacks, including NTP amplification attacks. Their approach leverages the UNWS-np-15 dataset, which captures the traffic patterns associated with NTP-based DDoS attacks. The system applies classification models to distinguish between attack and normal traffic based on key features such as unusual response sizes and high request frequencies. In their experiments, the Random Forest and XGBoost algorithms achieved impressive accuracy rates of approximately 89% and 90%, respectively. The results showed significant improvements over previous methods, with accuracy rates higher than those of earlier research by 4-11%. This study demonstrates how machine learning can effectively enhance the detection and mitigation of NTP amplification attacks, thereby providing a robust approach for real-time attack classification and prediction. Next, the authors in [268] improved the time synchronization accuracy of Network Time Protocol (NTP) systems, particularly the effects of software-related latencies and cryptographic protection (via Network Time Security or NTS) on timing precision. These findings highlight that software-induced delays in NTP messages contribute significantly to synchronization inaccuracies, especially in environments with cryptographic protection. The study decomposes these latencies into components and proposes methods to compensate for them through timestamp adjustments. For uncorrectable latencies, mitigation strategies are suggested to enhance

synchronization accuracy across various NTP implementations and hardware setups.

Furthermore, Kopp et al. [269] analyzed the persistence of NTP amplification attacks at major Internet Exchange Points (IXPs). Their research highlights the ongoing prevalence of such attacks, despite long-standing mitigation strategies. Using flow-based classification, traffic volumes are monitored from multiple sources, particularly through port 123 (the NTP port). Their findings revealed that even with filtering mechanisms and scrubbing services, a significant amount of NTP amplification traffic still manages to penetrate networks, causing infrastructure damage. The study underscores the need for continuous improvements in detection strategies and enhanced protocol security. Finally, Dobrin and Dimiter [270] proposed an SDN-based framework for detecting and mitigating NTP amplification attacks. Their solution integrates “*Virtual Security Functions (VSFs)*” with OpenFlow to monitor and analyze network traffic in real time. The system identifies suspicious flows linked to NTP attacks by observing OpenFlow tables and by using anomaly detection techniques. The programmability of SDN allows for dynamic responses, which automatically block malicious traffic as soon as it is detected. This study demonstrates how SDN’s centralized control and flexibility provide a robust framework to counter evolving amplification threats and minimize manual intervention.

B. DNS AMPLIFICATION

In an SDN environment, where network resources are dynamically allocated and controlled through software, Domain Name System (DNS) amplification attacks pose unique challenges. In a DNS amplification attack, an attacker sends small DNS queries to open DNS resolvers using a spoofed IP address belonging to the victim. The DNS resolver then responds to the spoofed address, sending back a significantly larger response. This results in a significant amplification of the traffic directed at the victim’s network, often leading to service disruption or complete unavailability. To counter these attacks, researchers have developed frameworks using techniques such as traffic filtering, DNS request-response tracking, and programmable network infrastructure. The proposed key frameworks are as follows.

- **DNSGuard** [271]: A defense framework for DNS attacks using programmable network switches. It aims to overcome limitations in processing power and storage of switches, allowing real-time detection and mitigation of both volumetric and non-volumetric DNS attacks. Key findings include the effectiveness of “DNSGuard’s” recursive parsing and multi-phase analysis model, enabling accurate attack detection with minimal latency. Experimental results show that “DNSGuard” maintains line-rate defense capabilities, adding only nanoseconds of delay under heavy network loads.
- **ICMP-Based Outgoing Traffic Analysis** [272]: detects attacks by analyzing the ICMP packets generated when

closed UDP ports are accessed during amplification. Using an SDN module, this method identifies malicious patterns in outgoing traffic and blocks attack sources without disrupting legitimate users.

- **DIDA** [273]: leverages programmable switches at ISP network edges to monitor DNS queries and responses in real time. It autonomously detects and mitigates attack traffic without relying on external scrubbing services, thereby minimizing the latency and operational overhead.
- **ANYway** [274]: focuses on limiting responses to ‘ANY’ DNS queries, which are commonly exploited for amplification. This method can reduce the attack volume by up to 69%, but the study noted that some highly potent domains might still bypass the mitigation, requiring additional defenses.
- **SNORT-Based DNS Intrusion Detection System (IDS)** [275]: integrates custom attack signatures into SNORT to detect and block amplification and other DNS-based attacks. The system generates alerts and enforces firewall rules in real time, ensuring effective mitigation while maintaining the integrity of DNS services.

C. SSDP AMPLIFICATION

Simple Service Discovery Protocol (SSDP) amplification is a specific type of DDoS attack that exploits SSDP, which is part of the Universal Plug and Play (UPnP) framework. This protocol allows devices in a local network to discover each other and communicate seamlessly. However, its design flaws render it susceptible to exploitation by attackers. Amplified traffic can congest the SDN controller and its associated data paths, leading to degraded performance for legitimate users. To defend against SSDP amplification attacks, researchers have implemented several strategies, including access control, network segmentation, firmware updates, and traffic monitoring. For example, Sassani et al. [276] utilized “*Access Control Lists (ACLs)*”, “*Reflexive ACLs (RACLs)*”, and firewalls to block malicious traffic and safeguard network performance. Similarly, Gurjar and Somani [277] applied network segmentation through container-level separation, hardware isolation, and packet limiting to ensure uninterrupted service availability during attacks. Likewise, Liu et al. [278] emphasized proactive defense with their “*Broad Learning-based Comprehensive Defense (BLCD)*”, which integrates multi-location defenses, automated firmware updates, and attack mitigation along the entire attack chain. In addition, Nawrocki et al. [279] focused on traffic monitoring and deployed honeypots to detect SSDP amplification attempts while refining the detection thresholds for better accuracy. Moreover, Tang et al. [280] employed programmable data planes through the “*Fake Amplification Phenomenon Monitor (FAPM)*”, which optimizes real-time traffic filtering with P4 switches to counter amplification traffic. Finally, Anagnostopoulos et al. [281] conducted an empirical study on SSDP amplifiers, highlighting the

importance of constant traffic monitoring, segmentation, and mitigation efforts to effectively address evolving amplification threats.

Further studies should explore advanced techniques such as adaptive ML models and dynamic mitigation frameworks tailored for evolving SSDP amplification attacks. In [276], the authors suggested refining access control mechanisms to handle new attack patterns, whereas in [278], the need for deeper research on hardware-based isolation to enhance scalability was emphasized. The authors of [278] highlighted the importance of improving firmware update automation to preemptively address the vulnerabilities in IoT devices. Additionally, Tang et al. [280] recommended expanding the use of programmable data planes to support multi-protocol attack detection, and [279] proposed optimizing honeypot deployments to enhance visibility and attack detection. Future research should conduct long-term monitoring and geographical analyses to better understand amplification trends across diverse regions.

D. SNMP AMPLIFICATION

Simple Network Management Protocol (SNMP) amplification attacks are a type of denial-of-service attack that exploit the SNMP protocol, which is commonly used for managing and monitoring network devices. In these attacks, an attacker sends a small SNMP request to multiple vulnerable devices, which then responds with much larger replies. This amplification effect allows attackers to flood the target system with a significantly higher volume of traffic than they initially sent, overwhelming the target's resources. SNMP-based DDoS attacks in SDN environments exploit the protocol's ability to generate amplified responses to small requests, particularly through features like "*GetBulkRequest*," which retrieves large sets of management data, overwhelming the target with spoofed traffic. These attacks are challenging to detect because of the difficulty in distinguishing malicious traffic from legitimate network-management operations. Several mitigation strategies have been proposed, including moving target defense to make networks less predictable, ACL to block known attack patterns, and "*Kullback-Leibler Distance (KLD)*" analysis to detect anomalies [282], [283], [284]. Situational awareness tools monitor indicators of compromise (IOCs) to provide early warnings and suggest countermeasures. These approaches are critical in SDN, where SNMP-based attacks can disrupt the control plane, necessitating real-time monitoring and control mechanisms to prevent service disruptions and maintain network stability. Addressing this attack is essential to maintain the reliability of SDN networks, as even brief disruptions can impact the critical infrastructure by relying on these networks. Future research should focus on refining detection algorithms to reduce false positives and developing lightweight solutions that can be integrated into resource-constrained environments. Further studies are needed to explore the impact of these mitigation strategies on network performance and scalability under high-traffic conditions.

E. SUMMARY AND INSIGHTS

The literature on amplification attacks, summarized in Table 7, focuses on the NTP, DNS, SSDP, and SNMP protocols, with most studies leveraging SDN technologies for mitigation. Research on NTP amplification attacks highlights the use of programmable switches, such as P4, virtual security functions, and honeypots to monitor traffic and identify amplification factors [267], [268], [270]. DNS-based attacks are mitigated through query control techniques, such as restricting ANY queries [274], and further detection enhancements are achieved using SNORT IDS and ICMP analysis [272], [275]. SSDP attacks, often linked to IoT devices, employ container-level separation, honeypots, and programmable switches to reduce their impact [277], [278], [280]. SNMP amplification studies explore advanced mitigation techniques, including MT strategies and RL approaches [282], [283].

Current research emphasizes the importance of proper server configuration and effective mitigation strategies, such as ingress filtering and rate limiting, and addresses the challenges posed by the volumetric nature of these attacks. A clear trend emerges toward integrating SDN control and data planes to enhance the flexibility and effectiveness of defense mechanisms [267], [270]. Simulated environments, such as Mininet, Docker, and honeypots, are essential tools for testing these solutions [270], [277], [278], though more empirical studies on real-world networks are needed to validate their practical effectiveness [281].

Through a database search of *IEEE Xplore*, *Google Scholar*, *ScienceDirect*, and similar platforms, we found no studies specifically focusing on CharGen amplification attacks. Originally designed for testing and debugging network connections, CharGen has been exploited as a vector for DoS attacks, owing to its ability to generate large volumes of traffic. Although CharGen was briefly mentioned in one study [269], it was not the primary focus, revealing a gap in the literature. Future research should explore CharGen's potential to cause disruptions and improve detection techniques through machine learning, establish standardized service configuration guidelines, and analyze the impact of emerging technologies.

Additionally, it is crucial to understand attacker behavior and foster collaboration between ISPs and organizations to strengthen collective defense mechanisms. With the growing prevalence of IoT devices, further research is needed to develop scalable mitigation strategies for SSDP-based attacks [280]. Addressing these underexplored areas and the challenges of real-world implementation will be essential for building comprehensive and robust DDoS defense mechanisms.

XII. REFLECTION ATTACKS

A. NTP REFLECTION

NTP reflection attacks are a type of denial-of-service attack that exploit vulnerabilities in the NTP protocol. These

TABLE 7. Summary of amplification-based DDoS attack detection techniques in SDN. Acronyms: CA: Classification Approaches, SA: Statistical and Analytical methods, TS: Traditional Security measures, CT: Clustering Techniques, NN: Nearest Neighbor approaches, GM: Generative Models, RL: Reinforcement Learning methods, MT: Moving Target defense strategies.

Reference	Amplif. Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C A	S A	T S	C T	N N	G M	R L	M T		
[267], 2022	NTP	Data,Control	ML improves NTP amplification attacks	UNWS-np-15	✓	✓								
[268], 2020		Data plane	Compensating latencies and timestamp adjustments.	Wireshark	✓	✓	✓							
[269], 2021		All planes	Flow-based traffic filtering amplification attacks	IXP flows	✓									
[270], 2021		All planes	VSF with OpenFlow to detect and mitigate attacks	Mininet,POX	✓									
[271], 2024	DNS	Data plane	“DNSGuard” defends DNS attacks efficiently.	Progr-Switch		✓								
[272], 2023		Control	ICMP traffic analysis via SDN module to block attack	Linux tools		✓								
[273], 2020		Data plane	Programmable switches blocking attack traffic	P4-switches	✓									
[274], 2021		Data,Control	Restricting responses to ‘ANY’ DNS queries.	Large-scale DNS		✓								
[275], 2023		Application	SNORT IDS with DNS signatures to detect attacks	Simulated	✓									
[276], 2022	SSDP	Application	ACLs, RACLs, and Firewalls	Physical setup			✓							
[277], 2023		Control	Container-level separation, packet limiting	Mininet, Docker		✓							✓	
[278], 2023		Control	Honeypots for traffic monitoring	Honeypot	✓	✓								
[279], 2023		Application	BLCD with multi-location defenses	Simulated IoT	✓				✓					
[280], 2024		Data plane	FAPM with P4 programmable switches	BMv2	✓					✓			✓	
[281], 2022		All planes	Empirical analysis of SSDP amplifiers	IP scans		✓		✓						
[282], 2020	SNMP	Data plane	Mitigation through protocol specific analysis	Wireshark		✓	✓							
[283], 2023		Control	SNMP monitoring with MTD approach	AmizonWeb	✓	✓	✓						✓	
[284], 2022		Control	Situational Awareness Tool for real-time detection	SimulatedDataset	✓	✓								

attacks are particularly dangerous because they can generate massive amounts of traffic directed at a target, overwhelming its resources and rendering it inaccessible. In SDN environments, programmability and centralized control of the network provide new opportunities to effectively mitigate these attacks. Existing studies have focused on leveraging SDN features such as programmable switches, OpenFlow rules, and in-network detection systems to filter malicious NTP traffic and ensure network stability. Key approaches include automating flow updates, minimizing the load on the control plane, and enhancing resource management during attacks. The following studies provide insights into various strategies for addressing NTP reflection attacks within SDN. Pashamokhtari et al. [285] explore NTP reflection attacks and their impact on network security. They proposed a mitigation framework that detects and blocks malicious NTP traffic by using real-time anomaly detection. Their approach focused on filtering spoofed traffic and reducing amplification effects, achieving high accuracy in attack identification. The method significantly enhances defense against reflection-based DDoS attacks, ensuring minimal disruption to network performance. The results demonstrate a reduction in attack volume with the proposed solution efficiently mitigating NTP-based threats.

Lin [286] investigates NTP-based reflection attacks in SDN environments, focusing on applying OpenFlow-based flow tables to dynamically filter malicious traffic at network boundaries. The study demonstrates how SDN controllers automate mitigation efforts, improving operational efficiency by minimizing manual intervention. The findings confirm

that SDN-based filtering strategies significantly reduce the impact of NTP reflection attacks, while maintaining the availability of legitimate network services. Wang and Wu [287] proposed the “*EC-NTD*” scheme, which is an advanced SDN-based solution for defending against DrDoS attacks, including NTP reflection attacks. The scheme combines Network Address Port Translation (NAPT) with a two-stage detection mechanism to differentiate between legitimate and attack traffic. Their approach ensures high detection accuracy with minimal controller overhead, maintaining service availability, while effectively mitigating malicious NTP reflection traffic within SDN networks.

B. DNS REFLECTION

A flood of amplified DNS traffic can overwhelm SDN controllers, leading to degraded performance and usage [288]. This can prevent legitimate users from accessing critical services and disrupting their normal network operations. Several studies have applied various approaches including traffic filtering, container separation, and enforcing secure DNS protocols. For instance, the authors of [277] presented three mitigation strategies to isolate attack traffic and improve service availability: container-level separation, packet limiting, and network hardware separation. The experiments showed that segmenting resources reduced the performance impact of DNS amplification and reflection attacks. These techniques can be dynamically applied to SDN to redirect malicious flows and maintain service continuity. From the perspective of the vulnerabilities of open DNS reflections conducted by Liu [289]. The authors recommend “*source*

address validation (SAV)” and *“response rate limiting (RRL)”* to prevent abuse in reflection attacks. Their findings revealed that many open resolvers remain vulnerable, stressing the need for improved safeguards. SDN can adopt these measures centrally to block spoofed traffic and dynamically control the response rates across the network. Furthermore, Sanjay [290] simulated DNS amplification and tunneling attacks to understand their impact and suggested mitigation techniques such as monitoring traffic patterns and limiting public DNS queries. This emphasizes blocking recursive DNS requests during attacks to curb amplification. In SDN, these detection frameworks can be enhanced by real-time monitoring and policy updates via the controller, thereby improving response capabilities. In other hand, Davis and Deccio [291] proposed DNS Protocol Advertisement Records (DPAR) to specify secure transport protocols (e.g., DNS-over-TCP) and prevent spoofed queries, reducing the risk of reflection attacks. They argued that DNS servers should enforce these protocols to ensure query authenticity. SDN can utilize DPAR to filter noncompliant traffic at entry points, ensuring efficient attack mitigation across the network.

C. SSDP REFLECTION

While both SSDP reflection and amplification attacks exploit vulnerabilities in SSDP-enabled devices, reflection attacks primarily reflect traffic from multiple sources to obscure an attacker’s identity. One such study on SSDP reflection attacks using the CICDDoS2019 dataset was conducted by the authors of [292]. SSDP is categorized as reflection-based DDoS attacks that exploit UDP protocols to generate amplified traffic for victims. This study focuses on developing ML models to detect various DDoS attacks, including SSDP reflections, in SDN environments. By leveraging the centralized control plane of SDN, the authors emphasize the importance of rapid feature selection and dimensionality reduction techniques to efficiently classify SSDP attacks. The research highlights the ability of SDN to facilitate real-time traffic monitoring and anomaly detection, enabling proactive mitigation of reflection attacks such as SSDP within large-scale networks. Another study evaluated SSDP reflection attacks within an SDN context by modeling their impact using attack trees [293]. The authors explored how SSDP attacks, when combined with IoT devices, could amplify traffic and overwhelm SDN-controlled networks. SSDP attacks are identified as a high-propensity threat owing to the protocol’s reliance on UDP, which allows attackers to generate large volumes of traffic with spoofed IPs. The hierarchical modeling of attack scenarios in this study demonstrates the ease with which SSDP reflections can disrupt SDN infrastructure by overloading network controllers. Additionally, this study emphasizes the role of SDN in isolating malicious IoT devices and reconfiguring network flows to mitigate the impact of SSDP reflection attacks, underscoring the potential of SDN for adaptive defense mechanisms against such threats.

D. SUMMARY AND INSIGHTS

Table 8 summarizes the literature work on reflection attacks. Reflection attacks in SDN environments have become increasingly sophisticated due to the unique architecture and operational characteristics of SDN. Attackers typically begin by probing the network to collect intelligence about its configuration, enabling them to craft and trigger control messages effectively. Recent frameworks, such as DIDA, address these challenges by monitoring downlink messages for anomalies based on host-application pairs. This approach not only prioritizes legitimate traffic during attacks but also minimizes latency for valid users [285]. Additionally, machine learning techniques have shown significant promise in enhancing detection capabilities by analyzing flow patterns to distinguish between legitimate traffic bursts and DDoS attacks [292]. However, most existing solutions predominantly focus on single-controller environments, often overlooking the complexities of multicontroller systems, where load balancing can obscure attack detection.

To address these pressing challenges, future research should concentrate on several key areas. Firstly, there is a critical need for robust detection mechanisms capable of accurately differentiating between DDoS attacks and legitimate traffic spikes in multi-controller SDN setups, potentially utilizing adaptive algorithms that adjust detection thresholds in real time. Furthermore, current defences frequently overlook vulnerabilities at the data and control layers, underscoring the necessity for studies aimed at strengthening these layers against exploitation. In particular, SNMP and CharGen reflection attacks require more attention, as recent studies on these specific threats are limited. Additionally, developing integrated defence strategies that amalgamate various detection methods could significantly enhance the protection against sophisticated reflection attacks. Finally, more empirical research is essential to validate proposed solutions in real-world scenarios, thereby ensuring their effectiveness across diverse conditions.

XIII. RESULTS AND DISCUSSIONS

Figure 14(a) shows the evaluation metrics used in DDoS detection studies for SDNs in seven categories to simplify their analysis: performance, error, specificity, evaluation, resource utilization, timing, and traffic-handling metrics. This grouping made it easier to understand the focus areas of the current research. Among the performance metrics, commonly used measures, such as accuracy, recall, precision, and F-measure, stand out for assessing how reliably detection systems classify traffic. In the error metrics category, the false positive rate is especially significant because it helps minimize disruptions caused by mistakenly flagged legitimate traffic. The specificity metrics group includes the specificity and true negative rate, which evaluate how effectively normal traffic is identified. Evaluation metrics, such as the ROC curve and AUC, provide a comprehensive overview of detection performance across

TABLE 8. Summary of reflection-based DDoS Attack detection techniques in SDN. Acronyms: **CA**:. Classification Approaches, **SA**:. Statistical and Analytical methods, **TS**:. Traditional Security measures, **CT**:. Clustering Techniques, **NN**:. Nearest Neighbor approaches, **GM**:. Generative models, **RL**:. Reinforcement Learning methods, **MT**:. Moving Target defense strategies.

Reference	Reflec. Attack Type	SDN Layer	Security Solutions Description	Evaluation and Tools	DDoS Detection Method									
					C	S	T	C	N	G	R	M		
					A	A	S	T	N	M	L	T		
[285], 2022	NTP	Data,Control	Proposed framework to defend against NTP attacks	AdIoTack		✓	✓							
[286], 2020		Control	OpenFlow rule for traffic filtering	Mininet	✓		✓							
[287], 2024		Data,Control	EC-NTD scheme with NAPT and two stage detection	Mininet	✓	✓								
[277], 2023	DNS	Data plane	Proposed packet limiting to reduce attacks.	HAProxy		✓	✓							
[289], 2024		Data,Control	Proposed siyrce address validation	Zmap	✓	✓	✓							
[290], 2020		Control	Traffic monitoring to reduce DNS attacks	DNSdros.c			✓		✓					
[291], 2021		Control	Proposed DPAR to enforce secure protocols	DNS server			✓							
[292], 2022	SSDP	Control	ML models to detect SSDP attacks.	CIICDDoS2019	✓	✓		✓	✓					
[293], 2021		Data,Control	Proposed tree modeling for monitoring attacks	Impact analysis		✓					✓	✓		

different thresholds. Resource utilization metrics, including CPU usage and system overhead, assess the efficiency of detection mechanisms and ensure that they can operate without overwhelming the network. Timing metrics, such as detection and training times, emphasize the need for rapid responses in real-world scenarios. Finally, traffic handling metrics, such as throughput and packet drop rate, were used to evaluate the impact of the system on network performance and user experience. This comprehensive categorization highlights the key metrics that researchers prioritize when analyzing and improving DDoS detection methods in SDNs.

Figure 14(b) categorizes the datasets commonly used in DDoS research within SDN layers. Dedicated DDoS datasets, such as CIC-DDoS2019 (2019) and CAIDA DDoS Attack Dataset (2007), are specifically tailored to study DDoS attacks in SDN environments. Although CIC-DDoS2019 provides modern traffic patterns suitable for contemporary attack scenarios, CAIDA (2007) is outdated and lacks relevance to the evolving dynamics of current DDoS threats. Intrusion detection datasets, including CIC-IDS2018 (2018), CIC-IDS2017 (2017), and UNSW-NB15 (2015), offer a broader scope for intrusion detection, but fall short in providing detailed DDoS-specific insights for SDN research. Older datasets, such as the NSL-KDD (2009), KDD-CUP99 (1999), and DARPA 1999 Dataset (1999), are increasingly unsuitable owing to their simplified attack models and outdated traffic characteristics. Synthetic or custom SDN datasets, including simulation-based datasets and SDA datasets, target specific attack scenarios but often lack the realism and consistency necessary for generalizability. The literature highlights a critical gap: the need for updated, standardized, and publicly accessible datasets that reflect modern attack scenarios, realistic SDN behaviors, and comprehensive feature sets. Such datasets are essential for improving the accuracy, relevance, and reproducibility of DDoS evaluation in SDN environments.

The distribution of DDoS detection methods across various attack categories reveals interesting trends in the application

of ML techniques and traditional methods for detecting different types of DDoS attacks. As shown in Figure 15, the study categorizes DDoS attacks into seven main types: Flooding, Application Layer, Protocol, Resource Exhaustion, Botnet-based, Amplification, and Reflection DDoS attacks, and evaluates how different detection techniques are employed in each.

- **Flooding DDoS Attacks:** With 29 papers, this category has a strong focus on SA methods, which were applied in 22 out of 29 studies. Additionally, CA approaches are widely used (11 papers), whereas methods such as RL, NN, and GM have limited applications. Notably, MT strategies were not used in this category, as shown in Figure 15.
- **Application Layer DDoS Attacks:** This category, with 27 papers, shows a notable preference for CA frameworks and TS, with 15 and 14 papers, respectively. Statistical SA systems also showed moderate usage (nine papers). The application of GM (eight studies) and MT (one study) is evident, reflecting a more varied approach for countering these attacks.
- **Protocol DDoS Attacks:** Out 27 studies, protocol DDoS attacks show a clear reliance on TS measures (25 papers). However, there is minimal use of other methods, such as CA and NN techniques. This suggests that more established classical methods are still dominant in detecting attacks that target network protocols.
- **Resource Exhaustion Attacks:** Balance was observed across detection methods for this category (24 papers). While the CA and SA methods were applied in 9 and 11 studies, TS (13 papers) was the most prominent. The use of RL and GM was very limited, whereas MT was used in two studies.
- **Botnet-Based DDoS Attacks:** This category is the most researched with 33 papers. It is dominated by the CA (23 papers) and SA methods (15 papers), showing that these methods are often preferred for detecting attacks involving botnets. There is also significant use of RL (five papers) and some applications of TS measures

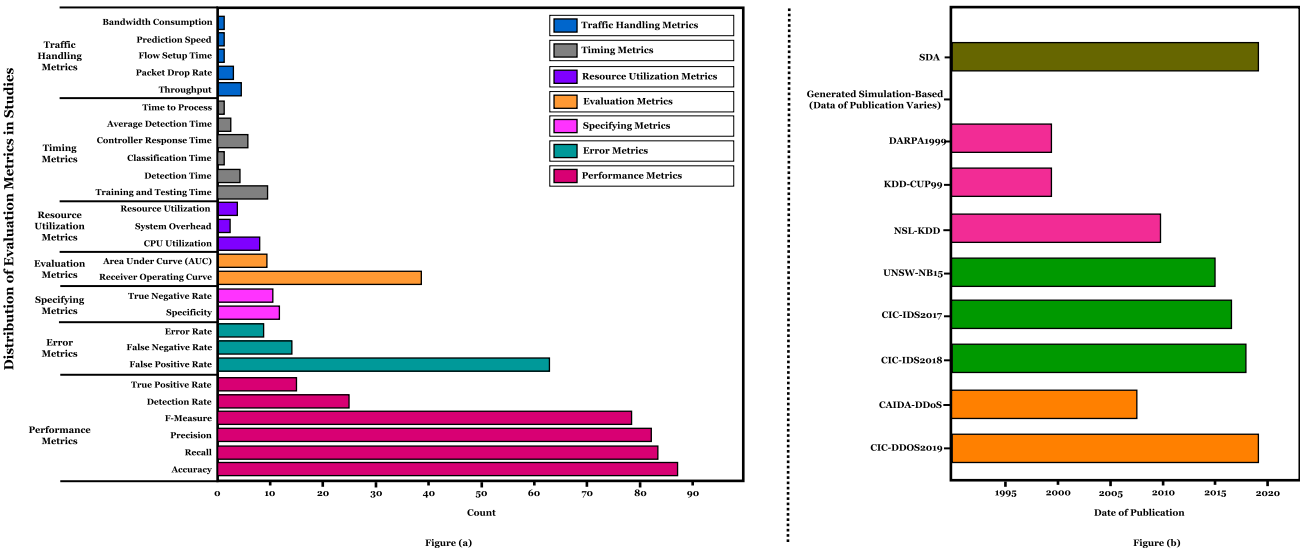


FIGURE 14. Evaluation metrics and dataset landscape in DDoS research. Figure (a) Distribution of evaluation metrics utilized in DDoS detection studies, categorized into traffic handling, timing, resource utilization, performance, error, and evaluation metrics, showcasing their prevalence and significance. Figure (b) Classification of datasets commonly employed in DDoS research, categorized into dedicated DDoS datasets, intrusion detection datasets with DDoS attacks, older or legacy datasets, and synthetic or custom SDN datasets, with analysis of their limitations and insights from current literature.

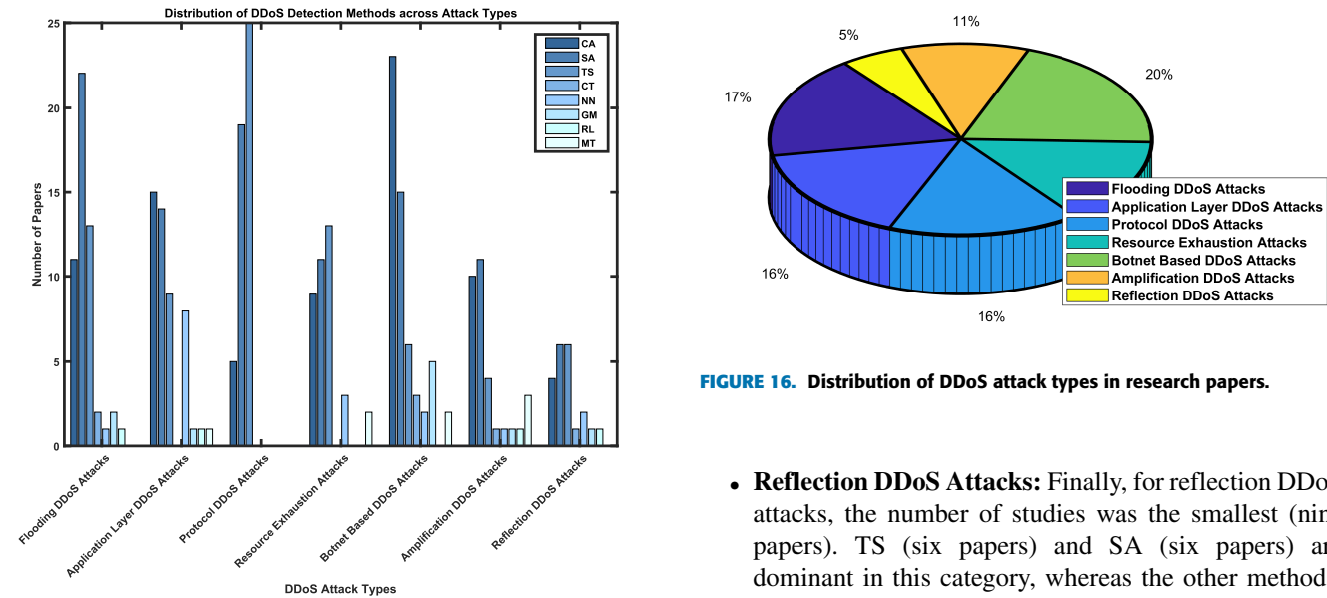


FIGURE 15. Distribution of DDoS detection methods (CA, SA, TS, CT, NN, GM, RL, MT) across various attack types (Flooding, Application Layer, Protocol, Resource Exhaustion, Botnet Based, Amplification, and Reflection) from research papers published between 2020 and 2024.

FIGURE 16. Distribution of DDoS attack types in research papers.

(six papers). However, the application of MT is limited.

- **Amplification DDoS Attacks:** In Amplification attacks were studied in 18 papers. SA is employed in 10 studies, while traditional security 11 papers are also commonly applied. Interestingly, MT was used in three studies, indicating a growing interest in the dynamic defense mechanisms in this domain.

- **Reflection DDoS Attacks:** Finally, for reflection DDoS attacks, the number of studies was the smallest (nine papers). TS (six papers) and SA (six papers) are dominant in this category, whereas the other methods, such as CA and RL, have relatively fewer applications.

Key insights from the analysis reveal that statistical and analytical methods dominate in most DDoS attack categories, while traditional security measures remain widely used, especially for protocol and reflection attacks. Classification approaches are prominent in botnet-based attacks, and emerging techniques like reinforcement learning and moving target defense are still underutilized. Overall, traditional methods continue to be preferred, but there is growing interest in integrating machine learning and dynamic defense strategies.

Figure 16 illustrates the distribution of research papers across the various categories of DDoS attacks in an SDN

environment. The most studied category is Botnet-Based DDoS Attacks, with 33 papers accounting for 20% of the total. This was followed by Flooding DDoS Attacks (29 papers, 17%) and Application Layer DDoS Attacks (27 papers, 16%). Protocol DDoS Attacks have also received significant attention, with 27 studies (16%). Resource-exhaustion attacks accounted for 24 papers (14%). Amplification DDoS Attacks and Reflection DDoS Attacks were less frequently studied, with 18 (11%) and 9 papers (5%), respectively. This suggests a stronger focus on botnets and flooding attacks than on the amplification and reflection types.

A. HYBRID DETECTION MODELS IN SDN ENVIRONMENTS

Hybrid detection models leverage the advantages of multiple techniques to enhance the accuracy, scalability, and robustness of DDoS detection mechanisms in SDN environments [8]. By combining ML algorithms, SA methods, and analytical frameworks, these frameworks address the limitations of the single-method approaches. Below, we discuss several studies that have successfully applied hybrid models to address various DDoS threats across different attack categories.

- **Flooding DDoS Attacks:** Sudar et al. [110] utilized a combination of the C4.5 decision tree algorithm and analytical methods to detect flooding DDoS attacks. This hybrid approach capitalizes on the decision tree's ability to classify traffic patterns and the analytical framework's capacity to detect subtle statistical anomalies.
- **Application Layer Attacks:** James et al. [105] proposed a hybrid model combining LSTM networks with CNN. This approach exploits the LSTM's ability to analyze sequential data and the CNN's feature extraction capabilities, providing a robust solution for detecting application-layer DDoS attacks. Similarly, Krishnan et al. [162] employed an LSTM and CNN hybrid to effectively classify and mitigate application-layer threats, achieving high detection accuracy while maintaining scalability in SDN environments.
- **Resource Exhaustion Attacks:** Nesha et al. [218] introduced a CNN and LSTM hybrid model to address resource exhaustion attacks. The CNN was used to extract key features from network traffic, while the LSTM identified temporal patterns, improving the detection of complex attack vectors.
- **Botnet DDoS Attacks:** Nguyen et al. [107] implemented a KNN algorithm combined with XGBoost to detect botnet-based DDoS attacks. This hybrid model leverages KNN's simplicity for initial classification and XGBoost's efficiency in handling high-dimensional data to refine detection accuracy. Varjrobo et al. [249] also combined LSTM and XGBoost to mitigate botnet-driven threats, demonstrating the effectiveness of ensemble models in dynamic and evolving attack scenarios.

TABLE 9. Summary of hybrid detection models in SDN for DDoS attacks.

Reference	Hybrid Models	DDoS Attack Category
[110], 2023	C4.5 + AM	Flooding DDoS Attacks
[105], 2021	LSTM + CNN	Application Layer Attacks
[107], 2020	KNN + XGBoost	Botnet DDoS Attacks
[162], 2022	LSTM + CNN	Application Layer Attacks
[218], 2022	CNN + LSTM	Resource Exhaustion Attacks
[249], 2024	LSTM + XGBoost	Botnet DDoS Attacks

Table 9 summarizes key studies employing hybrid techniques, their combinations, and their targeted attack categories. Hybrid methods demonstrate significant potential for enhancing DDoS detection and mitigation in SDN environments by combining the strengths of multiple methodologies. Techniques such as LSTM paired with CNN or XGBoost combined with KNN improve feature extraction, temporal pattern recognition, and data handling; reduce false positives; and improve scalability for dynamic traffic. These models enable real-time adaptability and can address evolving threats more effectively. Future studies should focus on lightweight solutions for resource-constrained environments, predictive analytics for proactive detection, and testing across diverse datasets to ensure robustness and scalability, making them vital for advancing SDN security.

XIV. CHALLENGES AND FUTURE WORKS

Although this review paper explores eight detection methods, including statistical techniques, traditional measures, and classification approaches, which show promise in addressing DDoS detection challenges in SDN, notable limitations and unresolved issues remain across these methods.

- One significant challenge is that statistical methods often rely on predefined distribution assumptions regarding network traffic [285], [287], [288], [289]. In an SDN dynamic environment, these assumptions may not hold, thereby leading to inaccurate detection results. Variability in traffic patterns makes it difficult for these methods to adapt to new attack types or changes in normal traffic behavior.
- Statistical approaches may not capture all the relevant features necessary for accurate DDoS detection, which can result in missed detections or false positives. This limitation arises when these methods fail to identify subtle anomalies, which may indicate an attack.
- Over time, the statistical properties of network traffic can change because of legitimate shifts in user behavior or infrastructure changes. AI-based models trained on historical data may become less effective as they encounter different traffic patterns, leading to a performance degradation known as model drift.
- The reviewed studies show that many classification approaches excel at detecting high-rate DDoS attacks but struggle with low-rate or stealthy attacks that blend

in with legitimate traffic. This limitation is significant as attackers increasingly use such tactics to evade detection systems.

- As discussed in Section IX, particularly in IX-A, the REA relies on traditional detection methods that frequently use fixed thresholds to identify anomalies. This approach can be problematic because it does not adapt well to changing network conditions or evolving attack patterns, leading to either missed detections or false positives.
- Although a few studies have applied clustering detection approaches to DDoS attacks, particularly botnet-based DDoS attacks, these strategies are continually evolving. This presents a challenge for static clustering models that do not quickly adapt to new patterns. Regular retraining or reconfiguration of clusters may be necessary, but it can be resource-intensive and difficult to implement in real-time.
- Based on our comprehensive survey, we found that a few studies, such as [105], [107], [120], [217], and [248], used KNN, which is sensitive to the scale of features used in distance calculations. If features are not normalized or standardized, it can lead to skewed results, where certain features dominate the distance metric. This sensitivity requires careful preprocessing of the data to ensure accurate detection.
- The effectiveness of generative models depends heavily on the availability of high-quality labelled datasets for training [247], [249]. However, obtaining comprehensive datasets that accurately represent both normal and attack traffic is challenging. Poorly labelled or imbalanced datasets may lead to biased models that fail to effectively generalize.
- RL algorithms must balance exploration (attempting new actions) with exploitation (using known actions that yield high rewards). This trade-off can lead to suboptimal performance if the model spends too much time exploring or exploiting ineffective strategies.
- While moving target strategies aim to confuse attackers by altering the attack surface, they may not be sufficient against sophisticated or novel DDoS attack vectors that do not rely on predictable patterns. Continuous adaptation is required, but it can be resource-intensive and may not keep pace with rapidly evolving threats.
- Many existing datasets for training detection models are based on traditional network environments and may not accurately reflect the characteristics of SDN architecture. While some studies have used outdated datasets such as DARPA199 and KDD99 (as shown in Figure 14), others have generated unrealistic datasets, most of which are not publicly available. This limitation can hinder the development of effective detection algorithms tailored to flooding and other DDoS attacks in SDN.
- Relying on a single controller can complicate the detection of anomalies indicative of DDoS attacks in the

SDN layers. Several studies have used a single controller in their experiments. The controller may not have access to comprehensive data from across the entire network, which limits its ability to identify subtle patterns that could signal an ongoing attack.

- Finally, to effectively combat evolving DDoS threats, detection systems must continuously learn from the new data. Implementing adaptive mechanisms within SDN architectures presents significant challenges, particularly with respect to resource allocation and model retraining.

This review emphasizes several key areas for future research on the strategic integration of AI into traditional and other DDoS anomaly detection methods within the SDN layers.

- Entropy methods are effective for measuring randomness and disorder in network traffic, enabling the identification of anomalies that may indicate DDoS attacks. There is growing interest in hybrid models that combine entropy-based detection with ML algorithms such as SVM and CNN. These models leverage the strengths of both statistical analysis and ML to improve the detection rates and reduce false positives [103], [105], [106].
- AI can automate the response to detected DDoS attacks, reducing the time required to mitigate threats. Traditional methods often require manual intervention, which can be slow or inefficient. The implementation of continuous learning mechanisms within AI models is vital for adapting to new attack vectors as they emerge. Future research can investigate adaptive algorithms that evolve based on new data inputs and changing network conditions.
- Many traditional methods are less effective against low-rate DDoS attacks owing to their subtle nature. Future studies should prioritize the development of detection techniques specifically designed to identify these types of threats without generating excessive false positives.
- DL models can process large volumes of data in real time, making them well suited for dynamic environments such as SDN. When combined with clustering techniques, these models can quickly adapt to changing traffic patterns and detect DDoS attacks as they occur, thereby minimizing potential damage. Techniques such as wavelet decomposition or advanced statistical analysis could be employed to derive meaningful features from raw traffic data before applying clustering algorithms
- KNN has been shown to achieve high accuracy in classifying network traffic, with studies indicating that it can achieve approximately 99.4% accuracy in DDoS detection compared with other traditional methods [105]. Developing hybrid models that combine DL for feature extraction with KNN for classification

TABLE 10. List of abbreviations.

Acronyms	Description	Acronyms	Description
ACK	Acknowledgment	AI	Artificial Intelligence
API	Application Programming Interface	ARP	Address Resolution Protocol
BGP	Border Gateway Protocol	CA	Classification Approaches
CNN	Convolutional Neural Network	CT	Clustering Techniques
DHCP	Dynamic Host Configuration Protocol	DL	Deep Learning
DNS	Domain Name System	DUID	DHCP Unique Identifier
DDoS	Distributed Denial-of-Service	eBPF	Extended Berkeley Packet Filter
FIN	Finish Flag in TCP Communication	FRE	Firewall Rule Explosion
GAN	Generative Adversarial Network	GM	Generative Models
GUI	Graphical User Interface	HTTP	Hypertext Transfer Protocol
ICMP	Internet Control Message Protocol	IDS	Intrusion Detection System
IoT	Internet of Things	IPS	Intrusion Prevention System
KLD	Kullback–Leibler Distance	LLDP	Link Layer Discovery Protocol
LSTM	Long Short-Term Memory	MAC	Media Access Control
ML	Machine Learning	MOHE	Ministry of Higher Education
MT	Moving Target Defense	NIDS	Network Intrusion Detection System
NN	Nearest Neighbor Approaches	NTP	Network Time Protocol
OFTO	OpenFlow Table Overflow	OPPF	OpenFlow Packet-In Flood
POD	Ping of Death	QoS	Quality of Service
RL	Reinforcement Learning	RNN	Recurrent Neural Network
RUDY	R-U-Dead-Yet	SA	Statistical and Analytical Methods
SNMP	Simple Network Management Protocol	SSDP	Simple Service Discovery Protocol
SVM	Support Vector Machine	SYN	Synchronize Flag in TCP Communication
TCP	Transmission Control Protocol	TS	Traditional Security Measures
UDP	User Datagram Protocol	UNWS	UNet Wireless Simulation
VSF	Virtual Security Functions	XDP	Express Data Path

could provide a more robust approach for DDoS detection.

- Generative techniques were used to model the distribution of normal network traffic. By training on legitimate traffic, these models can identify deviations that may signal a DDoS attack. However, they often face challenges related to high-dimensional data, such as increased computational complexity and difficulty in convergence during training. Future work should focus on developing techniques that mitigate these challenges while maintaining a high detection accuracy.
- Generating datasets enables researchers to conduct extensive testing of detection algorithms without the high cost associated with capturing live traffic under attack conditions. This cost-effectiveness allows for more iterative experimentation and refinement of the AI models. Future work should emphasize generating datasets that include low-rate DDoS attacks and those that target specific applications or services within SDN environments. This focus will help develop detection systems capable of identifying subtle attacks that traditional methods may miss.
- Multiple controllers can facilitate load balancing across the network, distribute traffic more evenly, and prevent any single controller from becoming a bottleneck.

Moreover, combining multiple controllers with advanced ML/DL techniques can significantly enhance detection capabilities. Future studies should explore how distributed learning models can be implemented across controllers to improve the anomaly detection and response times.

- Most existing studies primarily rely on emulation tools, such as Mininet, which, while useful, may not fully capture the complexities and dynamics of actual network conditions. Real-world environments provide authentic traffic patterns that can significantly enhance the accuracy of DDoS detection. Future work should involve testing detection algorithms under a variety of operational conditions, including different types of DDoS attacks (e.g., volumetric and application-layer), varying network loads, and diverse configurations of SDN components.
- Blockchain provides an immutable ledger that records all transactions and events within a network. This feature enhances data integrity and allows for the transparent auditing of network activities, making it easier to trace the origins of DDoS attacks and understand their impact. Future research should focus on developing robust frameworks that integrate blockchain technology with the existing SDN architectures for DDoS detection. This includes defining protocols for communication between

SDN components and the blockchain network to ensure seamless operations.

- Future research should focus on developing integrated frameworks that unify detection mechanisms across the data, control, and application planes. Such frameworks would define protocols for communication and data sharing among planes to enhance the detection capabilities.

XV. CONCLUSION

Since the emergence of Software-Defined Networking (SDN) technology, it has gained significant attention from both academia and industry. One of the primary reasons for this interest is SDN's centralized control architecture of SDN, which enables enhanced programmability, flexibility, and efficient network management. Despite these advantages, SDN is vulnerable to several attacks, particularly Distributed Denial of Service (DDoS) attacks. These attacks can target the controller, data plane devices, or northbound APIs, thereby leading to resource exhaustion and potential network outages. Therefore, ensuring the continued operation and reliability of SDN requires prompt identification of anomalous behavior in the network and effective mitigation of such threats. This article provides a comprehensive review of detection and mitigation methods to address the critical challenges of DDoS attacks in SDNs. We introduce a novel taxonomy of DDoS attacks in SDN, categorizing them by their mechanisms and impacts, such as flooding, application layer, protocol-based, and botnet-driven attacks. This structured framework enhances our understanding of diverse threat landscapes. We then analyze the detection methods for each attack type, covering approaches such as machine learning, statistical techniques, and reinforcement learning, while highlighting their strengths, limitations, and applicability. A visual guide was also presented to summarize and categorize these detection techniques, serving as a practical reference for researchers and practitioners. Furthermore, we discuss the scalability and effectiveness of various detection techniques to address different SDN-specific attack vectors. This survey is based on an extensive review of over 165 articles published in reputable journals and databases such as IEEE Xplore, ScienceDirect, and Springer. Finally, the study identifies key research gaps, including the need for hybrid frameworks and adaptive detection models, and suggests future directions that will drive advancements in DDoS detection systems. This study serves as a valuable resource for both novice and expert audiences, fostering continued innovation in SDN security.

APPENDIX

Table 10 provides references for the acronyms used in this study.

REFERENCES

- [1] K. S. Sahoo, D. Puthal, M. Tiwary, J. J. P. C. Rodrigues, B. Sahoo, and R. Dash, "An early detection of low rate DDoS attack to SDN based data center networks using information distance metrics," *Future Gener. Comput. Syst.*, vol. 89, pp. 685–697, Dec. 2018.
- [2] N. Indrason and G. Saha, "Exploring blockchain-driven security in SDN-based IoT networks," *J. Netw. Comput. Appl.*, vol. 224, Apr. 2024, Art. no. 103838.
- [3] Y. Al-Dunainawi, B. R. Al-Kaseem, and H. S. Al-Raweshdy, "Optimized artificial intelligence model for DDoS detection in SDN environment," *IEEE Access*, vol. 11, pp. 106733–106748, 2023.
- [4] N. H. Hussein, S. P. Koh, C. T. Yaw, S. K. Tiong, F. Benedict, T. Yusaf, K. Kadirgama, and T. C. Hong, "SDN-based VANET routing: A comprehensive survey on architectures, protocols, analysis, and future challenges," *IEEE Access*, early access, Jan. 17, 2024, doi: 10.1109/ACCESS.2024.3355313.
- [5] P. Shantharama, A. S. Thyagaturu, N. Karakoc, L. Ferrari, M. Reisslein, and A. Scaglione, "LayBack: SDN management of multi-access edge computing (MEC) for network access services and radio resource sharing," *IEEE Access*, vol. 6, pp. 57545–57561, 2018.
- [6] S. H. A. Kazmi, F. Qamar, R. Hassan, and K. Nisar, "Routing-based interference mitigation in SDN enabled beyond 5G communication networks: A comprehensive survey," *IEEE Access*, vol. 11, pp. 4023–4041, 2023.
- [7] G. N. Nurkahfi, A. Triwinarko, B. Prawara, N. Armi, T. Juhana, N. R. Syambas, E. Mulyana, E. Dogheche, and I. Dayoub, "On SDN to support the IEEE 802.11 and C-V2X-based vehicular communications use-cases and performance: A comprehensive survey," *IEEE Access*, vol. 12, pp. 95926–95958, 2024.
- [8] R. Amin, M. Reisslein, and N. Shah, "Hybrid SDN networks: A survey of existing approaches," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3259–3306, 4th Quart., 2018.
- [9] K. S. Sahoo, D. Puthal, M. Tiwary, M. Usman, B. Sahoo, Z. Wen, B. P. S. Sahoo, and R. Ranjan, "ESMLB: Efficient switch migration-based load balancing for multicontroller SDN in IoT," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 5852–5860, Jul. 2020.
- [10] T. T. Hieu and N. Kitsuwat, "Swift recovery: An innovative routing approach for multi-failure protection in software-defined networks," *IEEE Access*, vol. 12, pp. 158685–158702, 2024.
- [11] V. G. da Silva Ruffo, D. M. Lent, M. Komarchesqui, V. F. Schiavon, M. V. O. de Assis, L. F. Carvalho, and M. L. Proença, "Anomaly and intrusion detection using deep learning for software-defined networks: A survey," *Expert Syst. Appl.*, vol. 256, Dec. 2024, Art. no. 124982.
- [12] T. Dargahi, A. Caponi, M. Ambrosin, G. Bianchi, and M. Conti, "A survey on the security of stateful SDN data planes," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 3, pp. 1701–1725, 3rd Quart., 2017.
- [13] D. Adanza, L. Gifre, P. Alemany, J.-P. Fernández-Palacios, O. González-De-Dios, R. Muñoz, and R. Vilalta, "Enabling traffic forecasting with cloud-native SDN controller in transport networks," *Comput. Netw.*, vol. 250, Aug. 2024, Art. no. 110565.
- [14] S. Ibrahim, A. M. Youssef, M. Shoman, and S. Taha, "Intelligent SDN to enhance security in IoT networks," *Egyptian Informat. J.*, vol. 28, Dec. 2024, Art. no. 100564.
- [15] N. M. Yungaicela-Naula, C. Vargas-Rosales, J. A. Perez-Diaz, E. Jacob, and C. Martinez-Cagnazzo, "Physical assessment of an SDN-based security framework for DDoS attack mitigation: Introducing the SDN-SlowRate-DDoS dataset," *IEEE Access*, vol. 11, pp. 46820–46831, 2023.
- [16] A. A. Wabi, I. Idris, O. M. Olaniyi, and J. A. Ojienyi, "DDoS attack detection in SDN: Method of attacks, detection techniques, challenges and research gaps," *Comput. Secur.*, vol. 139, Apr. 2024, Art. no. 103652.
- [17] K. S. Sahoo, S. K. Panda, S. Sahoo, B. Sahoo, and R. Dash, "Toward secure software-defined networks against distributed denial of service attack," *J. Supercomput.*, vol. 75, no. 8, pp. 4829–4874, Aug. 2019.
- [18] P. K. Dey and M. Yuksel, "An economic analysis of cloud-assisted routing for wider area SDN," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 1, pp. 445–458, Mar. 2020.
- [19] K. S. Sahoo, B. K. Tripathy, K. Naik, S. Ramasubbareddy, B. Balusamy, M. Khari, and D. Burgos, "An evolutionary SVM model for DDOS attack detection in software defined networks," *IEEE Access*, vol. 8, pp. 132502–132513, 2020.
- [20] S. Chamoli and V. Mittal, "DDoS landscape: Insight into attacks and tools," in *Proc. Int. Conf. Electr. Electron. Comput. Technol. (ICEECT)*, Aug. 2024, pp. 1–6.
- [21] A. I. Martyshkin, R. A. Biktashev, and A. D. Karlyganov, "Research and analysis of well-known solutions and development of DNS server software in the infrastructure of the digital economy," in *Proc. Int. Russian Autom. Conf. Cham, Switzerland: Springer*, Jan. 2024, pp. 344–356.

- [22] A. K. Jain, H. Shukla, and D. Goel, "A comprehensive survey on DDoS detection, mitigation, and defense strategies in software-defined networks," *Cluster Comput.*, vol. 27, no. 9, pp. 13129–13164, Dec. 2024.
- [23] S. Kaur, K. Kumar, N. Aggarwal, and G. Singh, "A comprehensive survey of DDoS defense solutions in SDN: Taxonomy, research challenges, and future directions," *Comput. Secur.*, vol. 110, Nov. 2021, Art. no. 102423.
- [24] F. S. D. Silva, E. Silva, E. P. Neto, M. Lemos, A. J. V. Neto, and F. Esposito, "A taxonomy of DDoS attack mitigation approaches featured by SDN technologies in IoT scenarios," *Sensors*, vol. 20, no. 11, p. 3078, May 2020.
- [25] I. A. Valdovinos, J. A. Pérez-Díaz, K.-K.-R. Choo, and J. F. Botero, "Emerging DDoS attack detection and mitigation strategies in software-defined networks: Taxonomy, challenges and future directions," *J. Netw. Comput. Appl.*, vol. 187, Aug. 2021, Art. no. 103093.
- [26] R. Uddin, S. A. P. Kumar, and V. Chamola, "Denial of service attacks in edge computing layers: Taxonomy, vulnerabilities, threats and solutions," *Ad Hoc Netw.*, vol. 152, Jan. 2024, Art. no. 103322.
- [27] K. M. Sudar and P. Deepalakshmi, "Flow-based detection and mitigation of low-rate DDoS attack in SDN environment using machine learning techniques," in *IoT Analytics Sensor Networks*. Cham, Switzerland: Springer, Sep. 2021, pp. 193–205.
- [28] M. R. Kadri, A. Abdelli, J. Ben Othman, and L. Mokdad, "Survey and classification of dos and DDoS attack detection and validation approaches for IoT environments," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101021.
- [29] P. Shukla, C. R. Krishna, and N. V. Patil, "IoT traffic-based DDoS attacks detection mechanisms: A comprehensive review," *J. Supercomput.*, vol. 80, no. 7, pp. 9986–10043, May 2024.
- [30] M. Khalil and H. Bilal, "Enabling software defined networks: Precursors, evolution and prognosis," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, Jan. 2020, pp. 20–25.
- [31] A. Lara, A. Kolasani, and B. Ramamurthy, "Network innovation using OpenFlow: A survey," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 1, pp. 493–512, 1st Quart., 2014.
- [32] S. Badotra and S. N. Panda, "Evaluation and comparison of OpenDayLight and open networking operating system in software-defined networking," *Cluster Comput.*, vol. 23, no. 2, pp. 1281–1291, Jun. 2020.
- [33] W. Wang, L. Dong, B. Zhuge, M. Gao, F. Jia, R. Jin, J. Yu, and X. Wu, "Design and implementation of an open programmable router compliant to IETF ForCES specifications," in *Proc. 6th Int. Conf. Netw. (ICN)*, Apr. 2007, p. 82.
- [34] J. Singh and S. Behal, "Detection and mitigation of DDoS attacks in SDN: A comprehensive review, research challenges and future directions," *Comput. Sci. Rev.*, vol. 37, Aug. 2020, Art. no. 100279.
- [35] S. Barguil, V. Lopez, C. Manta-Caro, A. M. L. De Lerma, O. G. D. Dios, E. Echeverry, J. P. Fernandez-Palacios, J. Karvonen, J. Kemppainen, N. Maya, and R. Vilalta, "Field trial of programmable L3 VPN service deployment using SDN-based multi-domain service provisioning over IP/optical networks," *IEEE Netw.*, vol. 35, no. 6, pp. 217–224, Nov. 2021.
- [36] S. Ahmad and A. H. Mir, "Scalability, consistency, reliability and security in SDN controllers: A survey of diverse SDN controllers," *J. Netw. Syst. Manage.*, vol. 29, no. 1, pp. 1–59, Jan. 2021.
- [37] S. Murphy and A. Edmonds, "Realizing edge computing connectivity with open virtual networking," in *Proc. IEEE/ACM Int. Conf. Utility Cloud Comput. Companion (UCC Companion)*, Dec. 2018, pp. 128–133.
- [38] I. Ahmad, S. Namal, M. Ylianttila, and A. Gurtov, "Security in software defined networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2317–2346, 4th Quart., 2015.
- [39] C. Trois, M. D. Del Fabro, L. C. E. de Bona, and M. Martinello, "A survey on SDN programming languages: Toward a taxonomy," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 4, pp. 2687–2712, 4th Quart., 2016.
- [40] A. H. Abdi, L. Audah, A. Salh, M. A. Alhartomi, H. Rasheed, S. Ahmed, and A. Tahir, "Security control and data planes of SDN: A comprehensive review of traditional, AI, and MTD approaches to security solutions," *IEEE Access*, vol. 12, pp. 69941–69980, 2024.
- [41] D. E. Sarmiento, A. Lebre, L. Nussbaum, and A. Chari, "Decentralized SDN control plane for a distributed cloud-edge infrastructure: A survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 256–281, 1st Quart., 2021.
- [42] R. Wazirali, R. Ahmad, and S. Alhiyari, "SDN-OpenFlow topology discovery: An overview of performance issues," *Appl. Sci.*, vol. 11, no. 15, p. 6999, Jul. 2021.
- [43] E. Kaljic, A. Maric, P. Njemcevic, and M. Hadzialic, "A survey on data plane flexibility and programmability in software-defined networking," *IEEE Access*, vol. 7, pp. 47804–47840, 2019.
- [44] Q. Yan, F. R. Yu, Q. Gong, and J. Li, "Software-defined networking (SDN) and distributed denial of service (DDoS) attacks in cloud computing environments: A survey, some research issues, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 602–622, 1st Quart., 2016.
- [45] Y. Yu, X. Li, X. Leng, L. Song, K. Bu, Y. Chen, J. Yang, L. Zhang, K. Cheng, and X. Xiao, "Fault management in software-defined networking: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 349–392, 1st Quart., 2019.
- [46] M. Saquetti, R. Canofre, A. F. Lorenzon, F. D. Rossi, J. R. Azambuja, W. Cordeiro, and M. C. Luizelli, "Toward in-network intelligence: Running distributed artificial neural networks in the data plane," *IEEE Commun. Lett.*, vol. 25, no. 11, pp. 3551–3555, Nov. 2021.
- [47] M. Karakus and A. Duresi, "Quality of service (QoS) in software defined networking (SDN): A survey," *J. Netw. Comput. Appl.*, vol. 80, pp. 200–218, Feb. 2017.
- [48] J. W. Guck, A. Van Bemten, M. Reisslein, and W. Kellerer, "Unicast QoS routing algorithms for SDN: A comprehensive survey and performance evaluation," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 1, pp. 388–415, 1st Quart., 2018.
- [49] J. Buzzio-García, J. Vergara, S. Ríos-Guiral, C. Garzón, S. Gutiérrez, J. F. Botero, J. L. Quiroz-Arroyo, and J. A. Pérez-Díaz, "Exploring traffic patterns through network programmability: Introducing SDNFlow, a comprehensive OpenFlow-based statistics dataset for attack detection," *IEEE Access*, vol. 12, pp. 42163–42180, 2024.
- [50] F. Altheide, S. Buttgerit, and M. Rossberg, "Increasing resilience of SD-WAN by distributing the control plane [extended version]," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 3, pp. 2569–2581, Jun. 2024.
- [51] P. P. Shahrbabaki, R. W. L. Coutinho, and Y. R. Shayan, "SDN-LB: A novel server workload balancing algorithm for IoT video analytics," *Ad Hoc Netw.*, vol. 155, Mar. 2024, Art. no. 103398.
- [52] M. A. Ribeiro, M. S. Pereira Fonseca, and J. de Santi, "Detecting and mitigating DDoS attacks with moving target defense approach based on automated flow classification in SDN networks," *Comput. Secur.*, vol. 134, Nov. 2023, Art. no. 103462.
- [53] J. Kim, M. Seo, S. Lee, J. Nam, V. Yegneswaran, P. Porras, G. Gu, and S. Shin, "Enhancing security in SDN: Systematizing attacks and defenses from a penetration perspective," *Comput. Netw.*, vol. 241, Mar. 2024, Art. no. 110203.
- [54] Z. A. Bhuiyan, S. Islam, M. M. Islam, A. B. M. A. Ullah, F. Naz, and M. S. Rahman, "On the (in)security of the control plane of SDN architecture: A survey," *IEEE Access*, vol. 11, pp. 91550–91582, 2023.
- [55] D. Carrascal, E. Rojas, J. M. Arco, D. Lopez-Pajares, J. Alvarez-Horcajo, and J. A. Carral, "A comprehensive survey of in-band control in SDN: Challenges and opportunities," *Electronics*, vol. 12, no. 6, p. 1265, Mar. 2023.
- [56] Y. Liu, B. Zhao, Y. An, and J. Guo, "DACAS: Integration of attribute-based access control for northbound interface security in SDN," *World Wide Web*, vol. 26, no. 4, pp. 2143–2173, Jul. 2023.
- [57] U. Tupakula, K. K. Karmakar, V. Varadharajan, and B. Collins, "Implementation of techniques for enhancing security of southbound infrastructure in SDN," in *Proc. 13th Int. Conf. Netw. Future (NoF)*, Oct. 2022, pp. 1–5.
- [58] N. Hu, F. Luan, X. Tian, and C. Wu, "A novel SDN-based application-awareness mechanism by using deep learning," *IEEE Access*, vol. 8, pp. 160921–160930, 2020.
- [59] A. L. Aliyu, A. Aneiba, M. Patwary, and P. Bull, "A trust management framework for software defined network (SDN) controller and network applications," *Comput. Netw.*, vol. 181, Nov. 2020, Art. no. 107421.
- [60] N. V. R. Kumar and P. M. Kumar, "Application of SDN for secure communication in IoT environment," *Comput. Commun.*, vol. 151, pp. 60–65, Feb. 2020.
- [61] N. V. Ha, D. D. Quan, and T. T. T. Nguyen, "Graphical user interface for RYU software defined network controller," in *Proc. IEEE 8th Inf. Technol. Int. Seminar (ITIS)*, Oct. 2022, pp. 312–317.
- [62] C. Wang, H. Ni, and L. Liu, "An enhanced message distribution mechanism for northbound interfaces in the SDN environment," *Appl. Sci.*, vol. 11, no. 10, p. 4346, May 2021.

- [63] A. H. Abdi, L. Audah, A. M. Omar, and M. J. Abdiaziz, "Design and simulation of a secured enterprise network architecture for all departments at East Africa University (EAU), Somalia," in *Proc. 4th Int. Conf. Sci. Inf. Technol. Smart Admin. (ICSINTESA)*, Jul. 2024, pp. 552–557.
- [64] M. Abbasi, A. Shahraki, and A. Taherkordi, "Deep learning for network traffic monitoring and analysis (NTMA): A survey," *Comput. Commun.*, vol. 170, pp. 19–41, Mar. 2021.
- [65] A. Azab, M. Khasawneh, S. Alrabaei, K.-K.-R. Choo, and M. Sarsour, "Network traffic classification: Techniques, datasets, and challenges," *Digit. Commun. Netw.*, vol. 10, no. 3, pp. 676–692, Jun. 2024.
- [66] T. E. Ali, Y.-W. Chong, and S. Manickam, "Machine learning techniques to detect a DDoS attack in SDN: A systematic review," *Appl. Sci.*, vol. 13, no. 5, p. 3183, Mar. 2023.
- [67] B. Alhijawi, S. Almajali, H. Elgala, H. B. Salameh, and M. Ayyash, "A survey on DoS/DDoS mitigation techniques in SDNs: Classification, comparison, solutions, testing tools and datasets," *Comput. Electr. Eng.*, vol. 99, Apr. 2022, Art. no. 107706.
- [68] B. A. Khalaf, S. A. Mostafa, A. Mustapha, M. A. Mohammed, and W. M. Abdullah, "Comprehensive review of artificial intelligence and statistical approaches in distributed denial of service attack and defense methods," *IEEE Access*, vol. 7, pp. 51691–51713, 2019.
- [69] H. Lin, C. Wu, and M. Masdari, "A comprehensive survey of network traffic anomalies and DDoS attacks detection schemes using fuzzy techniques," *Comput. Electr. Eng.*, vol. 104, Dec. 2022, Art. no. 108466.
- [70] R. Vishwakarma and A. K. Jain, "A survey of DDoS attacking techniques and defence mechanisms in the IoT network," *Telecommun. Syst.*, vol. 73, no. 1, pp. 3–25, Jan. 2020.
- [71] F. Alwahedi, A. Aldaheri, M. A. Ferrag, A. Battah, and N. Tihanyi, "Machine learning techniques for IoT security: Current research and future vision with generative AI and large language models," *Internet Things Cyber-Phys. Syst.*, vol. 4, pp. 167–185, Jan. 2024.
- [72] I. Ortega-Fernandez and F. Liberati, "A review of denial of service attack and mitigation in the smart grid using reinforcement learning," *Energies*, vol. 16, no. 2, p. 635, Jan. 2023.
- [73] S. Sengupta, A. Chowdhary, A. Sabur, A. Alshamrani, D. Huang, and S. Kambhampati, "A survey of moving target defenses for network security," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 1909–1941, 3rd Quart., 2020.
- [74] N. Khan, R. B. Salleh, A. Koubaa, Z. Khan, M. K. Khan, and I. Ali, "Data plane failure and its recovery techniques in SDN: A systematic literature review," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 35, no. 3, pp. 176–201, Mar. 2023.
- [75] M. Mittal, K. Kumar, and S. Behal, "Deep learning approaches for detecting DDoS attacks: A systematic review," *Soft Comput.*, vol. 27, no. 18, pp. 13039–13075, Sep. 2023.
- [76] M. Paliwal, D. Shrimankar, and O. Tembhurne, "Controllers in SDN: A review report," *IEEE Access*, vol. 6, pp. 36256–36270, 2018.
- [77] M. S. Farooq, S. Riaz, and A. Alvi, "Security and privacy issues in software-defined networking (SDN): A systematic literature review," *Electronics*, vol. 12, no. 14, p. 3077, Jul. 2023.
- [78] S. T. Zargar, J. Joshi, and D. Tipper, "A survey of defense mechanisms against distributed denial of service (DDoS) flooding attacks," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 4, pp. 2046–2069, 4th Quart., 2013.
- [79] R. K. C. Chang, "Defending against flooding-based distributed denial-of-service attacks: A tutorial," *IEEE Commun. Mag.*, vol. 40, no. 10, pp. 42–51, Oct. 2002.
- [80] N. Sharmila Kumari, H. S. Vimala, C. N. Pruthvi, and J. Shreyas, "Holistic survey on security in IoT application layer: Attacks, protocols, and applications," *IEEE Access*, vol. 12, pp. 186957–187014, 2024.
- [81] D. Cevallos-Salas, J. Estrada-Jiménez, and D. S. Guzmán, "Application layer security for internet communications: A comprehensive review, challenges, and future trends," *Comput. Electr. Eng.*, vol. 119, Oct. 2024, Art. no. 109498.
- [82] A. Praseed and P. S. Thilagam, "DDoS attacks at the application layer: Challenges and research perspectives for safeguarding web applications," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 661–685, 1st Quart., 2019.
- [83] M. Singh, M. Singh, and S. Kaur, "Issues and challenges in DNS based botnet detection: A survey," *Comput. Secur.*, vol. 86, pp. 28–52, Sep. 2019.
- [84] I. Ali, A. I. A. Ahmed, A. Almogren, M. A. Raza, S. A. Shah, A. Khan, and A. Gani, "Systematic literature review on IoT-based botnet attack," *IEEE Access*, vol. 8, pp. 212220–212232, 2020.
- [85] M. Gelgi, Y. Guan, S. Arunachala, M. Samba Siva Rao, and N. Dragoni, "Systematic literature review of IoT botnet DDOS attacks and evaluation of detection techniques," *Sensors*, vol. 24, no. 11, p. 3571, Jun. 2024.
- [86] N. Hoque, D. K. Bhattacharyya, and J. K. Kalita, "Botnet in DDoS attacks: Trends and challenges," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2242–2270, 4th Quart., 2015.
- [87] R. R. Nuiaa, S. Manickam, and A. H. Alsaedi, "Distributed reflection denial of service attack: A critical review," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 11, no. 6, p. 5327, Dec. 2021.
- [88] B. Banerjee and S. Neogy, "A brief overview of security attacks and protocols in MANET," in *Proc. IEEE 18th India Council Int. Conf. (INDICON)*, Dec. 2021, pp. 1–6.
- [89] B. Al-Duwairi, E. Al-Quraan, and Y. AbdelQader, "ISDSN: Mitigating SYN flood attacks in software defined networks," *J. Netw. Syst. Manage.*, vol. 28, no. 4, pp. 1366–1390, Oct. 2020.
- [90] P. Goldschmidt and J. Kucera, "Defense against SYN flood DoS attacks using network-based mitigation techniques," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manage. (IM)*, May 2021, pp. 772–777.
- [91] D. Nashat and F. A. Hussain, "Multifractal detrended fluctuation analysis based detection for SYN flooding attack," *Comput. Secur.*, vol. 107, Aug. 2021, Art. no. 102315.
- [92] C.-H. Yang, J.-P. Wu, F.-Y. Lee, T.-Y. Lin, and M.-H. Tsai, "Detection and mitigation of SYN flooding attacks through SYN/ACK packets and black/white lists," *Sensors*, vol. 23, no. 8, p. 3817, Apr. 2023.
- [93] Z. Alarnaout, N. Mostafa, S. Alabed, W. H. F. Aly, and A. Shdefat, "RAPT: A robust attack path tracing algorithm to mitigate SYN-flood DDoS cyberattacks," *Sensors*, vol. 23, no. 1, p. 102, Dec. 2022.
- [94] M. Dimolianis, A. Pavlidis, and V. Maglaris, "SYN flood attack detection and mitigation using machine learning traffic classification and programmable data plane filtering," in *Proc. 24th Conf. Innov. Clouds, Internet Netw. Workshops (ICIN)*, Mar. 2021, pp. 126–133.
- [95] Y. Jia, F. Zhong, A. Alrawais, B. Gong, and X. Cheng, "FlowGuard: An intelligent edge defense mechanism against IoT DDoS attacks," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9552–9562, Oct. 2020.
- [96] S. Muthurajkumar, A. Geetha, S. Aravind, and H. B. Meharajnis, "UDP flooding attack detection using entropy in software-defined networking," in *Proc. Int. Conf. Commun. Comput. Technol. (ICCCCT)*. Cham, Switzerland: Springer, Sep. 2022, pp. 549–560.
- [97] Y. Wang, J. Ding, T. Zhang, Y. Xiao, and X. Hei, "From replay to regeneration: Recovery of UDP flood network attack scenario based on SDN," *Mathematics*, vol. 11, no. 8, p. 1897, Apr. 2023.
- [98] S. Javanmardi, M. Ghahramani, M. Shojafar, M. Alazab, and A. M. Caruso, "M-RL: A mobility and impersonation-aware IDS for DDoS UDP flooding attacks in IoT-fog networks," *Comput. Secur.*, vol. 140, May 2024, Art. no. 103778.
- [99] Z.-Y. Shen, M.-W. Su, Y.-Z. Cai, and M.-H. Tasi, "Mitigating SYN flooding and UDP flooding in P4-based SDN," in *Proc. 22nd Asia-Pacific Netw. Oper. Manage. Symp. (APNOMS)*, Sep. 2021, pp. 374–377.
- [100] A. Jaszcz and D. Połap, "AIMM: Artificial intelligence merged methods for flood DDoS attacks detection," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 10, pp. 8090–8101, Nov. 2022.
- [101] R. U. Rasool, H. Wang, U. Ashraf, K. Ahmed, Z. Anwar, and W. Rafique, "A survey of link flooding attacks in software defined network ecosystems," *J. Netw. Comput. Appl.*, vol. 172, Dec. 2020, Art. no. 102803.
- [102] A. Dhingra and V. Sindhu, "A review of DIS-flooding attacks in RPL based IoT network," in *Proc. Int. Conf. Commun., Comput. Internet Things (IC3IoT)*, Mar. 2022, pp. 1–6.
- [103] S. A. Varma and Karri. G. Reddy, "A review of DDoS attacks and its countermeasures in cloud computing," in *Proc. 5th Int. Conf. Inf. Syst. Comput. Netw. (ISCON)*, Oct. 2021, pp. 1–6.
- [104] I. Ko, D. Chambers, and E. Barrett, "Adaptable feature-selecting and threshold-moving complete autoencoder for DDoS flood attack mitigation," *J. Inf. Secur. Appl.*, vol. 55, Dec. 2020, Art. no. 102647.
- [105] J. D. Gadze, A. A. Bamfo-Asante, J. O. Agyemang, H. Nunoo-Mensah, and K. A.-B. Opare, "An investigation into the application of deep learning in the detection and mitigation of DDOS attack on SDN controllers," *Technologies*, vol. 9, no. 1, p. 14, Feb. 2021.

- [106] S.-H. Lee, Y.-L. Shiu, C.-H. Cheng, Y.-H. Li, and Y.-F. Huang, "Detection and prevention of DDoS attacks on the IoT," *Appl. Sci.*, vol. 12, no. 23, p. 12407, Dec. 2022.
- [107] N. N. Tuan, P. H. Hung, N. D. Nghia, N. V. Tho, T. V. Phan, and N. H. Thanh, "A DDoS attack mitigation scheme in ISP networks using machine learning based on SDN," *Electronics*, vol. 9, no. 3, p. 413, Feb. 2020.
- [108] E. M. Onyema, M. A. Kumar, S. Balasubramanian, S. Bharany, A. U. Rehman, E. T. Eldin, and M. Shafiq, "A security policy protocol for detection and prevention of internet control message protocol attacks in software defined networks," *Sustainability*, vol. 14, no. 19, p. 11950, Sep. 2022.
- [109] A. Hirsi, L. Audah, A. Salh, M. A. Alhartomi, and S. Ahmed, "Detecting DDoS threats using supervised machine learning for traffic classification in software defined networking," *IEEE Access*, vol. 12, pp. 166675–166702, 2024.
- [110] K. M. Sudar, P. Deepalakshmi, A. Singh, and P. N. Srinivasu, "TFAD: TCP flooding attack detection in software-defined networking using proxy-based and machine learning-based mechanisms," *Cluster Comput.*, vol. 26, no. 2, pp. 1461–1477, Apr. 2023.
- [111] X. Zhang, L. Chen, and J. Bai, "SYN flood attack detection and defense method based on extended Berkeley packet filter," in *Proc. Int. Conf. Natural Comput., Fuzzy Syst. Knowl. Discovery*. Cham, Switzerland: Springer, Jan. 2022, pp. 1416–1427.
- [112] M. A. Al-Naeem, "Prediction of re-occurrences of spoofed ACK packets sent to deflate a target wireless sensor network node by DDOS," *IEEE Access*, vol. 9, pp. 87070–87078, 2021.
- [113] S. Q. A. Shah, F. Z. Khan, and M. Ahmad, "Mitigating TCP SYN flooding based EDOS attack in cloud computing environment using binomial distribution in SDN," *Comput. Commun.*, vol. 182, pp. 198–211, Jan. 2022.
- [114] A. Hirsi, L. Audah, A. Salh, N. M. Sahar, S. Ahmed, and M. A. Alhartomi, "DDoS anomaly detection in software-defined networks: An evaluation of machine learning techniques for traffic classification and prediction," in *Proc. Int. Conf. Future Technol. Smart Soc. (ICFTSS)*, Aug. 2024, pp. 100–105.
- [115] I. Nedyalkov and G. Georgiev, "Application of IP network modeling platforms for cyber-attack research," in *Congress on Intelligent Systems*. Cham, Switzerland: Springer, 2023, pp. 229–245.
- [116] C.-L. Chen and J. L. Lai, "An experimental detection of distributed denial of service attack in CDX 3 platform based on snort," *Sensors*, vol. 23, no. 13, p. 6139, Jul. 2023.
- [117] K. B. Adedeji, A. M. Abu-Mahfouz, and A. M. Kurien, "DDoS attack and detection methods in internet-enabled networks: Concept, research perspectives, and challenges," *J. Sensor Actuator Netw.*, vol. 12, no. 4, p. 51, Jul. 2023.
- [118] M. A. Helmiawan, E. Julian, Y. Cahyan, and A. Saeppani, "Experimental evaluation of security monitoring and notification on network intrusion detection system for server security," in *Proc. 9th Int. Conf. Cyber IT Service Manage. (CITSM)*, Sep. 2021, pp. 1–6.
- [119] A. Abdollahi and M. Fathi, "An intrusion detection system on ping of death attacks in IoT networks," *Wireless Pers. Commun.*, vol. 112, no. 4, pp. 2057–2070, Jun. 2020.
- [120] O. M. Almorabea, T. J. S. Khanzada, M. A. Aslam, F. A. Hendi, and A. M. Almorabea, "IoT network-based intrusion detection framework: A solution to process ping floods originating from embedded devices," *IEEE Access*, vol. 11, pp. 119118–119145, 2023.
- [121] A. R. Hakim, J. Rinaldi, and M. Y. B. Setiadji, "Design and implementation of NIDS notification system using WhatsApp and telegram," in *Proc. 8th Int. Conf. Inf. Commun. Technol. (ICOICT)*, Jun. 2020, pp. 1–4.
- [122] N. K. Singh and S. K. B. J., "Detection and prevention of UDP protocol exploiting and Smurf attack in WSN using sequential probability ratio test algorithm," in *Proc. Int. Conf. Data Sci. Netw. Secur. (ICDSNS)*, Jul. 2023, pp. 1–6.
- [123] S. Shadrooh and K. Nøravåg, "SMoTeF: Smurf money laundering detection using temporal order and flow analysis," *Int. J. Speech Technol.*, vol. 54, nos. 15–16, pp. 7461–7478, Aug. 2024.
- [124] K. P. Vijayakumar, K. Pradeep, A. Balasundaram, and M. R. Prusty, "Enhanced cyber attack detection process for Internet of Health Things (IoHT) devices using deep neural network," *Processes*, vol. 11, no. 4, p. 1072, Apr. 2023.
- [125] N. Ravi, S. M. Shalinie, and D. Danyson Jose Theres, "BALANCE: Link flooding attack detection and mitigation via hybrid-SDN," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 3, pp. 1715–1729, Sep. 2020.
- [126] S. Usman, I. Winarno, and A. Sudarsono, "Implementation of SDN-based IDS to protect virtualization server against HTTP DoS attacks," in *Proc. Int. Electron. Symp. (IES)*, Sep. 2020, pp. 195–198.
- [127] J. I. C. Eunaicy and S. Suguna, "Web attack detection using deep learning models," *Mater. Today, Proc.*, vol. 62, pp. 4806–4813, Jan. 2022.
- [128] A. Dhanapal and P. Nithyanandam, "An OpenStack based cloud testbed framework for evaluating HTTP flooding attacks," *Wireless Netw.*, vol. 27, no. 8, pp. 5491–5501, Nov. 2021.
- [129] E. Chatzoglou, V. Kouliaridis, G. Kambourakis, G. Karopoulos, and S. Gritzalis, "A hands-on gaze on HTTP/3 security through the lens of HTTP/2 and a public dataset," *Comput. Secur.*, vol. 125, Feb. 2023, Art. no. 103051.
- [130] P. V. Razumov, O. A. Safaryan, I. A. Smirnov, V. M. Porksheyany, N. V. Boldyrikhin, D. A. Korochentsev, L. V. Cherkasova, and S. A. Osikov, "Developing of algorithm of HTTP FLOOD DDoS protection," in *Proc. 3rd Int. Conf. Comput. Appl. Inf. Secur. (ICCAIS)*, Mar. 2020, pp. 1–6.
- [131] L. D. Tsohdjou, S. Pierre, and A. Quintero, "An online entropy-based DDoS flooding attack detection system with dynamic threshold," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 2, pp. 1679–1689, Jun. 2022.
- [132] R. Mohammadi, C. Lal, and M. Conti, "HTTPScout: A machine learning based countermeasure for HTTP flood attacks in SDN," *Int. J. Inf. Secur.*, vol. 22, no. 2, pp. 367–379, Apr. 2023.
- [133] T. A. S. Srinivas and S. S. Manivannan, "Prevention of hello flood attack in IoT using combination of deep learning with improved rider optimization algorithm," *Comput. Commun.*, vol. 163, pp. 162–175, Nov. 2020.
- [134] R. Sanjeetha, K. N. A. Shastry, H. R. Chetan, and A. Kanavalli, "Mitigating HTTP GET FLOOD DDoS attack using an SDN controller," in *Proc. Int. Conf. Recent Trends Electron., Inf., Commun. Technol. (RTEICT)*, Nov. 2020, pp. 6–10.
- [135] A. S. M. Rizvi, J. Mirković, J. Heidemann, W. Hardaker, and R. Story, "Defending root DNS servers against DDoS using layered defenses," in *Proc. 15th Int. Conf. Commun. Syst. Netw. (COMSNETS)*, Jan. 2023, pp. 513–521.
- [136] V. Gupta, A. Kochar, S. Saharan, and R. Kulshrestha, "DNS amplification based DDoS attacks in SDN environment: Detection and mitigation," in *Proc. IEEE 4th Int. Conf. Comput. Commun. Syst. (ICCCS)*, Feb. 2019, pp. 473–478.
- [137] R. F. Fouladi, O. Ermiş, and E. Anarim, "A DDoS attack detection and countermeasure scheme based on DWT and auto-encoder neural network for SDN," *Comput. Netw.*, vol. 214, Sep. 2022, Art. no. 109140.
- [138] Z. A. El Houda, L. Khoukhi, and A. S. Hafid, "Bringing intelligence to software defined networks: Mitigating DDoS attacks," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 4, pp. 2523–2535, Dec. 2020.
- [139] B. Savenko, S. Lysenko, K. Bobrovnikova, O. Savenko, and G. Markowsky, "Detection DNS tunneling botnets," in *Proc. 11th IEEE Int. Conf. Intell. Data Acquisition Adv. Comput. Syst., Technol. Appl. (IDAACS)*, vol. 1, Sep. 2021, pp. 64–69.
- [140] Y. Dai, T. Huang, and S. Wang, "DAmpADF: A framework for DNS amplification attack defense based on Bloom filters and NAmKeeper," *Comput. Secur.*, vol. 139, Apr. 2024, Art. no. 103718.
- [141] T.-K. Luong, T.-D. Tran, and G.-T. Le, "DDoS attack detection and defense in SDN based on machine learning," in *Proc. 7th NAFOSTED Conf. Inf. Comput. Sci. (NICS)*, Nov. 2020, pp. 31–35.
- [142] R. Yazdani, R. van Rijswijk-Deij, M. Jonker, and A. Sperotto, "A matter of degree: Characterizing the amplification power of open DNS resolvers," in *Proc. Int. Conf. Passive Act. Netw. Meas.* Cham, Switzerland: Springer, Jan. 2022, pp. 293–318.
- [143] M. M. Nesary and A. Aydeger, "VDNS: Securing DNS from amplification attacks," in *Proc. IEEE Int. Black Sea Conf. Commun. Netw. (BlackSeaCom)*, Jun. 2022, pp. 102–106.
- [144] M. Sikora, A. Krivulcik, R. Fudjak, and P. Blazek, "Design of advanced slow denial of service attack generator," in *Proc. 12th Int. Congr. Ultra Modern Telecommun. Control Syst. Workshops (ICUMT)*, Oct. 2020, pp. 99–104.
- [145] P. Treglia, "PiSecurityCheck: Server security check in a hand," in *Proc. 47th MIPRO ICT Electron. Conv. (MIPRO)*, May 2024, pp. 1549–1554.

- [146] S. Sood and N. Hubballi, "SlowTrack: Detecting slow rate denial of service attacks against HTTP with behavioral parameters," *J. Supercomput.*, vol. 80, no. 2, pp. 1788–1817, Jan. 2024.
- [147] P. Oktivasari, A. R. Zain, M. Agustin, A. Kurniawan, F. A. Murad, and M. F. Anshor, "Analysis of effectiveness of iptables on web server from slowloris attack," in *Proc. 5th Int. Conf. Comput. Informat. Eng. (IC2IE)*, Sep. 2022, pp. 215–219.
- [148] V. D. S. Faria, J. A. Gonçalves, C. A. M. D. Silva, G. D. B. Vieira, and D. M. Mascarenhas, "SDToW: A slowloris detecting tool for WMNs," *Information*, vol. 11, no. 12, p. 544, Nov. 2020.
- [149] V. D. M. Rios, P. R. M. Inácio, D. Magoni, and M. M. Freire, "Detection and mitigation of low-rate denial-of-service attacks: A survey," *IEEE Access*, vol. 10, pp. 76648–76668, 2022.
- [150] B. Bala and S. Behal, "AI techniques for IoT-based DDoS attack detection: Taxonomies, comprehensive review and research challenges," *Comput. Sci. Rev.*, vol. 52, May 2024, Art. no. 100631.
- [151] S. V. J. Rani, I. I. Ioannou, P. Nagaradjane, C. Christophorou, V. Vassiliou, H. Yarramsetti, S. Shridhar, L. M. Balaji, and A. Pitsillides, "A novel deep hierarchical machine learning approach for identification of known and unknown multiple security attacks in a D2D communications network," *IEEE Access*, vol. 11, pp. 95161–95194, 2023.
- [152] R. Bocu and M. Iavich, "Enhanced detection of low-rate DDoS attack patterns using machine learning models," *J. Netw. Comput. Appl.*, vol. 227, Jul. 2024, Art. no. 103903.
- [153] O. Valikhani, "UAV networks DoS attacks detection using artificial intelligence based on weighted machine learning," *Results Control Optim.*, vol. 16, Sep. 2024, Art. no. 100457.
- [154] M. A. O. Rabah, H. Drid, Y. Medjadba, and M. Rahouti, "Detection and mitigation of distributed denial of service attacks using ensemble learning and honeypots in a novel SDN-UAV network architecture," *IEEE Access*, vol. 12, pp. 128929–128940, 2024.
- [155] M. Iavich, L. Mirtskhulava, G. Iashvili, and L. Globa, "5G laboratory for checking machine learning algorithms," in *Proc. IEEE Int. Conf. Inf. Telecommun. Technol. Radio Electron. (UkrMiCo)*, Nov. 2021, pp. 43–46.
- [156] C. Dantas, P. Pessoa, J. Ferreira, P. Maciel, and J. Dantas, "DDoS detection based on hardware performance counters selection," in *Proc. IEEE Int. Conf. Enabling Technol., Infrastructure Collaborative Enterprises (WETICE)*, Dec. 2023, pp. 1–6.
- [157] A. Fadhlillah, N. Karna, and A. Irawan, "IDS performance analysis using anomaly-based detection method for DOS attack," in *Proc. IEEE Int. Conf. Internet Things Intell. Syst. (IoTaIS)*, Jan. 2021, pp. 18–22.
- [158] Y.-X. Cai, T.-C. Chang, and C.-C. Wang, "A case study of detecting HULK-based DDoS traffic with random forest classifier," in *Proc. Int. Conf. Consum. Electron. Taiwan (ICCE-Taiwan)*, Jul. 2023, pp. 609–610.
- [159] R. Aliyev, "DDoS simulation: Empowering targets through simulated attacks," in *Proc. IEEE 17th Int. Conf. Appl. Inf. Commun. Technol. (AICT)*, Oct. 2023, pp. 1–4.
- [160] A. Sinha, S. Kumar, P. Mishra, U. Garg, and A. Aggarwal, "A review and case study on attacking and security tools at application-layer of IoT," in *Proc. Int. Conf. Internet Things Connected Technol.* Cham, Switzerland: Springer, Jan. 2021, pp. 51–63.
- [161] A. Mahmoud, T. Tambe, T. Aloui, D. Brooks, and G.-Y. Wei, "Golden-Eye: A platform for evaluating emerging numerical data formats in DNN accelerators," in *Proc. 52nd Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2022, pp. 206–214.
- [162] D. Krishnan, "Detection of denial-of-service attacks using stacked LSTM networks," in *Proceedings of Data Analytics and Management*, vol. 2. Cham, Switzerland: Springer, Nov. 2021, pp. 229–239.
- [163] C. Benzaïd, M. Boukhalfa, and T. Taleb, "Robust self-protection against application-layer (D)DoS attacks in SDN environment," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, May 2020, pp. 1–6.
- [164] D. Tang, Y. Yan, S. Zhang, J. Chen, and Z. Qin, "Performance and features: Mitigating the low-rate TCP-targeted DoS attack via SDN," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 428–444, Jan. 2022.
- [165] V. A. Kumar and D. Das, "Data sequence signal manipulation in multipath TCP (MPTCP): The vulnerability, attack and its detection," *Comput. Secur.*, vol. 103, Apr. 2021, Art. no. 102180.
- [166] A. Rahman, G. Mustafa, A. Q. Khan, M. Abid, and M. H. Durad, "Launch of denial of service attacks on the modbus/TCP protocol and development of its protection mechanisms," *Int. J. Crit. Infrastruct. Protection*, vol. 39, Dec. 2022, Art. no. 100568.
- [167] B. Hartpence and A. Kwasinski, "Combating TCP port scan attacks using sequential neural networks," in *Proc. Int. Conf. Comput., Netw. Commun. (ICNC)*, Feb. 2020, pp. 256–260.
- [168] X. Feng, Q. Li, K. Sun, C. Fu, and K. Xu, "Off-path TCP hijacking attacks via the side channel of downgraded IPID," *IEEE/ACM Trans. Netw.*, vol. 30, no. 1, pp. 409–422, Feb. 2022.
- [169] L. F. Eliyan and R. D. Pietro, "DeMi: A solution to detect and mitigate DoS attacks in SDN," *IEEE Access*, vol. 11, pp. 82477–82495, 2023.
- [170] J. Miguel-Alonso, "A research review of OpenFlow for datacenter networking," *IEEE Access*, vol. 11, pp. 770–786, 2023.
- [171] M. Kobayashi, S. Seetharaman, G. Parulkar, G. Appenzeller, J. Little, J. van Reijndam, P. Weissmann, and N. McKeown, "Maturing of OpenFlow and software-defined networking through deployments," *Comput. Netw.*, vol. 61, pp. 151–175, Mar. 2014.
- [172] A. Galal and X. Hesselbach, "Nano-networks communication architecture: Modeling and functions," *Nano Commun. Netw.*, vol. 17, pp. 45–62, Sep. 2018.
- [173] F. Pakzad, M. Portmann, W. L. Tan, and J. Indulska, "Efficient topology discovery in OpenFlow-based software defined networks," *Comput. Commun.*, vol. 77, pp. 52–61, Mar. 2016.
- [174] I. Z. Bholebawa and U. D. Dalal, "Performance analysis of SDN/OpenFlow controllers: POX versus floodlight," *Wireless Pers. Commun.*, vol. 98, no. 2, pp. 1679–1699, Jan. 2018.
- [175] J. Benabbou, K. Elbaamrani, and N. Idbouffer, "Security in OpenFlow-based SDN, opportunities and challenges," *Photonic Netw. Commun.*, vol. 37, no. 1, pp. 1–23, Feb. 2019.
- [176] J. Cao, M. Xu, Q. Li, K. Sun, and Y. Yang, "The LOFT attack: Overflowing SDN flow tables at a low rate," *IEEE/ACM Trans. Netw.*, vol. 31, no. 3, pp. 1416–1431, Jun. 2023.
- [177] Y. Liu, Y. Wang, and H. Feng, "POAGuard: A defense mechanism against preemptive table overflow attack in software-defined networks," *IEEE Access*, vol. 11, pp. 123659–123676, 2023.
- [178] D. Tang, Z. Zheng, C. Yin, B. Xiong, Z. Qin, and Q. Yang, "FTODEfender: An efficient flow table overflow attacks defending system in SDN," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121460.
- [179] D. Tang, C. Gao, W. Liang, J. Zhang, and K. Li, "FTMaster: A detection and mitigation system of low-rate flow table overflow attacks via SDN," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 4, pp. 5073–5084, Apr. 2023.
- [180] D. Tang, Z. Zheng, K. Li, C. Yin, W. Liang, and J. Zhang, "FTOP: An efficient flow table overflow preventing system for switches in SDN," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 3, pp. 2524–2536, May 2024.
- [181] A. Mudgal, A. Verma, M. Singh, K. S. Sahoo, E. Elmroth, and M. Bhuyan, "FloRa: Flow table low-rate overflow reconnaissance and detection in SDN," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6670–6683, Dec. 2024.
- [182] D. Tang, D. Zhang, Z. Qin, Q. Yang, and S. Xiao, "SFTO-guard: Real-time detection and mitigation system for slow-rate flow table overflow attacks," *J. Netw. Comput. Appl.*, vol. 213, Apr. 2023, Art. no. 103597.
- [183] D. Tang, Y. Yan, C. Gao, W. Liang, and W. Jin, "LIRFT: Mitigate the low-rate data plane DDoS attack with learning-to-rank enabled flow tables," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 3143–3157, 2023.
- [184] M. Soylu, L. Guillen, S. Izumi, T. Abe, and T. Suganuma, "NFV-GUARD: Mitigating flow table-overflow attacks in SDN using NFV," in *Proc. IEEE 7th Int. Conf. Netw. Softwarization (NetSoft)*, Jun. 2021, pp. 263–267.
- [185] S. K. Noh and M. Park, "HSST: Table-overflow attack defender with historical statistics based dynamic timeout in software defined networks," *Appl. Sci.*, vol. 13, no. 22, p. 12232, Nov. 2023.
- [186] N. Priyanka, T. R. Reshmi, and K. Murugan, "CEOF: Enhanced clustering-based entries optimization scheme to prevent flow table overflow," *Wireless Netw.*, vol. 28, no. 1, pp. 69–83, Jan. 2022.
- [187] S. Asgari and B. Akbari, "DDoS attack detection in OpenFlow based networks," in *Proc. 27th Int. Comput. Conf., Comput. Soc. Iran (CSICC)*, Feb. 2022, pp. 1–7.
- [188] D. Hu, P. Hong, and Y. Chen, "FADM: DDoS flooding attack detection and mitigation system in software-defined networking," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, Dec. 2017, pp. 1–7.
- [189] M. I. Kareem and M. N. Jasim, "Entropy-based distributed denial of service attack detection in software-defined networking," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 27, no. 3, p. 1542, Sep. 2022.

- [190] M. Erel-Özçevik and B. Canberk, "OFaaS: OpenFlow switch as a service for multi tenant slicing in SD-CDN," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 1, pp. 362–373, Mar. 2021.
- [191] A. Mishra, B. B. Gupta, D. Perakovic, S. Yamaguchi, and C.-H. Hsu, "Entropy based defensive mechanism against DDoS attack in SDN-cloud enabled online social networks," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2021, pp. 1–6.
- [192] F. Shoaib, Y.-W. Chow, and E. Vlahu-Gjorgjevska, "Preventing timing side-channel attacks in software-defined networks," in *Proc. IEEE Asia-Pacific Conf. Comput. Sci. Data Eng. (CSDE)*, Dec. 2021, pp. 1–6.
- [193] X. Huang, P. Shi, Y. Liu, and F. Xu, "Towards trusted and efficient SDN topology discovery: A lightweight topology verification scheme," *Comput. Netw.*, vol. 170, Apr. 2020, Art. no. 107119.
- [194] F. Hauser, M. Schmidt, M. Häberle, and M. Menth, "P4-MACsec: Dynamic topology monitoring and data layer protection with MACsec in P4-based SDN," *IEEE Access*, vol. 8, pp. 58845–58858, 2020.
- [195] S. Popic, M. Vuleta, P. Cvjetkovic, and B. M. Todorovic, "Secure topology detection in software-defined networking with network configuration protocol and link layer discovery protocol," in *Proc. Int. Symp. Ind. Electron. Appl. (INDEL)*, Nov. 2020, pp. 1–5.
- [196] I. A. Salti and N. Zhang, "LINK-GUARD: An effective and scalable security framework for link discovery in SDN networks," *IEEE Access*, vol. 10, pp. 130233–130252, 2022.
- [197] S. Soltani, M. Shojafar, H. Mostafaei, and R. Tafazolli, "Real-time link verification in software-defined networks," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 3, pp. 3596–3611, Jan. 2023.
- [198] K. Joseph, O. S. Eyobu, P. Kasyoka, and T. J. Oyana, "A link fabrication attack mitigation approach (LiFAMA) for software defined networks," *Electronics*, vol. 11, no. 10, p. 1581, May 2022.
- [199] M. W. Hussain, M. S. Khan, K. H. K. Reddy, and D. S. Roy, "Extended indirect controller-legacy switch forwarding for link discovery in hybrid multi-controller SDN," *Comput. Commun.*, vol. 189, pp. 148–157, May 2022.
- [200] S. Bagheri and A. Shameli-Sendi, "Dynamic firewall decomposition and composition in the cloud," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3526–3539, 2020.
- [201] A. Praseed and P. S. Thilagam, "HTTP request pattern based signatures for early application layer DDoS detection: A firewall agnostic approach," *J. Inf. Secur. Appl.*, vol. 65, Mar. 2022, Art. no. 103090.
- [202] S. Kim, S. Yoon, J. Naranituya, and H. Lim, "Secure collecting, optimizing, and deploying of firewall rules in software-defined networks," *IEEE Access*, vol. 8, pp. 15166–15177, 2020.
- [203] J. J. Praise, R. J. S. Raj, and J. V. B. Benifa, "Development of reinforcement learning and pattern matching (RLPM) based firewall for secured cloud infrastructure," *Wireless Pers. Commun.*, vol. 115, no. 2, pp. 993–1018, Nov. 2020.
- [204] K. Hayawi, Z. Tabelsi, S. Zeidan, and M. M. Masud, "Thwarting ICMP low-rate attacks against firewalls while minimizing legitimate traffic loss," *IEEE Access*, vol. 8, pp. 78029–78043, 2020.
- [205] A. Coscia, V. Dentamaro, S. Galantucci, A. Maci, and G. Pirlo, "An innovative two-stage algorithm to optimize firewall rule ordering," *Comput. Secur.*, vol. 134, Nov. 2023, Art. no. 103423.
- [206] S. Prabakaran, R. Ramar, I. Hussain, B. P. Kavin, S. S. Alshamrani, A. S. AlGhamdi, and A. Alshehri, "Predicting attack pattern via machine learning by exploiting stateful firewall as virtual network function in an SDN network," *Sensors*, vol. 22, no. 3, p. 709, Jan. 2022.
- [207] K. Aswal, A. Rajmohan, A. TRC, S. Mukund, V. J. Panicker, and J. P. Dhivvy, "Kavach: A machine learning based approach for enhancing the attack detection capability of firewalls," in *Proc. 12th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Jul. 2021, pp. 1–5.
- [208] F. M. Alotaibi and V. G. Vassilakis, "Toward an SDN-based web application firewall: Defending against SQL injection attacks," *Future Internet*, vol. 15, no. 5, p. 170, Apr. 2023.
- [209] Y. Shin, D. Koo, and J. Hur, "Inferring firewall rules by cache side-channel analysis in network function virtualization," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, Jul. 2020, pp. 1798–1807.
- [210] C. Togay, A. Kasif, C. Catal, and B. Tekinerdogan, "A firewall policy anomaly detection framework for reliable network security," *IEEE Trans. Rel.*, vol. 71, no. 1, pp. 339–347, Mar. 2022.
- [211] M. R. Amal and P. Venkadesh, "H-DOCTOR: Honeypot based firewall tuning for attack prevention," *Meas., Sensors*, vol. 25, Feb. 2023, Art. no. 100664.
- [212] D. Bruschi, A. Di Pasquale, S. Ghilardi, A. Lanzi, and E. Pagani, "A formal verification of ArpON—A tool for avoiding man-in-the-middle attacks in Ethernet networks," *IEEE Trans. Depend. Secure Comput.*, vol. 19, no. 6, pp. 4082–4098, Nov. 2022.
- [213] H. Y. Ibrahim, P. M. Ismael, A. A. Albabawat, and A. B. Al-Khalil, "A secure mechanism to prevent ARP spoofing and ARP broadcasting in SDN," in *Proc. Int. Conf. Comput. Sci. Softw. Eng. (CSASE)*, Apr. 2020, pp. 13–19.
- [214] S. M. Morsy and D. Nashat, "D-ARP: An efficient scheme to detect and prevent ARP spoofing," *IEEE Access*, vol. 10, pp. 49142–49153, 2022.
- [215] Y. Zhao, R. Guo, and P. Lv, "ARP spoofing analysis and prevention," in *Proc. 5th Int. Conf. Smart Grid Electr. Autom. (ICSGEA)*, Jun. 2020, pp. 572–575.
- [216] D. Majumdar and R. R. Chintala, "Shadow APR-ing for APR poisoning detection," in *Proc. 9th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, vol. 1, Mar. 2023, pp. 128–131.
- [217] Y. P. Atmojo, I. M. D. Susila, I. B. Suradarma, L. Yuningsih, E. S. Rini, and D. P. Hostyadi, "A new approach for ARP poisoning attack detection based on network traffic analysis," in *Proc. 4th Int. Seminar Res. Inf. Technol. Intell. Syst. (ISRITI)*, Dec. 2021, pp. 18–23.
- [218] N. Ahuja, G. Singal, D. Mukhopadhyay, and A. Nehra, "Ascertain the efficient machine learning approach to detect different ARP attacks," *Comput. Electr. Eng.*, vol. 99, Apr. 2022, Art. no. 107757.
- [219] D. R. Thomas, P. V. W. Nancy, G. Sowmiya, T. Adhithya, and V. Peroumal, "Detection and prevention of poisoning targets with ARP cache using scapy," in *Proc. Int. Conf. Intell. Innov. Technol. Comput., Electr. Electron. (IITCEE)*, Jan. 2024, pp. 1–6.
- [220] F. Mvuh, V. K. Tchendji, C. T. Djamegni, A. H. Anwar, D. K. Tosh, and C. Kamhoua, "GaTeBaSep: Game theory-based security protocol against ARP spoofing attacks in software-defined networks," *Int. J. Inf. Secur.*, vol. 23, no. 1, pp. 373–387, Feb. 2024.
- [221] C. Toprak, C. Turker, and A. T. Erman, "Detection of DHCP starvation attacks in software defined networks: A case study," in *Proc. 3rd Int. Conf. Comput. Sci. Eng. (UBMK)*, Sep. 2018, pp. 636–641.
- [222] H. U. Ishiaq, A. A. Bhutta, and A. N. Mian, "DHCP DoS and starvation attacks on SDN controllers and their mitigation," *J. Comput. Virol. Hacking Techn.*, vol. 20, no. 1, pp. 15–25, May 2023.
- [223] A. AbdulGhaffar, S. K. Paul, and A. Matrawy, "An analysis of DHCP vulnerabilities, attacks, and countermeasures," in *Proc. Biennial Symp. Commun. (BSC)*, Jul. 2023, pp. 119–124.
- [224] A. Jony and M. N. Islam, "An effective technique to automatically detect and neutralize rogue DHCP server," in *Proc. Int. Conf. Inf. Commun. Technol. Sustain. Develop. (ICICT4SD)*, Sep. 2023, pp. 244–248.
- [225] A. Jony, A. S. M. Miah, and M. N. Islam, "An effective method to detect DHCP starvation attack using port scanning," in *Proc. Int. Conf. Next-Generat. Comput., IoT Mach. Learn. (NCIM)*, Jun. 2023, pp. 1–6.
- [226] A. Jony and M. N. Islam, "A new technique to mitigate DHCPv6 starvation attack and authenticate clients using DUID," in *Proc. 3rd Int. Conf. Advancement Electr. Electron. Eng. (ICAEEE)*, Apr. 2024, pp. 1–6.
- [227] P. Shrestha and T. D. Sherpa, "Dynamic host configuration protocol attacks and its detection using Python scripts," in *Proc. Int. Conf. Artif. Intell. Knowl. Discovery Concurrent Eng. (ICECONF)*, Jan. 2023, pp. 1–5.
- [228] S. Syed, F. Khuhawar, and S. Talpur, "Machine learning approach for classification of DHCP DoS attacks in NIDS," in *Proc. IEEE 18th Int. Conf. Smart Communities, Improving Quality Life Using ICT, IoT AI (HONET)*, Oct. 2021, pp. 143–146.
- [229] M. S. Tok and M. Demirci, "Security analysis of SDN controller-based DHCP services and attack mitigation with DHCPguard," *Comput. Secur.*, vol. 109, Oct. 2021, Art. no. 102394.
- [230] F. Mvuh, V. K. Tchendji, C. T. Djamegni, A. H. Anwar, D. K. Tosh, and C. Kamhoua, "Countering ARP spoofing attacks in software-defined networks using a game-theoretic approach," *Comput. Secur.*, vol. 139, Apr. 2024, Art. no. 103696.
- [231] X. Liu and Y. Du, "Towards effective feature selection for IoT botnet attack detection using a genetic algorithm," *Electronics*, vol. 12, no. 5, p. 1260, Mar. 2023.
- [232] M. M. Alani, "BotStop: Packet-based efficient and explainable IoT botnet detection using machine learning," *Comput. Commun.*, vol. 193, pp. 53–62, Sep. 2022.
- [233] Q. A. Al-Haija and M. Al-Dala'Ien, "ELBA-IoT: An ensemble learning model for botnet attack detection in IoT networks," *J. Sensor Actuator Netw.*, vol. 11, no. 1, p. 18, Mar. 2022.

- [234] G. L. Nguyen, B. Dumba, Q.-D. Ngo, H.-V. Le, and T. N. Nguyen, "A collaborative approach to early detection of IoT botnet," *Comput. Electr. Eng.*, vol. 97, Jan. 2022, Art. no. 107525.
- [235] M. Al-Sarem, F. Saeed, E. H. Alkhamash, and N. S. Alghamdi, "An aggregated mutual information based feature selection with machine learning methods for enhancing IoT botnet attack detection," *Sensors*, vol. 22, no. 1, p. 185, Dec. 2021.
- [236] M. Panda, A. A. A. Mousa, and A. E. Hassanien, "Developing an efficient feature engineering and machine learning model for detecting IoT-botnet cyber attacks," *IEEE Access*, vol. 9, pp. 91038–91052, 2021.
- [237] C. Okur and M. Dener, "Detecting IoT botnet attacks using machine learning methods," in *Proc. Int. Conf. Inf. Secur. Cryptol. (ISCTURKEY)*, Dec. 2020, pp. 31–37.
- [238] M. Alqahtani, H. Mathkour, and M. M. Ben Ismail, "IoT botnet attack detection based on optimized extreme gradient boosting and feature selection," *Sensors*, vol. 20, no. 21, p. 6336, Nov. 2020.
- [239] Z. Shao, S. Yuan, and Y. Wang, "Adaptive online learning for IoT botnet detection," *Inf. Sci.*, vol. 574, pp. 84–95, Oct. 2021.
- [240] O. Habibi, M. Chemmakha, and M. Lazaar, "Imbalanced tabular data modelization using CTGAN and machine learning to improve IoT botnet attacks detection," *Eng. Appl. Artif. Intell.*, vol. 118, Feb. 2023, Art. no. 105669.
- [241] S. Sriram, R. Vinayakumar, M. Alazab, and S. KP, "Network flow based IoT botnet attack detection using deep learning," in *Proc. IEEE INFOCOM Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Jul. 2020, pp. 189–194.
- [242] R. Li, Q. Li, Y. Huang, W. Zhang, P. Zhu, and Y. Jiang, "IoTEnsemble: Detection of botnet attacks on Internet of Things," in *Proc. Eur. Symp. Res. Comput. Secur.* Cham, Switzerland: Springer, Jan. 2022, pp. 569–588.
- [243] F. Hussain, S. G. Abbas, I. M. Pires, S. Tanveer, U. U. Fayyaz, N. M. Garcia, G. A. Shah, and F. Shahzad, "A two-fold machine learning approach to prevent and detect IoT botnet attacks," *IEEE Access*, vol. 9, pp. 163412–163430, 2021.
- [244] M. Ge, N. F. Syed, X. Fu, Z. Baig, and A. Robles-Kelly, "Towards a deep learning-driven intrusion detection approach for Internet of Things," *Comput. Netw.*, vol. 186, Feb. 2021, Art. no. 107784.
- [245] A. Affinito, S. Zinno, G. Stanco, A. Botta, and G. Ventre, "The evolution of Mirai botnet scans over a six-year period," *J. Inf. Secur. Appl.*, vol. 79, Dec. 2023, Art. no. 103629.
- [246] X. Zhang, O. Upton, N. L. Beebe, and K.-K.-R. Choo, "IoT botnet forensics: A comprehensive digital forensic case study on Mirai botnet servers," *Forensic Sci. Int., Digit. Invest.*, vol. 32, Apr. 2020, Art. no. 300926.
- [247] M. Nakip and E. Gelenbe, "MIRAI botnet attack detection with auto-associative dense random neural network," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2021, pp. 1–6.
- [248] A. R. S. A. Cruz, R. L. Gomes, and M. P. Fernandez, "An intelligent mechanism to detect cyberattacks of Mirai botnet in IoT networks," in *Proc. 17th Int. Conf. Distrib. Comput. Sensor Syst. (DCOSS)*, Jul. 2021, pp. 236–243.
- [249] V. Vajrobol, B. B. Gupta, A. Gaurav, and H.-M. Chuang, "Adversarial learning for Mirai botnet detection based on long short-term memory and XGBoost," *Int. J. Cognit. Comput. Eng.*, vol. 5, pp. 153–160, Jan. 2024.
- [250] R. G. Azhari, V. Suryani, R. R. Pahlevi, and A. A. Wardana, "The detection of Mirai botnet attack on the Internet of Things (IoT) device using support vector machine (SVM) model," in *Proc. 10th Int. Conf. Inf. Commun. Technol. (ICoICT)*, Aug. 2022, pp. 397–401.
- [251] C. Kelly, N. Pitropakis, S. McKeown, and C. Lambrinoudakis, "Testing and hardening IoT devices against the Mirai botnet," in *Proc. Int. Conf. Cyber Secur. Protection Digit. Services (Cyber Secur.)*, Jun. 2020, pp. 1–8.
- [252] A. Borys, A. Kamruzzaman, H. N. Thakur, J. C. Brickley, M. L. Ali, and K. Thakur, "An evaluation of IoT DDoS cryptojacking malware and mirai botnet," in *Proc. IEEE World AI IoT Congr. (AIoT)*, Jun. 2022, pp. 725–729.
- [253] M. Snehi and A. Bhandari, "Apprehending Mirai botnet philosophy and smart learning models for IoT-DDoS detection," in *Proc. 8th Int. Conf. Comput. Sustain. Global Develop. (INDIACom)*, Mar. 2021, pp. 501–505.
- [254] M. S. Pour, J. Khoury, and E. Bou-Harb, "HoneyComb: A darknet-centric proactive deception technique for curating IoT malware forensic artifacts," in *Proc. NOMS/IEEE/IFIP Netw. Oper. Manage. Symp.*, Apr. 2022, pp. 1–9.
- [255] A. Pham and Y. Nakamura, "Examination of investigation method of malware spreading state," in *Proc. Int. Conf. Ind. Netw. Intell. Syst.* Cham, Switzerland: Springer, Jan. 2021, pp. 128–141.
- [256] S. P. Anh and Y. Nakamura, "A baseline investigation into the evolution and prevalence of Mirai and Hajime utilizing a network telescope," *IEEE Access*, vol. 12, pp. 103789–103809, 2024.
- [257] D. Makiyara and S. Yamaguchi, "A proposal of patrol function by white-hat worm in botnet defense system," in *Proc. IEEE Int. Conf. Consum. Electron.-Asia (ICCE-Asia)*, Nov. 2021, pp. 1–5.
- [258] K. Kobayashi, "Markov chain-based modeling of malicious botnet spread for botnet defense systems," in *Proc. IEEE/SICE Int. Symp. Syst. Integr. (SII)*, Jan. 2023, pp. 1–5.
- [259] M. S. Pour, A. Mangino, K. Friday, M. Rathbun, E. Bou-Harb, F. Iqbal, S. Samtani, J. Crichigno, and N. Ghani, "On data-driven curation, learning, and analysis for inferring evolving Internet-of-Things (IoT) botnets in the wild," *Comput. Secur.*, vol. 91, Apr. 2020, Art. no. 101707.
- [260] D. S. Kim, K. O. Chee, and M. Ge, "A novel graphical security model for evolving cyber attacks in Internet of Things," in *Proc. 50th Annu. IEEE-IFIP Int. Conf. Dependable Syst. Networks-Supplemental Volume (DSN-S)*, Jun. 2020, pp. 57–58.
- [261] B. R. Chandavarkar, "Hardcoded credentials and insecure data transfer in IoT: National and international status," in *Proc. 11th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2020, pp. 1–7.
- [262] B. Susilo and R. F. Sari, "Intrusion detection in IoT networks using deep learning algorithm," *Information*, vol. 11, no. 5, p. 279, May 2020.
- [263] S. Kakati, D. Chouhan, A. Nag, and S. Panja, "Survey on recent malware detection techniques for IoT," in *Pattern Recognition and Data Analysis With Applications*. Cham, Switzerland: Springer, 2022, pp. 647–659.
- [264] M. Baz, "SEHIDS: Self evolving host-based intrusion detection system for IoT networks," *Sensors*, vol. 22, no. 17, p. 6505, Aug. 2022.
- [265] S.-Y. Hwang and J.-N. Kim, "A malware distribution simulator for the verification of network threat prevention tools," *Sensors*, vol. 21, no. 21, p. 6983, Oct. 2021.
- [266] S. Ismail, H. R. Hassen, M. Just, and H. Zantout, "A review of amplification-based distributed denial of service attacks and their mitigation," *Comput. Secur.*, vol. 109, Oct. 2021, Art. no. 102380.
- [267] Ismail, M. I. Mohmand, H. Hussain, A. A. Khan, U. Ullah, M. Zakarya, A. Ahmed, M. Raza, I. U. Rahman, and M. Haleem, "A machine learning-based classification and prediction technique for DDoS attacks," *IEEE Access*, vol. 10, pp. 21443–21454, 2022.
- [268] M. Langer, K. Heine, R. Bermbach, and D. Sibold, "Analysis and compensation of latencies in NTS-secured NTP time synchronization," in *Proc. Joint Conf. IEEE Int. Freq. Control Symp. Int. Symp. Appl. Ferroelectr. (IFCS-ISAF)*, Jul. 2020, pp. 1–10.
- [269] D. Kopp, C. Dietzel, and O. Hohlfeld, "DDoS never dies? An IXP perspective on DDoS amplification attacks," in *Proc. Int. Conf. Passive Act. Netw. Meas.* Cham, Switzerland: Springer, Jan. 2021, pp. 284–301.
- [270] D. Dobrin and A. Dimiter, "DDoS attack identification based on SDN," in *Proc. IEEE 20th Int. Symp. Netw. Comput. Appl. (NCA)*, Nov. 2021, pp. 1–8.
- [271] G. Duan, Q. Li, Z. Zhang, D. Zhao, G. Xie, Y. Yang, Z. Yuan, Y. Jiang, and M. Xu, "DNSGuard: In-network defense against DNS attacks," *IEEE Trans. Depend. Secure Comput.*, vol. 22, no. 1, pp. 597–613, Jan. 2025.
- [272] E. Sagatov, S. Mayhoub, A. Sukhov, and P. Calyam, "Countering DNS amplification attacks based on analysis of outgoing traffic," *J. Commun. Inf. Netw.*, vol. 8, no. 2, pp. 111–121, Jun. 2023.
- [273] X. Z. Khooi, L. Csikor, D. M. Divakaran, and M. S. Kang, "DIDA: Distributed in-network defense architecture against amplified reflection DDoS attacks," in *Proc. 6th IEEE Conf. Netw. Softwarization (NetSoft)*, Jun. 2020, pp. 277–281.
- [274] O. van der Toorn, J. Krupp, M. Jonker, R. van Rijswijk-Deij, C. Rossow, and A. Sperotto, "ANYway: Measuring the amplification DDoS potential of domains," in *Proc. 17th Int. Conf. Netw. Service Manage. (CNSM)*, Oct. 2021, pp. 500–508.
- [275] S. Adiwali, B. Rajendran, D. P. Shetty, and S. D. Sudarsan, "DNS intrusion detection (DID)—A SNORT-based solution to detect DNS amplification and DNS tunneling attacks," *Franklin Open*, vol. 2, Mar. 2023, Art. no. 100010.
- [276] B. A. Sassani, A. Palle, S. Dhakal, S. Bobuwala, and A. David, "Analysis of SSDP DRDoS attack's performance effects and mitigation techniques," in *Proc. Int. Conf. Futuristic Technol. (INCOFT)*, Nov. 2022, pp. 1–5.

- [277] H. S. Gurjar and G. Somani, "Amplification/reflection attack suppression using victim separation," *IEEE Neww. Lett.*, vol. 5, no. 2, pp. 140–143, Apr. 2023.
- [278] X. Liu, L. Zheng, S. Helal, W. Zhang, C. Jia, and J. Zhou, "A broad learning-based comprehensive defence against SSDP reflection attacks in IoTs," *Digit. Commun. Netw.*, vol. 9, no. 5, pp. 1180–1189, Oct. 2023.
- [279] M. Nawrocki, J. Kristoff, R. Hiesgen, C. Kanich, T. C. Schmidt, and M. Wählisch, "SoK: A data-driven view on methods to detect reflective amplification DDoS attacks using honeypots," in *Proc. IEEE 8th Eur. Symp. Secur. Privacy (EuroS&P)*, Jul. 2023, pp. 576–591.
- [280] D. Tang, X. Wang, K. Li, C. Yin, W. Liang, and J. Zhang, "FAPM: A fake amplification phenomenon monitor to filter DRDoS attacks with P4 data plane," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6703–6715, Dec. 2024.
- [281] M. Anagnostopoulos, S. Lagos, and G. Kambourakis, "Large-scale empirical evaluation of DNS and SSDP amplification attacks," *J. Inf. Secur. Appl.*, vol. 66, May 2022, Art. no. 103168.
- [282] J. J. C. Gondim, R. de Oliveira Albuquerque, and A. L. S. Orozco, "Mirror saturation in amplified reflection distributed denial of service: A case of study using SNMP, SSDP, NTP and DNS protocols," *Future Gener. Comput. Syst.*, vol. 108, pp. 68–81, Jul. 2020.
- [283] R. Gayathri, S. Usharani, M. Mahdal, R. Vezhavendhan, R. Vincent, M. Rajesh, and M. Elangovan, "Detection and mitigation of IoT-based attacks using SNMP and moving target defense techniques," *Sensors*, vol. 23, no. 3, p. 1708, Feb. 2023.
- [284] J. C. Merlino, M. Asiri, and N. Saxena, "DDoS cyber-incident detection in smart grids," *Sustainability*, vol. 14, no. 5, p. 2730, Feb. 2022.
- [285] A. Pashamokhtari, G. Batista, and H. H. Gharakheili, "AdIoTack: Quantifying and refining resilience of decision tree ensemble inference models against adversarial volumetric attacks on IoT networks," *Comput. Secur.*, vol. 120, Sep. 2022, Art. no. 102801.
- [286] S. Lin, "Research on security of UDP reflection attack in civil aviation information network based on SDN," in *Proc. IEEE 2nd Int. Conf. Civil Aviation Saf. Inf. Technol. (ICCSIT)*, Oct. 2020, pp. 210–216.
- [287] Y.-C. Wang and C.-Y. Wu, "EC-NTD: Efficient countermeasure against DrDoS attacks with NAPT and two-stage detection in SDN-based networks," *Comput. Netw.*, vol. 250, Aug. 2024, Art. no. 110594.
- [288] B. J. S. Kumar and V. S. S. Gowda, "Detection and prevention of UDP reflection amplification attack in WSN using cumulative sum algorithm," in *Proc. IEEE Int. Conf. Data Sci. Inf. Syst. (ICDSIS)*, Jul. 2022, pp. 1–5.
- [289] K. Liu, Q. Wang, J. Yin, L. Du, and T. Zang, "Revisiting open DNS resolver vulnerabilities to reflection-based DDoS threats," in *Proc. 27th Int. Conf. Comput. Supported Cooperat. Work Design (CSCWD)*, May 2024, pp. 3158–3163.
- [290] B. Rajendran and P. Shetty D., "DNS amplification & DNS tunneling attacks simulation, detection and mitigation approaches," in *Proc. Int. Conf. Inventive Comput. Technol. (ICICT)*, Feb. 2020, pp. 230–236.
- [291] J. Davis and C. Deccio, "Advertising DNS protocol use to mitigate DDoS attacks," in *Proc. IEEE 29th Int. Conf. Netw. Protocols (ICNP)*, Nov. 2021, pp. 1–11.
- [292] E. Navruzov and A. Kabulov, "Detection and analysis types of DDoS attack," in *Proc. IEEE Int. IoT, Electron. Mechatronics Conf. (IEMTRONICS)*, Jun. 2022, pp. 1–7.
- [293] R. Maciel, J. Araujo, C. Melo, P. Pereira, J. Dantas, J. Mendonça, and P. Maciel, "Impact evaluation of DDoS attacks using IoT devices," in *Proc. IEEE Int. Syst. Conf. (SysCon)*, Apr. 2021, pp. 1–8.



ABDINASIR HIRSI (Graduate Student Member, IEEE) received the B.S. degree in telecommunication engineering from Mohammad Ali Jinnah University, Karachi, Pakistan, in 2019, and the M.S. degree in electrical engineering, specializing in communication engineering from Bahria University, Karachi, in 2021. He is currently pursuing the Ph.D. degree in electrical engineering with the Universiti Tun Hussein Onn Malaysia (UTHM), Johor, Malaysia. He is also a Graduate Research

Assistant (GRA) with the Advance Telecommunication Research Center (ATRC), FKKEE, UTHM. His research interests include SDN security, DDoS attack detection and mitigation, cybersecurity, and AI techniques, such as machine learning and deep learning techniques for network intrusion detection. He is an Active Reviewer of IEEE ACCESS, having reviewed over 13 articles in related fields.



MOHAMMED A. ALHARTOMI (Member, IEEE) received the Ph.D. degree in electronic and electrical engineering from Leeds University, U.K., in 2016. He is currently an Assistant Professor with the Department of Electrical Engineering, University of Tabuk. His research interests include wireless and mobile communications, signal processing, optical wireless systems design, and visible light communications.



LUKMAN AUDAH (Member, IEEE) received the Bachelor of Engineering degree in telecommunications from the Universiti Teknologi Malaysia, in 2005, and the M.Sc. degree in communication networks and software and the Ph.D. degree in electronic engineering from the University of Surrey, U.K. He is currently a Senior Lecturer with the Communication Engineering Department, Universiti Tun Hussein Onn Malaysia. His research interests include wireless and mobile communications, internet traffic engineering, network system management, data security, and satellite communications.



ADEB SALH (Member, IEEE) received the bachelor's degree in electrical and electronic engineering from IBB University, Yemen, in 2007, and the master's and Ph.D. degrees in electrical and electronic engineering from the Universiti Tun Hussein Onn Malaysia, in 2015 and 2020, respectively. From 2007 to 2012, he was a Lecturer Assistant with the Yareem Community College. From 2020 to 2023, he was a Postdoctoral Researcher with UTHM and UTM. Currently, he is an Assistant Professor with the Faculty of Information and Communication Technology, Universiti Tunku Abdul Rahman. His research interests include 5G, 6G wireless communications, massive MIMO, artificial intelligence (AI), and the Internet of Things (IoT).



NAN MAD SAHAR received the Ph.D. degree in electrical engineering, specializing in neural network computing and computer engineering from Okayama University. He is a Senior Lecturer with the Department of Electronic Engineering, Faculty of Electrical and Electronic Engineering, Universiti Tun Hussein Onn Malaysia (UTHM). He is actively involved as a Principal Researcher with UTHM, focusing on computational intelligence, deep learning, and image processing. With extensive experience in supervising postgraduate research and contributing to academia, his work extends into consultation, technical training, and industry-relevant projects across various engineering fields.



SALMAN AHMED (Graduate Student Member, IEEE) received the bachelor's degree in electronic engineering from the Mehran University of Engineering and Technology, Jamshoro, Pakistan, in 2015, and the Master of Engineering degree in industrial automation and control from Quaid-e-Awam University of Science and Technology, Nawabshah, Pakistan, in 2021. He is currently pursuing the Ph.D. degree with the Faculty of Electrical and Electronics Engineering, Universiti Tun Hussein Onn Malaysia (UTHM). He is also a Graduate Research Assistant (GRA) with the Faculty of Electrical and Electronics Engineering, UTHM. He was a Lecturer in electronics engineering with Sukkur IBA University, Pakistan, from 2017 to 2022. His research interests include network security, cryptographic, and resource-constrained Internet of Things devices.



GODWIN OKON ANSA (Member, IEEE) received the B.Sc. degree in computer science from the University of Uyo, Nigeria, in 2003, and the M.Sc. degree in communications networks and software and the Ph.D. degree in electronic engineering (network security) from the University of Surrey, Guildford, U.K., in 2008 and 2013, respectively. He is a Senior Lecturer and the Postgraduate Program Coordinator with the Department of Computer Science, Akwa Ibom State University, Nigeria. Previously, he was the Ag. Head of the Department of Computer Science, Akwa Ibom State University, from 2013 to 2015 and from 2017 to 2020. His research interests include network security, artificial intelligence, delay tolerant networks, cloud computing security, and the Internet of Things. He is a member of Nigeria Computer Society (NCS),

the Computer Professionals [Registration Council] of Nigeria (CPN), and the International Association of Engineers (IAENG). He is a member of the Editorial Advisory Board of *Journal of ICT Development, Applications and Research*; and *Researchers Journal of Science and Technology* (REJOST).



ABDULLAHI FARAH received the Bachelor of Science degree in computer science from Mogadishu University, Somalia, in 2012, and the Master of Science degree in information technology from Open University Malaysia, in 2015. He is a distinguished telecommunications professional with over 12 years of comprehensive experience in the industry. He is currently the Technical Operations Manager with Somtel Telecom, Somalia, where he oversees and supervises technical teams across critical domains, including intelligent networks, network operation centers, radio development centers, transmission, IP, power systems, and various other technical projects. His expertise spans advanced telecommunications technologies, such as 4G and 5G, bolstered by specialized training workshops conducted by industry leaders, such as Huawei and Nokia. Over the years, he has played a pivotal role in transformative projects, including spearheading the fiber optic cabling project in Somalia, where Somtel connected the entire country through an extensive fiber optic networks. Starting his career at Somtel, he has risen through various roles, eventually becoming a Project Planning Manager, a position in which he demonstrated strong leadership, adaptability, and innovation. His consistent delivery of exceptional results and his ability to lead complex projects have solidified his reputation as a key figure in Somalia's telecommunications industry. He remains committed to driving technological advancements and fostering growth in the sector.

...